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United States Patent	12411408
Kind Code	B2
Date of Patent	September 09, 2025
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Resist composition and method of forming resist pattern

Abstract

A resist composition containing a resin component (A1) having a constitutional unit (a01) derived from a compound represented by General Formula (a0-1) and containing a compound (D0) represented by General Formula (d0), W.sup.01 represents a polymerizable group-containing group, Ya.sup.01 represents a single bond or the like, Rx.sup.01 represents an acid dissociable group, M.sup.m+ represents an m-valent organic cation, and R.sup.d0 represents a substituent
##STR00001##

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Appl. No.: 17/450150

Filed: October 06, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220121118 A1	Apr. 21, 2022

Foreign Application Priority Data

JP	2020-175441	Oct. 19, 2020
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Publication Classification

Int. Cl.: G03F7/038 (20060101); G03F7/20 (20060101); G03F7/32 (20060101)

U.S. Cl.:

CPC G03F7/038 (20130101); G03F7/2004 (20130101); G03F7/32 (20130101);

Field of Classification Search

USPC: None

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a resist composition and a method of forming a resist pattern.

(2) Priority is claimed on Japanese Patent Application No. 2020-175441, filed on Oct. 19, 2020, the content of which is incorporated herein by reference.

Description of Related Art

(3) In recent years, in the production of semiconductor elements and liquid crystal display elements, advances in lithography techniques have led to a rapid progress in the field of pattern miniaturization. Typically, these miniaturization techniques involve shortening the wavelength (increasing the energy) of the light source for exposure.

(4) Resist materials for use with these types of light sources for exposure require lithography characteristics such as a high resolution capable of reproducing a fine-sized pattern, and a high level of sensitivity to these types of light sources for exposure.

(5) As a resist material that satisfies these requirements, a chemically amplified resist composition which contains a base material component exhibiting changed solubility in a developing solution under action of acid, and an acid generator component that generates acid upon exposure has been used in the related art.

(6) In the formation of the resist pattern, the behavior of acid generated from an acid generator component upon exposure is considered as one factor that has a great influence on lithography characteristics.

(7) On the other hand, a chemically amplified resist composition having both an acid generator component and an acid diffusion controlling agent that controls the diffusion of acid generated from the acid generator component upon exposure has been proposed.

(8) For example, Patent Document 1 discloses a resist composition containing a resin component exhibiting changed solubility in a developing solution under action of acid, an acid generator component, and a photoreactive quencher having a specific structure, as an acid diffusion controlling agent. This photoreactive quencher is said to be a component that exhibits a quenching effect by causing an ion exchange reaction with an acid generated from an acid generator component. In a case where such a photoreactive quencher is blended, the diffusion of the acid generated from the acid generator component from exposed portions to unexposed portions of the resist film is controlled, and the lithography characteristics are improved.

CITATION LIST

Patent Document

(9) [Patent Document 1] Japanese Unexamined Patent Application, First Publication No. 2013-200560.

SUMMARY OF THE INVENTION

(10) With the further progress of lithography technology and the miniaturization of a resist pattern, for example, it is aimed to form a fine pattern of several tens of nanometers in lithography by EUV and EB. As the resist pattern size becomes smaller as described above, it is required that the resist material improves each of the lithography characteristics such as sensitivity, resolution, and roughness reduction property without the trade-off thereof. In addition, it is also required that a resist pattern having high rectangularity can be formed.

- (11) The present invention has been made in consideration of the above circumstances, and an object of the present invention is to provide a resist composition having good sensitivity, good resolution, good roughness reducing property, and good pattern shape, and a method of forming a resist pattern using the resist composition.
- (12) In order to achieve the above-described object, the present invention employs the following configurations.
- (13) That is, the first aspect according to the present invention is a resist composition which generates acid upon exposure and exhibits changed solubility in a developing solution under action of acid, which contains a resin component (A1) exhibiting changed solubility in a developing solution under action of acid and a compound (D0) represented by General Formula (d0), where the resin component (A1) has a constitutional unit (a01) derived from a compound represented by General Formula (a0-1).
- (14) ##STR00002##
- (15) [In the formula, M^{m+} represents an m-valent organic cation. m represents an integer of 1 or more. R^{d0} represents a substituent. p_0 represents an integer in a range of 0 to 3. In a case where p_0 represents 2 or 3, a plurality of R^{d0} 's may be the same or different from each other. q_0 represents an integer in a range of 0 to 3. n represents an integer of 1 or more. However, $n_0 + p_0 \leq (q_0 \times 2) + 5$ is satisfied.]
- (16) ##STR00003##
- (17) [In the formula, W^{01} represents a polymerizable group-containing group. Y^{01} represents a single bond or a divalent linking group. R^{01} represents an acid dissociable group. q represents an integer in a range of 0 to 3. n represents an integer of 1 or more. However, $n \leq q \times 2 + 4$ is satisfied.]
- (18) The second aspect according to the present invention is a method of forming a resist pattern, including a step of forming a resist film on a support using the resist composition according to the first aspect, a step of exposing the resist film, and a step of developing the exposed resist film to form a resist pattern.
- (19) According to the present invention, it is possible to provide a resist composition having good sensitivity, good resolution, good roughness reducing property, and good pattern shape, and a method of forming a resist pattern using the resist composition.
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Description

DETAILED DESCRIPTION OF THE INVENTION

- (1) In the present specification and the scope of the present claims, the “aliphatic” is a relative concept used with respect to the “aromatic” and defines a group or compound that has no aromaticity.
- (2) The “alkyl group” includes linear, branched or cyclic, monovalent saturated hydrocarbon group, unless otherwise specified. The same applies to the alkyl group in an alkoxy group.
- (3) The “alkylene group” includes linear, branched or cyclic, divalent saturated hydrocarbon group, unless otherwise specified.
- (4) Examples of the “halogen atom” include a fluorine atom, a chlorine atom, a bromine atom, and an iodine atom.
- (5) The “constitutional unit” indicates a monomer unit that constitutes the formation of a polymeric compound (a resin, a polymer, or a copolymer).
- (6) In a case where “may have a substituent” is described, both of a case where a hydrogen atom (—H) is substituted with a monovalent group and a case where a methylene group (—CH₂—) is substituted with a divalent group are included.
- (7) The “exposure” is used as a general concept that includes irradiation with any form of radiation.

(8) The “acid decomposable group” indicates a group having acid decomposability, in which at least part of bonds in the structure of the acid decomposable group can be cleaved under action of acid.

(9) Examples of the acid decomposable group having a polarity which is increased under action of acid include groups which are decomposed under action of acid to generate a polar group.

(10) Examples of the polar group include a carboxy group, a hydroxyl group, an amino group, and a sulfo group ($\text{—SO}_3\text{H}$).

(11) More specific examples of the acid decomposable group include a group (for example, a group obtained by protecting a hydrogen atom of the OH-containing polar group with an acid dissociable group) obtained by protecting the above-described polar group with an acid dissociable group.

(12) The “acid dissociable group” indicates any one of (i) a group in which a bond between the acid dissociable group and an atom adjacent to the acid dissociable group can be cleaved under action of acid; and (ii) a group in which part of bonds are cleaved under action of acid, and then a decarboxylation reaction occurs, thereby cleaving the bond between the acid dissociable group and the atom adjacent to the acid dissociable group.”

(13) It is necessary that the acid dissociable group that constitutes the acid decomposable group is a group that exhibits a lower polarity than the polar group generated by the dissociation of the acid dissociable group. Thus, in a case where the acid dissociable group is dissociated under action of acid, a polar group exhibiting a higher polarity than the acid dissociable group is generated, whereby the polarity increases. As a result of the above, the polarity of the entire component (A1) is increased. By the increase in the polarity, the solubility in a developing solution relatively changes. The solubility in a developing solution is increased in a case where the developing solution is an alkali developing solution, whereas the solubility in a developing solution is decreased in a case where the developing solution is an organic developing solution.

(14) The “base material component” is an organic compound having a film-forming ability. The organic compounds used as the base material component are roughly classified into a non-polymer and a polymer. As the non-polymer, those having a molecular weight of 500 or more and less than 4,000 are usually used. Hereinafter, a “low molecular weight compound” refers to a non-polymer having a molecular weight of 500 or more and less than 4,000. As the polymer, those having a molecular weight of 1,000 or more are usually used. Hereinafter, a “resin”, a “polymeric compound”, or a “polymer” refers to a polymer having a molecular weight of 1,000 or more. As the molecular weight of the polymer, a polystyrene-equivalent weight average molecular weight determined by gel permeation chromatography (GPC) is used.

(15) The “constitutional unit derived from” means a constitutional unit that is formed by the cleavage of a multiple bond between carbon atoms, for example, an ethylenic double bond.

(16) In the “acrylic acid ester”, the hydrogen atom bonded to the carbon atom at the α -position may be substituted with a substituent. The substituent ($\text{R}^{\text{sup.}\alpha}$) that is substituted for the hydrogen atom bonded to the carbon atom at the α -position is an atom other than the hydrogen atom, or a group. Further, an itaconic acid diester in which the substituent ($\text{R}^{\text{sup.}\alpha}$) is substituted with a substituent having an ester bond or an α -hydroxyacryl ester in which the substituent ($\text{R}^{\text{sup.}\alpha}$) is substituted with a hydroxyalkyl group or a group obtained by modifying a hydroxyl group of the hydroxyalkyl group can be included in the acrylic acid ester. A carbon atom at the α -position of acrylic acid ester indicates the carbon atom bonded to the carbonyl group of acrylic acid unless otherwise specified.

(17) Hereinafter, an acrylic acid ester obtained by substituting a hydrogen atom bonded to the carbon atom at the α -position with a substituent is also referred to as an α -substituted acrylic acid ester.

(18) The “derivative” includes a compound obtained by substituting a hydrogen atom at the α -position of an object compound with another substituent such as an alkyl group or a halogenated alkyl group; and a derivative thereof. Examples of the derivative thereof include a derivative

obtained by substituting the hydrogen atom of a hydroxyl group of an object compound in which a hydrogen atom at the α -position may be substituted with a substituent, with an organic group; and a derivative obtained by bonding a substituent other than the hydroxyl group to an object compound in which a hydrogen atom at the α -position may be substituted with a substituent. The α -position refers to the first carbon atom adjacent to the functional group unless otherwise specified.

(19) Examples of the substituent that is substituted for the hydrogen atom at the α -position of hydroxystyrene include the same one as R.sup. α x.

(20) In the present specification and the scope of the present claims, asymmetric carbon atoms may be present, and thus enantiomers or diastereomers may be present depending on the structures represented by the chemical formula. In that case, these isomers are represented by one chemical formula. These isomers may be used alone or in the form of a mixture.

(21) (Resist Composition)

(22) The resist composition according to the present embodiment is a resist composition that generates acid upon exposure and exhibits changed solubility in a developing solution under action of acid.

(23) Such a resist composition contains a base material component (A) (hereinafter, also referred to as a “component (A)”) exhibiting changed solubility in a developing solution under action of acid, and a compound (D0) (hereinafter, also referred to as a “compound (D0)”) represented by General Formula (d0).

(24) In a case where a resist film is formed using the resist composition according to the present embodiment and the formed resist film is subjected to selective exposure, an acid is generated at exposed portions of the resist film, and the generated acid acts on the component (A) to change the solubility of the component (A) in a developing solution, whereas the solubility of the component (A) in a developing solution is not changed at unexposed portions of the resist film, which generates the difference in solubility in the developing solution between the exposed portion and unexposed portions of the resist film.

(25) The resist composition according to the present embodiment may be a positive-tone resist composition or a negative-tone resist composition.

(26) Further, in the formation of a resist pattern, the resist composition according to the present embodiment may be applied to an alkali developing process using an alkali developing solution in the developing treatment, or a solvent developing process using an organic developing solution in the developing treatment.

(27) That is, the resist composition according to the present embodiment is a “positive-tone resist composition for an alkali developing process” that forms a positive-tone resist pattern in an alkali developing process and is a “negative-tone resist composition for a solvent developing process” that forms a negative-tone resist pattern in a solvent developing process.

(28) <Component (A)>

(29) In the resist composition according to the present embodiment, the component (A) contains a resin component (A1) (hereinafter, also referred to as a “component (A1)”) exhibiting changed solubility in a developing solution under action of acid, and the resin component (A1) has a constitutional unit (a01) derived from a compound represented by General Formula (a0-1).

(30) As the component (A), at least the component (A1) is used, and at least one of another polymeric compound and another low molecular weight compound may be used in combination with the component (A1).

(31) In the resist composition according to the present embodiment, the component (A) may be used alone or in a combination of two or more kinds thereof.

(32) In Regard to Component (a1):

(33) The component (A1) has a constitutional unit (a01).

(34) <<Constitutional Unit (a01)>>

(35) The constitutional unit (a01) is a constitutional unit derived from a compound represented by

General Formula (a0-1).

(36) ##STR00004##

(37) [In the formula, W^{sup.01} represents a polymerizable group-containing group. Ya^{sup.01} represents a single bond or a divalent linking group. Rx^{sup.01} represents an acid dissociable group. q represents an integer in a range of 0 to 3. n represents an integer of 1 or more. However, $n \leq q \times 2 + 4$ is satisfied.]

(38) In General Formula (a0-1), W^{sup.01} represents a polymerizable group-containing group.

(39) The “polymerizable group” as W^{sup.01} is a group that enables a compound having the polymerizable group to be polymerized by radical polymerization or the like, and includes a group containing a multiple bond between carbon atoms, such as an ethylenic double bond.

(40) In the constitutional unit (a01), the multiple bonds in the polymerizable group are cleaved to form a main chain.

(41) Examples of the polymerizable group as W^{sup.01} include a vinyl group, an allyl group, acryloyl group, a methacryloyl group, a fluorovinyl group, a difluorovinyl group, a trifluorovinyl group, a difluorotrifluoromethylvinyl group, a trifluoroallyl group, a perfluoroallyl group, a trifluoromethylacryloyl group, a nonylfluorobutylacryloyl group, a vinyl ether group, a fluorine-containing vinyl ether group, an allyl ether group, a fluorine-containing allyl ether group, a styryl group, and a vinylnaphthyl group, a fluorine-containing styryl group, a fluorine-containing vinylnaphthyl group, a norbornyl group, a fluorine-containing norbornyl group, and a silyl group.

(42) The “polymerizable group-containing group” as W^{sup.01} may be a group composed of only a polymerizable group, or a group composed of a polymerizable group and a group other than the polymerizable group. Examples of the group other than the polymerizable group include a divalent hydrocarbon group which may have a substituent and a divalent linking group containing a hetero atom.

(43) Divalent Hydrocarbon Group which May have Substituent:

(44) In a case where the group other than the polymerizable group represents a divalent hydrocarbon group which may have a substituent, the hydrocarbon group may be an aliphatic hydrocarbon group or an aromatic hydrocarbon group.

(45) Aliphatic Hydrocarbon Group as a Group Other than the Polymerizable Group

(46) The aliphatic hydrocarbon group indicates a hydrocarbon group that has no aromaticity. The aliphatic hydrocarbon group may be saturated or unsaturated. In general, it is preferable that the aliphatic hydrocarbon group is saturated.

(47) Examples of the aliphatic hydrocarbon group include a linear or branched aliphatic hydrocarbon group, and an aliphatic hydrocarbon group containing a ring in the structure thereof.

(48) Linear or Branched Aliphatic Hydrocarbon Group

(49) The linear aliphatic hydrocarbon group described above preferably has 1 to 10 carbon atoms, more preferably 1 to 6 carbon atoms, still more preferably 1 to 4 carbon atoms, and most preferably 1 to 3 carbon atoms.

(50) The linear aliphatic hydrocarbon group is preferably a linear alkylene group, and specific examples thereof include a methylene group [—CH_{sub.2}—], an ethylene group [—

(CH_{sub.2})_{sub.2}—], a trimethylene group [—(CH_{sub.2})_{sub.3}—], a tetramethylene group [—(CH_{sub.2})_{sub.4}—], and a pentamethylene group [—(CH_{sub.2})_{sub.5}—].

(51) The branched aliphatic hydrocarbon group described above preferably has 2 to 10 carbon atoms, more preferably 3 to 6 carbon atoms, still more preferably 3 or 4 carbon atoms, and most preferably 3 carbon atoms.

(52) The branched aliphatic hydrocarbon group is preferably a branched alkylene group, and specific examples thereof include alkylalkylene groups, for example, alkylmethylene groups such as —CH(CH_{sub.3})—, —CH(CH_{sub.2}CH_{sub.3})—, —C(CH_{sub.3})_{sub.2}—, —C(CH_{sub.3})

(CH_{sub.2}CH_{sub.3})—, —C(CH_{sub.3})(CH_{sub.2}CH_{sub.2}CH_{sub.3})—, and —

C(CH_{sub.2}CH_{sub.3})_{sub.2}—; alkylethylene groups such as —CH(CH_{sub.3})CH_{sub.2}—, —

CH(CH.sub.3)CH(CH.sub.3)—, —C(CH.sub.3).sub.2CH.sub.2—, —CH(CH.sub.2CH.sub.3)CH.sub.2—, and —C(CH.sub.2CH.sub.3).sub.2—CH.sub.2—; alkyltrimethylene groups such as —CH(CH.sub.3)CH.sub.2CH.sub.2—, and —CH.sub.2CH(CH.sub.3)CH.sub.2—; and alkyltetramethylene groups such as —CH(CH.sub.3)CH.sub.2CH.sub.2CH.sub.2—, and —CH.sub.2CH(CH.sub.3)CH.sub.2CH.sub.2—.

The alkyl group in the alkylalkylene group is preferably a linear alkyl group having 1 to 5 carbon atoms.

(53) The linear or branched aliphatic hydrocarbon group may have or may not have a substituent. Examples of the substituent include a fluorine atom, a fluorinated alkyl group having 1 to 5 carbon atoms, which has been substituted with a fluorine atom, and a carbonyl group.

(54) Aliphatic Hydrocarbon Group Containing Ring in Structure Thereof

(55) Examples of the aliphatic hydrocarbon group containing a ring in the structure thereof include a cyclic aliphatic hydrocarbon group which may have a substituent containing a hetero atom in the ring structure thereof (a group obtained by removing two hydrogen atoms from an aliphatic hydrocarbon ring), a group obtained by bonding the cyclic aliphatic hydrocarbon group to the terminal of a linear or branched aliphatic hydrocarbon group, and a group obtained by interposing the cyclic aliphatic hydrocarbon group in a linear or branched aliphatic hydrocarbon group. Examples of the linear or branched aliphatic hydrocarbon group include the same ones as those described above.

(56) The cyclic aliphatic hydrocarbon group preferably has 3 to 20 carbon atoms and more preferably 3 to 12 carbon atoms.

(57) The cyclic aliphatic hydrocarbon group may be a polycyclic group or a monocyclic group. The monocyclic alicyclic hydrocarbon group is preferably a group obtained by removing two hydrogen atoms from a monocycloalkane. The monocycloalkane preferably has 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane. The polycyclic alicyclic hydrocarbon group is preferably a group obtained by removing two hydrogen atoms from a polycycloalkane, and the polycycloalkane is preferably a group having 7 to 12 carbon atoms.

(58) Specific examples of the polycyclic alicyclic hydrocarbon group include adamantane, norbornane, isobornane, tricyclodecane, and tetracyclododecane.

(59) The cyclic aliphatic hydrocarbon group may have or may not have a substituent. Examples of the substituent include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, and a carbonyl group.

(60) The alkyl group as the substituent is preferably an alkyl group having 1 to 5 carbon atoms, and most preferably a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group.

(61) The alkoxy group as the substituent is preferably an alkoxy group having 1 to 5 carbon atoms, more preferably a methoxy group, an ethoxy group, an n-propoxy group, an iso-propoxy group, an n-butoxy group, or a tert-butoxy group, and most preferably a methoxy group or an ethoxy group.

(62) Examples of the halogen atom for the substituent include a fluorine atom, a chlorine atom, a bromine atom, and an iodine atom, and a fluorine atom is preferable.

(63) Examples of the halogenated alkyl group as the substituent include a group obtained by substituting part or all hydrogen atoms in the above-described alkyl groups with the above-described halogen atoms.

(64) In the cyclic aliphatic hydrocarbon group, part of carbon atoms constituting the ring structure thereof may be substituted with a substituent containing a hetero atom. The substituent containing a hetero atom is preferably —O—, —C(=O)—O—, —S—, —S(=O).sub.2—, or —S(=O).sub.2—O—.

(65) Aromatic Hydrocarbon Group as a Group Other than the Polymerizable Group

(66) The aromatic hydrocarbon group is a hydrocarbon group having at least one aromatic ring.

(67) The aromatic ring is not particularly limited as long as it is a cyclic conjugated system having

($4n+2$) π electrons, and may be monocyclic or polycyclic. The aromatic ring preferably has 5 to 30 carbon atoms, more preferably 5 to 20 carbon atoms, still more preferably 6 to 15 carbon atoms, and particularly preferably 6 to 12 carbon atoms. Here, the number of carbon atoms in a substituent is not included in the number of carbon atoms. Specific examples of the aromatic ring include aromatic hydrocarbon rings such as benzene, naphthalene, anthracene, and phenanthrene; and an aromatic heterocyclic ring obtained by substituting part of carbon atoms constituting the above-described aromatic hydrocarbon ring with a hetero atom. Examples of the hetero atom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom. Specific examples of the aromatic heterocyclic ring include a pyridine ring and a thiophene ring.

(68) Specific examples of the aromatic hydrocarbon group include a group (an arylene group or a heteroarylene group) obtained by removing two hydrogen atoms from the above-described aromatic hydrocarbon ring or the above-described aromatic heterocyclic ring; a group obtained by removing two hydrogen atoms from an aromatic compound (for example, biphenyl or fluorene) having two or more aromatic rings; and a group (for example, a group obtained by further removing one hydrogen atom from an aryl group in the arylalkyl group such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, a 1-naphthylethyl group, or a 2-naphthylethyl group) obtained by substituting one hydrogen atom of a group (an aryl group or a heteroaryl group) obtained by removing one hydrogen atom from the above aromatic hydrocarbon ring or the above aromatic heterocyclic ring, with an alkylene group. The alkylene group bonded to the aryl group or the heteroaryl group preferably has 1 to 4 carbon atoms, more preferably 1 or 2 carbon atoms, and particularly preferably 1 carbon atom.

(69) With respect to the aromatic hydrocarbon group, the hydrogen atom contained in the aromatic hydrocarbon group may be substituted with a substituent. For example, the hydrogen atom bonded to the aromatic ring in the aromatic hydrocarbon group may be substituted with a substituent. Examples of substituents include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, and a hydroxyl group.

(70) The alkyl group as the substituent is preferably an alkyl group having 1 to 5 carbon atoms, and most preferably a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group.

(71) Examples of the alkoxy group, the halogen atom, and the halogenated alkyl group, as the substituent, include the same groups as those exemplified as the substituent that is substituted for a hydrogen atom which the cyclic aliphatic hydrocarbon group has.

(72) Divalent Linking Group Containing Hetero Atom

(73) In a case where the group other than the polymerizable group represents a divalent linking group containing a hetero atom, preferred examples of the linking group include —O— , —C(=O)— , —O—C(=O)— , —O—C(=O)—O— , —C(=O)—NH— , —NH— , —NH—C(=NH)— (H may be substituted with a substituent such as an alkyl group, an acyl group, or the like), —S— , —S(=O).sub.2— , —S(=O).sub.2—O— , and a group represented by General Formula $\text{—Y.sup.21—O—Y.sup.22—}$, —Y.sup.21—O— , $\text{—Y.sup.21—C(=O)—O—Y.sup.21—}$, $[\text{Y.sup.21—C(=O)—O}]_{\text{sub.m''}}\text{—Y.sup.22—}$, $\text{—Y.sup.21—O—C(=O)—Y.sup.22—}$ or $\text{—Y.sup.21—S(=O).sub.2—O—Y.sup.22—}$ [in the formulae, Y.sup.21 and Y.sup.22 each independently represent a divalent hydrocarbon group which may have a substituent, 0 represents an oxygen atom, and m'' represents an integer in a range of 0 to 3].

(74) In a case where the divalent linking group containing a hetero atom is —C(=O)—NH— , —C(=O)—NH—C(=O)— , —NH— , or —NH—C(=NH)— , H may be substituted with a substituent such as an alkyl group, an acyl group, or the like. The substituent (an alkyl group, an acyl group, or the like) preferably has 1 to 10 carbon atoms, more preferably 1 to 8 carbon atoms, and particularly preferably 1 to 5 carbon atoms.

(75) In General Formulae $\text{—Y.sup.21—O—Y.sup.22—}$, —Y.sup.21—O— , —Y.sup.21C(=O)—O— , $\text{—C(=O)—O—Y.sup.21—}$, $\text{—[Y.sup.21—C(=O)—O]_{sub.m''}—Y.sup.22—}$, —Y.sup.21—O—

$C(=O)-Y_{sup.22}-$, and $-Y_{sup.21}-S(=O)_{sub.2}-O-Y_{sup.22}-$, $Y_{sup.21}$, and $Y_{sup.22}$ each independently represent a divalent hydrocarbon group which may have a substituent. Examples of the divalent hydrocarbon group include the same one as the divalent hydrocarbon groups which may have a substituent, mentioned in the explanation of the above-described divalent linking group.

(76) $Y_{sup.21}$ is preferably a linear aliphatic hydrocarbon group, more preferably a linear alkylene group, still more preferably a linear alkylene group having 1 to 5 carbon atoms, and particularly preferably a methylene group or an ethylene group.

(77) $Y_{sup.22}$ is preferably a linear or branched aliphatic hydrocarbon group and more preferably a methylene group, an ethylene group, or an alkylmethylene group. The alkyl group in the alkylmethylene group is preferably a linear alkyl group having 1 to 5 carbon atoms, more preferably a linear alkyl group having 1 to 3 carbon atoms, and most preferably a methyl group.

(78) In the group represented by General Formula $-[Y_{sup.21}-C(=O)-O]_{sub.m''}-Y_{sup.22}-$, m'' represents an integer in a range of 0 to 3, preferably an integer in a range of 0 to 2, more preferably 0 or 1, and particularly preferably 1. In other words, it is particularly preferable that the group represented by Formula $-[Y_{sup.21}-C(=O)-O]_{sub.m''}-Y_{sup.22}-$ represents a group represented by Formula $-Y_{sup.21}-C(=O)-O-Y_{sup.22}-$. Among the above, a group represented by Formula $-(CH_{sub.2})_{sub.a'}-C(=O)-O-(CH_{sub.2})_{sub.b'}-$ is preferable. In the formula, a' represents an integer in a range of 1 to 10, preferably an integer in a range of 1 to 8, more preferably an integer in a range of 1 to 5, still more preferably 1 or 2, and most preferably 1. b' represents an integer in a range of 1 to 10, preferably an integer in a range of 1 to 8, more preferably an integer in a range of 1 to 5, still more preferably 1 or 2, and most preferably 1.

(79) Suitable examples of $W_{sup.01}$ include a group represented by a chemical formula:

$C(R_{sup.X11})(R_{sup.X12})=C(R_{sup.X13})-Y_{a.sup.x0}$.

(80) In the chemical formula, $R_{sup.X11}$, $R_{sup.X12}$, and $R_{sup.X13}$ each represent a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, and $Y_{a.sup.x0}$ represents a single bond or a divalent linking group.

(81) The alkyl group having 1 to 5 carbon atoms as $R_{sup.X11}$, $R_{sup.X12}$, and $R_{sup.X13}$ is preferably a linear or branched alkyl group having 1 to 5 carbon atoms, and specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, and a neopentyl group. The halogenated alkyl group having 1 to 5 carbon atoms is a group obtained by substituting part or all hydrogen atoms in the alkyl group having 1 to 5 carbon atoms with a halogen atom. The halogen atom is particularly preferably a fluorine atom.

(82) Among these, $R_{sup.x11}$ and $R_{sup.X12}$ are each preferably a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a fluorinated alkyl group having 1 to 5 carbon atoms, and in terms of industrial availability, a hydrogen atom or a methyl group is more preferable, and a hydrogen atom is particularly preferable.

(83) In addition, $R_{sup.x13}$ is preferably a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a fluorinated alkyl group having 1 to 5 carbon atoms, and in terms of industrial availability, a hydrogen atom or a methyl group is more preferable, and a hydrogen atom is particularly preferable.

(84) The divalent linking group as $Y_{a.sup.x0}$ is not particularly limited, and suitable examples thereof include a divalent hydrocarbon group which may have a substituent and a divalent linking group having a hetero atom, each of which is the same as that described above.

(85) Among the above, $Y_{a.sup.x0}$ is preferably an ester bond $[-C(=O)-O-$ or $-O-C(=O)-]$, an ether bond $(-O-)$, a linear or branched alkylene group, an aromatic hydrocarbon group, or a combination thereof, or a single bond. Among the above, an ester bond $[-C(=O)-O-$, $-O-C(=O)-]$, a combination of an ester bond $[-C(=O)-O-$, $-O-C(=O)-]$ and a linear alkylene group, or a single bond is more preferable, and a single bond is still more preferable.

(86) In General Formula (a0-1), Ya.sup.01 represents a single bond or a divalent linking group. The divalent linking group as Ya.sup.01 is not particularly limited, and suitable examples thereof include a divalent hydrocarbon group which may have a substituent and a divalent linking group having a hetero atom, each of which is the same as that described above.

(87) In General Formula (a0-1), Ya.sup.01 is preferably, among the above, an ester bond [—C(=O)—O— or —O—C(=O)—], an ether bond (—O—), a linear or branched alkylene group, an aromatic hydrocarbon group, or a combination thereof, or a single bond. Among the above, Ya.sup.01 is more preferably a combination of an ester bond [—C(=O)—O— or —O—C(=O)—] and a linear alkylene group, or a single bond, and still more preferably a single bond.

(88) In General Formula (a0-1), Rx.sup.01 represents an acid dissociable group.

(89) Specific examples of the acid dissociable group include an “acetal-type acid dissociable group” and a “tertiary alkyl ester-type acid dissociable group” described below.

(90) Acetal-Type Acid Dissociable Group:

(91) Examples of the acid dissociable group for protecting a carboxy group or a hydroxyl group include the acid dissociable group represented by General Formula (a0-r-1) shown below (hereinafter, also referred to as an “acetal-type acid dissociable group”).

(92) ##STR00005##

(93) [In General Formula (a0-r-1), Ra.sup.01 and Ra.sup.02 each independently represent a hydrogen atom or an alkyl group. Ra.sup.03 represents a hydrocarbon group and may be bonded to any one of Ra.sup.01 or Ra.sup.02 to form a ring. * represents a bonding site].

(94) In General Formula (a0-r-1), Ra.sup.01 and Ra.sup.02 each independently represent a hydrogen atom or an alkyl group.

(95) In General Formula (a0-r-1), it is preferable that at least one of Ra.sup.01 and Ra.sup.02 represents a hydrogen atom and more preferable that both Ra.sup.01 and Ra.sup.02 represent a hydrogen atom.

(96) In a case where Ra.sup.01 and Ra.sup.02 represent an alkyl group, the alkyl group is preferably an alkyl group having 1 to 5 carbon atoms. Specific examples thereof preferably include a linear or branched alkyl group. More specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, and a neopentyl group. Among these, a methyl group or an ethyl group is preferable, and a methyl group is particularly preferable.

(97) In General Formula (a0-r-1), Ra.sup.01 and Ra.sup.02 are each independently, among the above, preferably a hydrogen atom or an alkyl group having 1 to 5 carbon atoms and more preferably a hydrogen atom or a methyl group.

(98) In General Formula (a0-r-1), examples of the hydrocarbon group as Ra.sup.03 include a linear or branched alkyl group and a cyclic hydrocarbon group.

(99) The linear alkyl group has preferably 1 to 5 carbon atoms, more preferably 1 to 4 carbon atoms, and still more preferably 1 or 2 carbon atoms. Specific examples thereof include a methyl group, an ethyl group, an n-propyl group, an n-butyl group, and an n-pentyl group. Among these, a methyl group, an ethyl group, or an n-butyl group is preferable, and a methyl group or an ethyl group is more preferable.

(100) The branched alkyl group has preferably 3 to 10 carbon atoms and more preferably 3 to 5 carbon atoms. Specific examples thereof include an isopropyl group, an isobutyl group, a tert-butyl group, an isopentyl group, a neopentyl group, a 1,1-diethylpropyl group, and a 2,2-dimethylbutyl group, and an isopropyl group is preferable.

(101) In a case where Ra.sup.03 represents a cyclic hydrocarbon group, the hydrocarbon group may be an aliphatic hydrocarbon group or an aromatic hydrocarbon group and may be a polycyclic group or a monocyclic group.

(102) The aliphatic hydrocarbon group which is a monocyclic group is preferably a group obtained by removing one hydrogen atom from a monocycloalkane. The monocycloalkane preferably has 3

to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane.

(103) The aliphatic hydrocarbon group which is a polycyclic group is preferably a group obtained by removing one hydrogen atom from a polycycloalkane. The polycycloalkane preferably has 7 to 12 carbon atoms, and specific examples thereof include adamantane, norbornane, isobornane, tricyclodecane, and tetracyclododecane.

(104) In a case where the cyclic hydrocarbon group as Ra.sup.03 represents an aromatic hydrocarbon group, the aromatic hydrocarbon group is a hydrocarbon group having at least one aromatic ring.

(105) The aromatic ring is not particularly limited as long as it is a cyclic conjugated system having $(4n+2)$ π electrons, and may be monocyclic or polycyclic. The aromatic ring preferably has 5 to 30 carbon atoms, more preferably 5 to 20 carbon atoms, still more preferably 6 to 15 carbon atoms, and particularly preferably 6 to 12 carbon atoms.

(106) Specific examples of the aromatic ring include aromatic hydrocarbon rings such as benzene, naphthalene, anthracene, and phenanthrene; and an aromatic heterocyclic ring obtained by substituting part of carbon atoms constituting the above-described aromatic hydrocarbon ring with a hetero atom. Examples of the hetero atom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom. Specific examples of the aromatic heterocyclic ring include a pyridine ring and a thiophene ring.

(107) Specific examples of the aromatic hydrocarbon group as Ra.sup.03 include a group obtained by removing one hydrogen atom from the above-described aromatic hydrocarbon ring or aromatic heterocyclic ring (an aryl group or a heteroaryl group); a group obtained by removing one hydrogen atom from an aromatic compound having two or more aromatic rings (biphenyl, fluorene or the like); and a group obtained by substituting one hydrogen atom of the above-described aromatic hydrocarbon ring or aromatic heterocyclic ring with an alkylene group (an arylalkyl group such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, a 1-naphthylethyl group, or a 2-naphthylethyl group). The number of carbon atoms in the alkylene group bonded to the aromatic hydrocarbon ring or aromatic heterocyclic ring is preferably in a range of 1 to 4, more preferably 1 or 2, and particularly preferably 1.

(108) The cyclic hydrocarbon group as Ra.sup.03 may have a substituent. Examples of the substituent include, —R.sup.P1, —R.sup.P2—O—R.sup.P1, —R.sup.P2—CO—R.sup.P1, —R.sup.P2—CO—OR.sup.P1, —R.sup.P2—O—CO—R.sup.P1, —R.sup.P2—OH, —R.sup.P2—CN, and —R.sup.P2—COOH (hereinafter, these substituents are also collectively referred to as “Ra.sup.x5”).

(109) Here, R.sup.P1 represents a monovalent chain-like saturated hydrocarbon group having 1 to 10 carbon atoms, a monovalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms, or a monovalent aromatic hydrocarbon group having 6 to 30 carbon atoms. In addition, R.sup.P2 represents a single bond, a divalent chain-like saturated hydrocarbon group having 1 to 10 carbon atoms, a divalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms, or a divalent aromatic hydrocarbon group having 6 to 30 carbon atoms. However, part or all hydrogen atoms contained in the chain-like saturated hydrocarbon group, the aliphatic cyclic saturated hydrocarbon group, and the aromatic hydrocarbon group of R.sup.P1 and R.sup.P2 may be substituted with a fluorine atom. In the aliphatic cyclic hydrocarbon group, one or more of the above-described substituents may be included as a single kind, or a plurality of the above-described substituents may be included as one or more respective kinds among the above-described substituents.

(110) Examples of the monovalent chain-like saturated hydrocarbon group having 1 to 10 carbon atoms include a methyl group, an ethyl group, a propyl group, a butyl group, a pentyl group, a hexyl group, a heptyl group, an octyl group, and a decyl group.

(111) Examples of the monovalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms include monocyclic aliphatic saturated hydrocarbon groups such as a cyclopropyl

group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, a cyclodecyl group, and cyclododecyl group; and polycyclic aliphatic saturated hydrocarbon groups such as a bicyclo[2.2.2]octanyl group, a tricyclo[5.2.1.0^{2,6}]decanyl group, a tricyclo[3.3.1.1^{3,7}]decanyl group, a tetracyclo[6.2.1.1^{3,6}.0^{2,7}]dodecanyl group, and an adamantyl group.

(112) Examples of the monovalent aromatic hydrocarbon group having 6 to 30 carbon atoms include groups obtained by removing one hydrogen atom from an aromatic hydrocarbon ring such as benzene, biphenyl, fluorene, naphthalene, anthracene, and phenanthrene.

(113) In a case where Ra.sup.03 is bonded to any one of Ra.sup.03 or Ra.sup.02 to form a ring, the cyclic group is preferably a 4-membered to 7-membered ring, and more preferably a 4-membered to 6-membered ring. Specific examples of the cyclic group include a tetrahydropyranyl group and a tetrahydrofuranyl group.

(114) Specific examples of the acetal-type acid dissociable group are shown below. represents a bonding site.

(115) ##STR00006##

(116) Tertiary alkyl ester-type acid dissociable group:

(117) Examples of the acid dissociable group for protecting the carboxy group include the acid dissociable group represented by General Formula (a0-r-2) shown below.

(118) Among the acid dissociable groups represented by General Formula (a0-r-2), for convenience, a group which is constituted of alkyl groups is referred to as a “tertiary alkyl ester-type acid dissociable group”.

(119) ##STR00007##

(120) [In General Formula (a0-r-2), Ra.sup.04 to Ra.sup.06 each independently represent a hydrocarbon group, where Ra.sup.05 and Ra.sup.06 may be bonded to each other to form a ring. * represents a bonding site].

(121) Examples of the hydrocarbon group as Ra.sup.04 include a linear or branched alkyl group, a chain-like or cyclic alkenyl group, and a cyclic hydrocarbon group.

(122) Examples of the linear or branched alkyl group and the cyclic hydrocarbon group (the aliphatic hydrocarbon group which is a monocyclic group, the aliphatic hydrocarbon group which is a polycyclic group, or the aromatic hydrocarbon group) as Ra.sup.04 include the same one as Ra.sup.03 described above.

(123) The chain-like or cyclic alkenyl group as Ra.sup.04 is preferably an alkenyl group having 2 to 10 carbon atoms.

(124) Examples of the hydrocarbon group as Ra.sup.05 to Ra.sup.06 include the same one as Ra.sup.03.

(125) In a case where Ra.sup.05 and Ra.sup.06 are bonded to each other to form a ring, suitable examples thereof include groups represented by General Formula (a0-r2-01), General Formula (a0-r2-02), and General Formula (a0-r2-03).

(126) On the other hand, in a case where Ra.sup.04 to Ra.sup.06 are not bonded to each other and represent an independent hydrocarbon group, suitable examples thereof include a group represented by General Formula (a0-r2-04).

(127) ##STR00008##

(128) [In General Formula (a0-r2-01), Ra.sup.001 represents a linear or branched alkyl group which may have a substituent. Yaa.sup.0 represents a carbon atom. Xaa.sup.0 represents a group that forms a cyclic hydrocarbon group together with Yaa.sup.0. Part or all hydrogen atoms contained in this cyclic hydrocarbon group may be substituted, and part of carbon atoms constituting a ring may be substituted a hetero atom. * represents a bonding site.

(129) In General Formula (a0-r2-02), Yab.sup.0 represents a carbon atom. Xab.sup.0 represents a group that forms a cyclic hydrocarbon group together with Yab.sup.0. Part or all hydrogen atoms contained in this cyclic hydrocarbon group may be substituted, and part of carbon atoms

constituting a ring may be substituted a hetero atom. Ra.sup.001 to Ra.sup.004 each independently represent a hydrogen atom, a monovalent chain-like saturated hydrocarbon group having 1 to 10 carbon atoms, or a monovalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms. Part or all hydrogen atoms contained in the chain-like saturated hydrocarbon group and the aliphatic cyclic saturated hydrocarbon group may be substituted. Two or more of Ra.sup.001 to Ra.sup.004 may be bonded to each other to form a cyclic structure. * represents a bonding site.

(130) In General Formula (a0-r2-03), Yac.sup.0 represents a carbon atom. Xac.sup.0 represents a group that forms a cyclic hydrocarbon group together with Yac.sup.0. Part or all hydrogen atoms contained in this cyclic hydrocarbon group may be substituted, and part of carbon atoms constituting a ring may be substituted a hetero atom. Ra.sup.005 represents an aromatic hydrocarbon group. Part or all hydrogen atoms contained in the aromatic hydrocarbon group may be substituted, and part of carbon atoms constituting a ring may be substituted with a hetero atom. * represents a bonding site.

(131) In General Formula (a0-r2-04), Ra.sup.006 and Ra.sup.007 each independently represent a monovalent chain-like saturated hydrocarbon group having 1 to 10 carbon atoms. Part or all hydrogen atoms contained in the chain-like saturated hydrocarbon group may be substituted. Ra.sup.008 represents a hydrocarbon group which may have a substituent. * represents a bonding site].

(132) In General Formula (a0-r2-01), Ra.sup.001 represents a linear or branched alkyl group which may have a substituent.

(133) The linear alkyl group has preferably 1 to 5 carbon atoms, more preferably 1 to 4 carbon atoms, and still more preferably 1 or 2 carbon atoms. Specific examples thereof include a methyl group, an ethyl group, an n-propyl group, an n-butyl group, and an n-pentyl group. Among these, a methyl group, an ethyl group, or an n-butyl group is preferable, and a methyl group or an ethyl group is more preferable.

(134) The branched alkyl group has preferably 3 to 10 carbon atoms and more preferably 3 to 5 carbon atoms. Specific examples thereof include an isopropyl group, an isobutyl group, a tert-butyl group, an isopentyl group, a neopentyl group, a 1,1-diethylpropyl group, and a 2,2-dimethylbutyl group, and a tert-butyl group is preferable.

(135) Examples of the substituent which may be contained in the linear or branched alkyl group as Ra.sup.001 include Ra.sup.x5 described above.

(136) In General Formula (a0-r2-01), Ra.sup.001 represents, among the above, preferably a linear alkyl group having 1 to 5 carbon atoms or a branched alkyl group having 3 to 10 carbon atoms, and more preferably a linear alkyl group having 1 to 4 carbon atoms or a branched alkyl group having 3 to 5 carbon atoms.

(137) In General Formula (a0-r2-01), Yaa.sup.0 represents a carbon atom, and Xaa.sup.0 represents a group that forms a cyclic hydrocarbon group together with Yaa.sup.0.

(138) The cyclic hydrocarbon group may be an aliphatic hydrocarbon group, may be a condensed cyclic hydrocarbon group of an aliphatic hydrocarbon group with an aromatic hydrocarbon group, or a may be a polycyclic group or a monocyclic group.

(139) The aliphatic hydrocarbon group which is a monocyclic group is preferably a group obtained by removing two or more hydrogen atoms from a monocycloalkane. The monocycloalkane has preferably 3 to 6 carbon atoms and more preferably 5 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane.

(140) The aliphatic hydrocarbon group which is a polycyclic group is preferably a group obtained by removing two or more hydrogen atoms from a polycycloalkane, where the polycycloalkane preferably has 7 to 12 carbon atoms. Specific examples thereof include adamantane, norbornane, isobornane, tricyclodecane, and tetracyclododecane.

(141) The aromatic hydrocarbon group in the condensed cyclic hydrocarbon group of an aliphatic hydrocarbon group with an aromatic hydrocarbon group is a hydrocarbon group having at least one

aromatic ring.

(142) The aromatic ring is not particularly limited as long as it is a cyclic conjugated system having $(4n+2)$ π electrons, and may be monocyclic or polycyclic. The aromatic ring preferably has 5 to 30 carbon atoms, more preferably 5 to 20 carbon atoms, still more preferably 6 to 15 carbon atoms, and particularly preferably 6 to 12 carbon atoms.

(143) Specific examples of the aromatic ring include aromatic hydrocarbon rings such as benzene, naphthalene, anthracene, and phenanthrene; and an aromatic heterocyclic ring obtained by substituting part of carbon atoms constituting the above-described aromatic hydrocarbon ring with a hetero atom. Examples of the hetero atom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom. Specific examples of the aromatic heterocyclic ring include a pyridine ring and a thiophene ring.

(144) Specific examples of the condensed cyclic hydrocarbon group of an aliphatic hydrocarbon group with an aromatic hydrocarbon group are shown below.

(145) ##STR00009##

(146) Part or all hydrogen atoms contained in the cyclic hydrocarbon group may be substituted, and part of carbon atoms constituting a ring may be substituted with a hetero atom.

(147) Specific examples of the substituent that is substituted for part or all hydrogen atoms contained in the cyclic hydrocarbon group include $R_{a.sup.x5}$ described above. In a case where part of carbon atoms constituting a ring is substituted with a hetero atom, the hetero atom includes an oxygen atom, a sulfur atom, and a nitrogen atom.

(148) In General Formula (a0-r2-01), the cyclic hydrocarbon group formed by $X_{a.sup.0}$ and $Y_{a.sup.0}$ is, among the above, preferably an aliphatic hydrocarbon group which is a monocyclic group or a polycyclic group, more preferably an aliphatic hydrocarbon group which is a monocyclic group, and still more preferably an aliphatic hydrocarbon group which is a monocyclic group having 5 or 6 carbon atoms.

(149) In General Formula (a0-r2-02), $Y_{a.sup.0}$ represents a carbon atom, and $X_{a.sup.0}$ represents a group that forms a cyclic hydrocarbon group together with $Y_{a.sup.0}$. Examples of this cyclic hydrocarbon group include the same one as the cyclic hydrocarbon group formed by $X_{a.sup.0}$ and $Y_{a.sup.0}$ described above.

(150) In General Formula (a0-r2-02), $R_{a.sup.001}$ to $R_{a.sup.004}$ each independently represent a hydrogen atom, a monovalent chain-like saturated hydrocarbon group having 1 to 10 carbon atoms, or a monovalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms.

(151) Examples of the monovalent chain-like saturated hydrocarbon group having 1 to 10 carbon atoms, as $R_{a.sup.001}$ to $R_{a.sup.004}$, include a methyl group, an ethyl group, a propyl group, a butyl group, a pentyl group, a hexyl group, a heptyl group, an octyl group, and a decyl group.

(152) Examples of the monovalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms, as $R_{a.sup.002}$ to $R_{a.sup.004}$, include monocyclic aliphatic saturated hydrocarbon groups such as a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, a cyclodecyl group, and cyclododecyl group; and polycyclic aliphatic saturated hydrocarbon groups such as a bicyclo[2.2.2]octanyl group, a tricyclo[5.2.1.0^{2,6}]decanyl group, a tricyclo[3.3.1.1^{3,7}]decanyl group, a tetracyclo[6.2.1.1^{3,6}.0^{2,7}]dodecanyl group, and an adamantyl group.

(153) Part or all hydrogen atoms contained in the chain-like saturated hydrocarbon group and the aliphatic cyclic saturated hydrocarbon group, as $R_{a.sup.002}$ to $R_{a.sup.004}$, may be substituted. Specific examples of the substituent that is substituted for part or all hydrogen atoms contained in the chain-like saturated hydrocarbon group and the aliphatic cyclic saturated hydrocarbon group include $R_{a.sup.x5}$ described above. In a case where part of carbon atoms constituting a ring is substituted with a hetero atom, the hetero atom includes an oxygen atom, a sulfur atom, and a nitrogen atom.

(154) Examples of the group containing a carbon-carbon double bond generated by forming a

cyclic structure, which is obtained by bonding two or more of Ra.sup.002 to Ra.sup.004 to each other, include a cyclopentenyl group, a cyclohexenyl group, a methylcyclopentenyl group, a methylcyclohexenyl group, a cyclopentylideneethenyl group, and a cyclohexylideneethenyl group. Among these, a cyclopentenyl group, a cyclohexenyl group, and a cyclopentylideneethenyl group are preferable from the viewpoint of easy synthesis.

(155) In General Formula (a0-r2-02), Ra.sup.002 to Ra.sup.004 are each independently, among the above, preferably a hydrogen atom or a monovalent chain-like saturated hydrocarbon group having 1 to 10 carbon atoms, more preferably a hydrogen atom, a methyl group, or an ethyl group, and still more preferably a hydrogen atom.

(156) In General Formula (a0-r2-03), Yac.sup.0 represents a carbon atom, and Xac.sup.0 represents a group that forms a cyclic hydrocarbon group together with Yac.sup.0. Examples of this cyclic hydrocarbon group include the same one as the cyclic hydrocarbon group formed by Xaa.sup.0 and Yaa.sup.0 described above.

(157) In General Formula (a0-r2-03), Ra.sup.005 as an aromatic hydrocarbon group is preferably a group obtained by removing one or more hydrogen atoms from an aromatic hydrocarbon ring having 6 to 15 carbon atoms, more preferably a group obtained by removing one or more hydrogen atoms from benzene, naphthalene, anthracene, or phenanthrene, still more preferably a group obtained by removing one or more hydrogen atoms from benzene or naphthalene, particularly preferably a group obtained by removing one or more hydrogen atoms from benzene.

(158) Part or all hydrogen atoms contained in the aromatic hydrocarbon group may be substituted, or part of carbon atoms constituting a ring may be substituted with a hetero atom.

(159) Specific examples of the substituent that is substituted for part or all hydrogen atoms contained in the aromatic hydrocarbon group include Ra.sup.x5 described above. In a case where part of carbon atoms constituting a ring is substituted with a hetero atom, the hetero atom includes an oxygen atom, a sulfur atom, and a nitrogen atom.

(160) In General Formula (a0-r2-04), Ra.sup.006 and Ra.sup.007 each independently represent a monovalent chain-like saturated hydrocarbon group having 1 to 10 carbon atoms.

(161) Examples of the monovalent chain-like saturated hydrocarbon group having 1 to 10 carbon atoms, as Ra.sup.006 and Ra.sup.007, include the same one as the monovalent chain-like saturated hydrocarbon group having 1 to 10 carbon atoms as Ra.sup.002 to Ra.sup.004 described above.

(162) In General Formula (a0-r2-04), Ra.sup.006 and Ra.sup.007 are, among the above, preferably an alkyl group having 1 to 5 carbon atoms, more preferably a methyl group or an ethyl group, and still more preferably a methyl group.

(163) In a case where the chain-like saturated hydrocarbon groups represented by Ra.sup.006 and Ra.sup.007 are substituted, examples of the substituent include the same group as Ra.sup.x5 described above.

(164) In General Formula (a0-r2-04), Ra.sup.008 represents a hydrocarbon group which may have a substituent. Examples of the hydrocarbon group as Ra.sup.008 include a linear or branched alkyl group and a cyclic hydrocarbon group.

(165) The linear alkyl group as Ra.sup.008 is preferably 1 to 5 carbon atoms, more preferably 1 to 4 carbon atoms, and still more preferably 1 or 2 carbon atoms. Specific examples thereof include a methyl group, an ethyl group, an n-propyl group, an n-butyl group, and an n-pentyl group. Among these, a methyl group, an ethyl group, or an n-butyl group is preferable, and a methyl group or an ethyl group is more preferable.

(166) The branched alkyl group as Ra.sup.008 preferably has 3 to 10 carbon atoms and more preferably 3 to 5 carbon atoms. Specific examples thereof include an isopropyl group, an isobutyl group, a tert-butyl group, an isopentyl group, a neopentyl group, a 1,1-diethylpropyl group, and a 2,2-dimethylbutyl group, and an isopropyl group is preferable.

(167) In a case where Ra.sup.008 represents a cyclic hydrocarbon group, the hydrocarbon group may be an aliphatic hydrocarbon group or an aromatic hydrocarbon group and may be a polycyclic

group or a monocyclic group.

(168) The aliphatic hydrocarbon group which is a monocyclic group is preferably a group obtained by removing one hydrogen atom from a monocycloalkane. The monocycloalkane preferably has 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane.

(169) The aliphatic hydrocarbon group which is a polycyclic group is preferably a group obtained by removing one hydrogen atom from a polycycloalkane. The polycycloalkane preferably has 7 to 12 carbon atoms, and specific examples thereof include adamantane, norbornane, isobornane, tricyclodecane, and tetracyclododecane.

(170) Examples of the aromatic hydrocarbon group as Ra.sup.008 include the same one as the aromatic hydrocarbon group as Ra.sup.005. Among them, Ra.sup.008 is preferably a group in which one or more hydrogen atoms have been removed from an aromatic hydrocarbon ring having 6 to 15 carbon atoms, more preferably a group in which one or more hydrogen atoms have been removed from benzene, naphthalene, anthracene, or phenanthrene, still more preferably a group in which one or more hydrogen atoms have been removed from benzene, naphthalene, or anthracene, particularly preferably a group in which one or more hydrogen atoms have been removed from naphthalene or anthracene, and most preferably a group in which one or more hydrogen atoms have been removed from naphthalene.

(171) Examples of the substituent which may be contained in Ra.sup.008 include the same one as the substituent which may be contained in Ra.sup.005.

(172) In a case where Ra.sup.008 in General Formula (a0-r2-04) is a naphthyl group, the position at which the tertiary carbon atom in General Formula (a1-r2-4) is bonded may be any of the 1-position and the 2-position of the naphthyl group.

(173) In a case where Ra.sup.008 in Formula (a0-r2-04) is an anthryl group, the position at which the tertiary carbon atom in Formula (a0-r2-04) is bonded may be any of the 1-position, the 2-position, and 9-position of the anthryl group.

(174) Specific examples of the group represented by General Formula (a0-r2-01) are shown below.

(175) ##STR00010## ##STR00011## ##STR00012## ##STR00013##

(176) Specific examples of the group represented by General Formula (a0-r2-02) are shown below.

(177) ##STR00014## ##STR00015## ##STR00016## ##STR00017## ##STR00018##
##STR00019## ##STR00020##

(178) Specific examples of the group represented by General Formula (a0-r2-03) are shown below.

(179) ##STR00021##

(180) Specific examples of the group represented by General Formula (a0-r2-04) are shown below.

(181) ##STR00022##

(182) In General Formula (a0-1), the acid dissociable group as Rx.sup.01 is preferably a tertiary alkyl ester-type acid dissociable group, more preferably an acid dissociable group represented by any one of General Formula (a0-r2-01) to (a0-r2-03), and still more preferably an acid dissociable group represented by General Formula (a0-r2-01).

(183) In General Formula (a0-1), q represents an integer in a range of 0 to 3. A case where q is 0 gives a benzene structure, a case where q is 1 gives a naphthalene structure, a case where q is 2 gives an anthracene structure, and a case where q is 3 gives a tetracene structure.

(184) In General Formula (a0-1), n represents an integer of 1 or more, preferably an integer in a range of 1 to 5, more preferably an integer in a range of 1 to 3, and still more preferably an integer of 1 or 2.

(185) In General Formula (a0-1), $n \leq q \times 2 + 4$ is satisfied.

(186) For example, in a case where q is 1, which gives a naphthalene structure, all hydrogen atoms in the naphthalene except for a hydrogen atom substituted with a polymerizable group-containing group (W.sup.01) and —Ya.sup.01—(C=O)—O—Rx.sup.01 group may be substituted with a hydroxy group. Further, in the naphthalene, the substitution positions of the polymerizable group-containing group (W.sup.01), —Ya.sup.01—(C=O)—O—Rx.sup.01 group, and the hydroxy group

are not particularly limited.

(187) Among the above, the constitutional unit (a01) is preferably a constitutional unit represented by General Formula (a0-1-1).

(188) ##STR00023##

(189) [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. Ya.sup.001 represents a single bond or a divalent linking group. Ya.sup.01 represents a single bond or a divalent linking group. Rax.sup.01 represents the acid dissociable group represented by General Formula (a0-r-1) or (a0-r-2) described above. q represents an integer in a range of 0 to 3. n represents an integer of 1 or more. However, $n \leq q \times 2 + 4$ is satisfied.]

(190) In General Formula (a0-1-1), the alkyl group having 1 to 5 carbon atoms as R is preferably a linear or branched alkyl group having 1 to 5 carbon atoms, and specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, and a neopentyl group. The halogenated alkyl group having 1 to 5 carbon atoms is a group obtained by substituting part or all hydrogen atoms in the alkyl group having 1 to 5 carbon atoms with a halogen atom. The halogen atom is particularly preferably a fluorine atom.

(191) R is preferably a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a fluorinated alkyl group having 1 to 5 carbon atoms, and most preferably a hydrogen atom or a methyl group in terms of industrial availability.

(192) In General Formula (a0-1-1), Ya.sup.001 represents a single bond or a divalent linking group.

(193) The divalent linking group as Ya.sup.001 is not particularly limited, and suitable examples thereof include a divalent hydrocarbon group which may have a substituent and a divalent linking group having a hetero atom, which are each the same as the divalent hydrocarbon group as W.sup.01 and the divalent linking group containing a hetero atom.

(194) Among the above, Ya.sup.001 is preferably an ester bond [—C(=O)—O— or —O—C(=O)—], an ether bond (—O—), a linear or branched alkylene group, an aromatic hydrocarbon group, or a combination thereof, or a single bond. Among the above, Ya.sup.001 is more preferably an ester bond [—C(=O)—O— , —O—C(=O)—], a combination of an ester bond [—C(=O)—O— , —O—C(=O)—] and a linear alkylene group, or a single bond, and still more preferably a single bond.

(195) In General Formula (a0-1-1), Ya.sup.01 is the same as Ya.sup.01 in General Formula (a0-1).

(196) In General Formula (a0-1-1), Rax.sup.01 represents the acid dissociable group represented by General Formula (a0-r-1) or (a0-r-2) described above.

(197) In General Formula (a0-1-1), Rax.sup.01 is, among the above, preferably an acid dissociable group represented by any one of General Formulae (a0-r2-01) to (a0-r2-03) and more preferably an acid dissociable group represented by General Formula (a0-r2-01).

(198) In General Formula (a0-1-1), q and n are the same as q and n in General Formula (a0-1).

(199) Specific examples of the constitutional unit (a01) are shown below. In each of the formulae shown below, R.sup. α represents a hydrogen atom, a methyl group, or a trifluoromethyl group.

(200) ##STR00024## ##STR00025## ##STR00026## ##STR00027## ##STR00028##
##STR00029## ##STR00030## ##STR00031##

(201) The constitutional unit (a01) in the resist composition according to the present embodiment is, among the above, preferably a constitutional unit represented by any one of General Formula (a01-1a-1) to (a01-1a-46), more preferably a constitutional unit represented by any one of General Formulae (a01-1a-1) to (a01-1a-40), and still more preferably a constitutional unit represented by any one of General Formulae (a01-1a-1), (a01-1a-2), (a01-1a-4), (a01-1a-8), (a01-1a-17), (a01-1a-28), (a01-1a-31), (a01-1a-34), (a01-1a-37) and (a01-1a-40).

(202) The constitutional unit (a01) contained in the component (A1) may be one kind or may be two or more kinds.

(203) With respect to the total (100% by mole) of all constitutional units constituting the component (A1), the proportion of the constitutional unit (a01) in the component (A1) is preferably 20% by mole or more, is more preferably 40% by mole or more, and may be 100% by mole, that is, may be a homopolymer.

(204) On the other hand, in the case of the copolymer having the constitutional unit (a01), the proportion of the constitutional unit (a01) in the component (A1) is more preferably in a range of 20% to 90% by mole and still more preferably in a range of 40% to 60% by mole.

(205) In a case where the proportion of the constitutional unit (a01) is set within the above preferred range, sensitivity can be further improved without reducing resolution. In addition, since the solubility of the developing solution can be suitably ensured, the fine resolution can be further improved.

(206) <<Other Constitutional Units>>

(207) The component (A1) may have other constitutional units as necessary in addition to the constitutional unit (a01).

(208) Examples of other constitutional units include a constitutional unit (a1) containing an acid decomposable group having a polarity which is increased under action of acid (provided that a constitutional unit corresponding to the constitutional unit (a01) is excluded); a constitutional unit (a10) represented by General Formula (a10-1) described later; a constitutional unit (a2) containing a lactone-containing cyclic group, a —SO₂—containing cyclic group, or a carbonate-containing cyclic group (provided that a constitutional unit corresponding to the constitutional unit (a01) or the constitutional unit (a1) is excluded); a constitutional unit (a3) containing a polar group-containing aliphatic hydrocarbon group (provided that a constitutional unit corresponding to the constitutional unit (a01), the constitutional unit (a1), or the constitutional unit (a2) is excluded); a constitutional unit (a4) containing an acid non-dissociable aliphatic cyclic group; and a constitutional unit (st) derived from styrene or a derivative thereof.

(209) In regard to Constitutional Unit (a1):

(210) The component (A1) may further have a constitutional unit (a1) in addition to the constitutional unit (a01).

(211) The constitutional unit (a1) is a constitutional unit (provided that a constitutional unit corresponding to the constitutional unit (a01) is excluded) containing an acid decomposable group having a polarity which is increased under action of acid.

(212) Examples of the acid dissociable group are the same as those which have been proposed so far as acid dissociable groups for the base resin for a chemically amplified resist composition.

(213) Specific examples of the acid dissociable group of the base resin proposed for a chemically amplified resist composition include the “acetal-type acid dissociable group” and the “tertiary alkyl ester-type acid dissociable group” described above, and a “tertiary alkyloxycarbonyl acid dissociable group” described below.

(214) Tertiary Alkyloxycarbonyl Acid Dissociable Group:

(215) Examples of the acid dissociable group for protecting a hydroxyl group include an acid dissociable group (hereinafter, for convenience, also referred to as a “tertiary alkyloxycarbonyl acid dissociable group”) represented by General Formula (a1-r-3) shown below.

(216) ##STR00032##

(217) [In the formula, Ra⁷ to Ra⁹ each represent an alkyl group.]

(218) In General Formula (a1-r-3), Ra⁷ to Ra⁹ are each preferably an alkyl group having 1 to 5 carbon atoms and more preferably an alkyl group having 1 to 3 carbon atoms.

(219) Further, the total number of carbon atoms in each of the alkyl groups is preferably in a range of 3 to 7, more preferably in a range of 3 to 5, and most preferably 3 or 4.

(220) Examples of the constitutional unit (a1) include a constitutional unit derived from acrylic acid ester in which the hydrogen atom bonded to the carbon atom at the α-position may be substituted with a substituent; a constitutional unit derived from acrylamide; a constitutional unit in

which at least part of hydrogen atoms in a hydroxyl group of a constitutional unit derived from hydroxystyrene or a hydroxystyrene derivative are protected by a substituent including an acid decomposable group; and a constitutional unit in which at least part of hydrogen atoms in —C(=O)—OH of a constitutional unit derived from vinylbenzoic acid or a vinylbenzoic acid derivative are protected by the substituent including an acid decomposable group.

(221) Among the above, the constitutional unit (a1) is preferably a constitutional unit derived from acrylic acid ester in which the hydrogen atom bonded to the carbon atom at the α -position may be substituted with a substituent.

(222) Preferred specific examples of such a constitutional unit (a1) include constitutional units represented by General Formula (a1-1) or (a1-2).

(223) ##STR00033##

(224) [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. Va.sup.1 represents a divalent hydrocarbon group which may have an ether bond. n.sub.a1 represents an integer in a range of 0 to 2. Ra.sup.1 is the acid dissociable group represented by General Formula (a0-r-1) or (a0-r-2) described above. Wa.sup.1 represents an ($\text{n.sub.a2}+1$)-valent hydrocarbon group, n.sub.a2 represents an integer in a range of 1 to 3, and Ra.sup.2 represents the acid dissociable group represented by General Formula (a0-r-1) or (a1-r-3) described above.]

(225) In General Formula (a1-1), the alkyl group having 1 to 5 carbon atoms as R is preferably a linear or branched alkyl group having 1 to 5 carbon atoms, and specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, and a neopentyl group. The halogenated alkyl group having 1 to 5 carbon atoms is a group obtained by substituting part or all hydrogen atoms in the alkyl group having 1 to 5 carbon atoms with a halogen atom. The halogen atom is particularly preferably a fluorine atom.

(226) R is preferably a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a fluorinated alkyl group having 1 to 5 carbon atoms, and most preferably a hydrogen atom or a methyl group in terms of industrial availability.

(227) In General Formula (a1-1), the divalent hydrocarbon group as Va.sup.1 may be an aliphatic hydrocarbon group or an aromatic hydrocarbon group.

(228) The aliphatic hydrocarbon group as the divalent hydrocarbon group represented by Va.sup.1 may be saturated or unsaturated. In general, it is preferable that the aliphatic hydrocarbon group is saturated.

(229) Specific examples of the aliphatic hydrocarbon group include a linear or branched aliphatic hydrocarbon group, and an aliphatic hydrocarbon group containing a ring in the structure thereof.

(230) The linear aliphatic hydrocarbon group preferably has 1 to 10 carbon atoms, more preferably 1 to 6 carbon atoms, still more preferably 1 to 4 carbon atoms, and most preferably 1 to 3 carbon atoms.

(231) The linear aliphatic hydrocarbon group is preferably a linear alkylene group, and specific examples thereof include a methylene group [—CH.sub.2—], an ethylene group [$\text{—(CH.sub.2).sub.2—}$], a trimethylene group [$\text{—(CH.sub.2).sub.3—}$], a tetramethylene group [$\text{—(CH.sub.2).sub.4—}$], and a pentamethylene group [$\text{—(CH.sub.2).sub.5—}$].

(232) The branched aliphatic hydrocarbon group has preferably 2 to 10 carbon atoms, more preferably 3 to 6 carbon atoms, still more preferably 3 or 4 carbon atoms, and most preferably 3 carbon atoms.

(233) The branched aliphatic hydrocarbon group is preferably a branched alkylene group, and specific examples thereof include alkylalkylene group groups, for example, alkylmethylene groups such as —CH(CH.sub.3)— , $\text{—CH(CH.sub.2CH.sub.3)—}$, $\text{—C(CH.sub.3).sub.2—}$, $\text{—C(CH.sub.3)(CH.sub.2CH.sub.3)—}$, $\text{—C(CH.sub.3)(CH.sub.2CH.sub.2CH.sub.3)—}$, and $\text{—C(CH.sub.2CH.sub.3).sub.2—}$; alkylethylene groups such as $\text{—CH(CH.sub.3)CH.sub.2—}$, —

CH(CH.sub.3)CH(CH.sub.3)—, —C(CH.sub.3).sub.2CH.sub.2—, —CH(CH.sub.2CH.sub.3)CH.sub.2—, and —C(CH.sub.2CH.sub.3).sub.2—CH.sub.2—; alkyltrimethylene groups such as —CH(CH.sub.3)CH.sub.2CH.sub.2—, and —CH.sub.2CH(CH.sub.3)CH.sub.2—; and alkyltetramethylene groups such as —CH(CH.sub.3)CH.sub.2CH.sub.2CH.sub.2—, and —CH.sub.2CH(CH.sub.3)CH.sub.2CH.sub.2—. The alkyl group in the alkylalkylene group is preferably a linear alkyl group having 1 to 5 carbon atoms.

(234) Examples of the aliphatic hydrocarbon group containing a ring in the structure thereof include an alicyclic hydrocarbon group (a group obtained by removing two hydrogen atoms from an aliphatic hydrocarbon ring), a group obtained by bonding the alicyclic hydrocarbon group to the terminal of a linear or branched aliphatic hydrocarbon group, and a group obtained by interposing the alicyclic hydrocarbon group in a linear or branched aliphatic hydrocarbon group. Examples of the linear or branched aliphatic hydrocarbon group include the same one as the above-described linear aliphatic hydrocarbon group or the above-described branched aliphatic hydrocarbon group.

(235) The alicyclic hydrocarbon group preferably has 3 to 20 carbon atoms and more preferably 3 to 12 carbon atoms.

(236) The alicyclic hydrocarbon group may be monocyclic or polycyclic. The monocyclic alicyclic hydrocarbon group is preferably a group obtained by removing two hydrogen atoms from a monocycloalkane. The monocycloalkane preferably has 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane. The polycyclic alicyclic hydrocarbon group is preferably a group obtained by removing two hydrogen atoms from a polycycloalkane, and the polycycloalkane is preferably a group having 7 to 12 carbon atoms. Specific examples thereof include adamantane, norbornane, isobornane, tricyclodecane, and tetracyclododecane.

(237) The aromatic hydrocarbon group as the divalent hydrocarbon group represented by Va.sup.1 is a hydrocarbon group having an aromatic ring.

(238) The aromatic hydrocarbon group preferably has 3 to 30 carbon atoms, more preferably 5 to 30 carbon atoms, still more preferably 5 to 20 carbon atoms, particularly preferably 6 to 15 carbon atoms, and most preferably 6 to 12 carbon atoms. Here, the number of carbon atoms in a substituent is not included in the number of carbon atoms.

(239) Specific examples of the aromatic ring contained in the aromatic hydrocarbon group include aromatic hydrocarbon rings such as benzene, biphenyl, fluorene, naphthalene, anthracene, and phenanthrene; and an aromatic heterocyclic ring obtained by substituting part of carbon atoms constituting the above-described aromatic hydrocarbon rings with a hetero atom. Examples of the hetero atom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom.

(240) Specific examples of the aromatic hydrocarbon group include a group in which two hydrogen atoms have been removed from the above-described aromatic hydrocarbon ring (an arylene group); and a group in which one hydrogen atom of a group (an aryl group) formed by removing one hydrogen atom from the aromatic hydrocarbon ring has been substituted with an alkylene group (for example, a group in which one or more hydrogen atoms have been removed from an aryl group in arylalkyl groups such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, a 1-naphthylethyl group, or a 2-naphthylethyl group). The alkylene group (an alkyl chain the arylalkyl group) preferably has 1 to 4 carbon atoms, more preferably 1 or 2 carbon atoms, and particularly preferably 1 carbon atom.

(241) In General Formula (a1-1), Ra.sup.1 represents an acid dissociable group represented by General Formula (a0-r-1) or (a0-r-2), and it is preferably an acid dissociable group represented by General Formula (a0-r-2) and more preferably an acid dissociable group represented by General Formula (a0-r2-01).

(242) In General Formula (a1-2), the (n.sub.a2+1)-valent hydrocarbon group as Wa.sup.1 may be an aliphatic hydrocarbon group or an aromatic hydrocarbon group. The aliphatic hydrocarbon

group indicates a hydrocarbon group that has no aromaticity and may be saturated or unsaturated. In general, it is preferable that the aliphatic hydrocarbon group is saturated. Examples of the aliphatic hydrocarbon group include a linear or branched aliphatic hydrocarbon group, an aliphatic hydrocarbon group containing a ring in the structure thereof, and a combination of the linear or branched aliphatic hydrocarbon group and the aliphatic hydrocarbon group containing a ring in the structure thereof.

(243) The valency of (n.sub.a2+1) is preferably divalent, trivalent, or tetravalent, and more preferably divalent or trivalent.

(244) In General Formula (a1-2), Ra.sup.2 is an acid dissociable group represented by General Formula (a0-r-1) or (a1-r-3).

(245) Specific examples of the constitutional unit represented by General Formula (a1-1) are shown below. In each of the formulae shown below, R.sup.α represents a hydrogen atom, a methyl group, or a trifluoromethyl group.

(246) ##STR00034## ##STR00035## ##STR00036## ##STR00037## ##STR00038##
##STR00039## ##STR00040## ##STR00041## ##STR00042## ##STR00043## ##STR00044##
##STR00045## ##STR00046##

(247) The constitutional unit (a1) contained in the component (A1) may be one kind or may be two or more kinds.

(248) Since it is easy to enhance the characteristics (the sensitivity, the shape, or the like) of lithography with an electron beam or EUV, the constitutional unit (a1) is preferably a constitutional unit represented by General Formula (a1-1) and more preferably a constitutional unit represented by General Formula (a1-1-1), where Ra.sup.1 in General Formula (a1-1) is an acid dissociable group represented by General Formula (a0-r2-01).

(249) ##STR00047##

(250) In General Formula (a1-1-1), R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. Va.sup.1 represents a divalent hydrocarbon group which may have an ether bond. n.sub.a1 represents an integer in a range of 0 to 2. Ra.sup.10 represents an acid dissociable group represented by General Formula (a0-r2-01).

(251) ##STR00048##

(252) [In General Formula (a0-r2-01), Ra.sup.001 represents a linear or branched alkyl group which may have a substituent. Yaa.sup.0 represents a carbon atom. Xaa.sup.0 represents a group that forms a cyclic hydrocarbon group together with Yaa.sup.0. Part or all hydrogen atoms contained in this cyclic hydrocarbon group may be substituted, and part of carbon atoms constituting a ring may be substituted a hetero atom. * represents a bonding site.]

(253) The proportion of the constitutional unit (a1) in the component (A1) is preferably in a range of 10% to 80% by mole and more preferably in a range of 40% to 60% by mole with respect to the total (100% by mole) of all constitutional units constituting the component (A1).

(254) In a case where the proportion of the constitutional unit (a1) is equal to or larger than the lower limit value of the preferred range described above, lithography characteristics such as sensitivity, resolution, and roughness amelioration are improved. On the other hand, in a case where it is equal to or smaller than the upper limit value of the above preferred range, balance with other constitutional units can be obtained, and various lithography characteristics are improved.

(255) In regard to constitutional unit (a10):

(256) The component (A1) may further have a constitutional unit (a10) represented by General Formula (a10-1).

(257) ##STR00049##

(258) [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. Ya.sup.x1 represents a single bond or a divalent linking group. Wa.sup.x1 represents an aromatic hydrocarbon group which may have a

substituent. n.sub.ax1 represents an integer of 1 or more.]

(259) In General Formula (a10-1), R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms.

(260) The alkyl group having 1 to 5 carbon atoms as R is preferably a linear or branched alkyl group having 1 to 5 carbon atoms, and specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, and a neopentyl group.

(261) The halogenated alkyl group having 1 to 5 carbon atoms as R is a group obtained by substituting part or all hydrogen atoms of an above-described alkyl group having 1 to 5 carbon atoms with a halogen atom. The halogen atom is particularly preferably a fluorine atom.

(262) R is preferably a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a fluorinated alkyl group having 1 to 5 carbon atoms, and in terms of industrial availability, R is more preferably a hydrogen atom, a methyl group, or trifluoromethyl group, still more preferably a hydrogen atom or a methyl group, and particularly preferably a methyl group.

(263) In General Formula (a10-1), Ya.sup.x1 represents a single bond or a divalent linking group.

(264) In the chemical formulae described above, the divalent linking group as Ya.sup.x1 is not particularly limited, and suitable examples thereof include a divalent hydrocarbon group which may have a substituent and a divalent linking group having hetero atoms.

(265) Divalent Hydrocarbon Group which May have Substituent:

(266) In a case where Ya.sup.x1 represents a divalent hydrocarbon group which may have a substituent, the hydrocarbon group may be an aliphatic hydrocarbon group or an aromatic hydrocarbon group.

(267) Aliphatic Hydrocarbon Group as Ya

(268) The aliphatic hydrocarbon group indicates a hydrocarbon group that has no aromaticity. The aliphatic hydrocarbon group may be saturated or unsaturated. In general, it is preferable that the aliphatic hydrocarbon group is saturated.

(269) Examples of the aliphatic hydrocarbon group include a linear or branched aliphatic hydrocarbon group, and an aliphatic hydrocarbon group containing a ring in the structure thereof.

(270) Linear or Branched Aliphatic Hydrocarbon Group

(271) The linear aliphatic hydrocarbon group preferably has 1 to 10 carbon atoms, more preferably 1 to 6 carbon atoms, still more preferably 1 to 4 carbon atoms, and most preferably 1 to 3 carbon atoms.

(272) The linear aliphatic hydrocarbon group is preferably a linear alkylene group, and specific examples thereof include a methylene group [$\text{—CH}_{\text{sub.2}}\text{—}$], an ethylene group [$\text{—(CH}_{\text{sub.2}}\text{)}_{\text{sub.2}}\text{—}$], a trimethylene group [$\text{—(CH}_{\text{sub.2}}\text{)}_{\text{sub.3}}\text{—}$], a tetramethylene group [$\text{—(CH}_{\text{sub.2}}\text{)}_{\text{sub.4}}\text{—}$], and a pentamethylene group [$\text{—(CH}_{\text{sub.2}}\text{)}_{\text{sub.5}}\text{—}$].

(273) The branched aliphatic hydrocarbon group has preferably 2 to 10 carbon atoms, more preferably 3 to 6 carbon atoms, still more preferably 3 or 4 carbon atoms, and most preferably 3 carbon atoms.

(274) The branched aliphatic hydrocarbon group is preferably a branched alkylene group, and specific examples thereof include alkylalkylene group groups, for example, alkylmethylene groups such as $\text{—CH(CH}_{\text{sub.3}}\text{)—}$, $\text{—CH(CH}_{\text{sub.2}}\text{CH}_{\text{sub.3}}\text{)—}$, $\text{—C(CH}_{\text{sub.3}}\text{)}_{\text{sub.2}}\text{—}$, $\text{—C(CH}_{\text{sub.3}}\text{)(CH}_{\text{sub.2}}\text{CH}_{\text{sub.3}}\text{)—}$, $\text{—C(CH}_{\text{sub.3}}\text{)(CH}_{\text{sub.2}}\text{CH}_{\text{sub.2}}\text{CH}_{\text{sub.3}}\text{)—}$, and $\text{—C(CH}_{\text{sub.2}}\text{CH}_{\text{sub.3}}\text{)}_{\text{sub.2}}\text{—}$; alkylethylene groups such as $\text{—CH(CH}_{\text{sub.3}}\text{)CH}_{\text{sub.2}}\text{—}$, $\text{—CH(CH}_{\text{sub.3}}\text{)CH(CH}_{\text{sub.3}}\text{)—}$, $\text{—C(CH}_{\text{sub.3}}\text{)}_{\text{sub.2}}\text{CH}_{\text{sub.2}}\text{—}$, $\text{—CH(CH}_{\text{sub.2}}\text{CH}_{\text{sub.3}}\text{)CH}_{\text{sub.2}}\text{—}$, and $\text{—C(CH}_{\text{sub.2}}\text{CH}_{\text{sub.3}}\text{)}_{\text{sub.2}}\text{—CH}_{\text{sub.2}}\text{—}$; alkyltrimethylene groups such as $\text{—CH(CH}_{\text{sub.3}}\text{)CH}_{\text{sub.2}}\text{CH}_{\text{sub.2}}\text{—}$, and $\text{—CH}_{\text{sub.2}}\text{CH(CH}_{\text{sub.3}}\text{)CH}_{\text{sub.2}}\text{—}$; and alkyltetramethylene groups

such as $\text{—CH(CH.sub.3)CH.sub.2CH.sub.2CH.sub.2—}$, and $\text{—CH.sub.2CH(CH.sub.3)CH.sub.2CH.sub.2—}$. The alkyl group in the alkylalkylene group is preferably a linear alkyl group having 1 to 5 carbon atoms.

(275) The linear or branched aliphatic hydrocarbon group may have or may not have a substituent. Examples of the substituent include a fluorine atom, a fluorinated alkyl group having 1 to 5 carbon atoms, which has been substituted with a fluorine atom, and a carbonyl group.

(276) Aliphatic Hydrocarbon Group Containing Ring in Structure Thereof

(277) Examples of the aliphatic hydrocarbon group containing a ring in the structure thereof include a cyclic aliphatic hydrocarbon group which may have a substituent containing a hetero atom in the ring structure thereof (a group obtained by removing two hydrogen atoms from an aliphatic hydrocarbon ring), a group obtained by bonding the cyclic aliphatic hydrocarbon group to the terminal of a linear or branched aliphatic hydrocarbon group, and a group obtained by interposing the cyclic aliphatic hydrocarbon group in a linear or branched aliphatic hydrocarbon group. Examples of the linear or branched aliphatic hydrocarbon group include the same ones as those described above.

(278) The cyclic aliphatic hydrocarbon group preferably has 3 to 20 carbon atoms and more preferably 3 to 12 carbon atoms.

(279) The cyclic aliphatic hydrocarbon group may be a polycyclic group or a monocyclic group. The monocyclic alicyclic hydrocarbon group is preferably a group obtained by removing two hydrogen atoms from a monocycloalkane. The monocycloalkane preferably has 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane. The polycyclic alicyclic hydrocarbon group is preferably a group obtained by removing two hydrogen atoms from a polycycloalkane, and the polycycloalkane is preferably a group having 7 to 12 carbon atoms. Specific examples thereof include adamantane, norbornane, isobornane, tricyclodecane, and tetracyclododecane.

(280) The cyclic aliphatic hydrocarbon group may have or may not have a substituent. Examples of the substituent include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, and a carbonyl group.

(281) The alkyl group as the substituent is preferably an alkyl group having 1 to 5 carbon atoms, and more preferably a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group.

(282) The alkoxy group as the substituent is preferably an alkoxy group having 1 to 5 carbon atoms, more preferably a methoxy group, an ethoxy group, an n-propoxy group, an iso-propoxy group, an n-butoxy group, or a tert-butoxy group, and still more preferably a methoxy group or an ethoxy group.

(283) The halogen atom as the substituent is preferably a fluorine atom.

(284) Examples of the halogenated alkyl group as the substituent include groups obtained by substituting part or all hydrogen atoms in the above-described alkyl groups with the above-described halogen atoms.

(285) In the cyclic aliphatic hydrocarbon group, part of carbon atoms constituting the ring structure thereof may be substituted with a substituent containing a hetero atom. The substituent containing a hetero atom is preferably —O— , —C(=O)—O— , —S— , —S(=O).sub.2— , or —S(=O).sub.2—O— .

(286) Aromatic Hydrocarbon Group as Ya.sup.x1

(287) The aromatic hydrocarbon group is a hydrocarbon group having at least one aromatic ring.

(288) The aromatic ring is not particularly limited as long as it is a cyclic conjugated system having $(4n+2)$ π electrons, and may be monocyclic or polycyclic. The aromatic ring preferably has 5 to 30 carbon atoms, more preferably 5 to 20 carbon atoms, still more preferably 6 to 15 carbon atoms, and particularly preferably 6 to 12 carbon atoms. Here, the number of carbon atoms in a substituent is not included in the number of carbon atoms.

(289) Specific examples of the aromatic ring include aromatic hydrocarbon rings such as benzene, naphthalene, anthracene, and phenanthrene; and an aromatic heterocyclic ring obtained by substituting part of carbon atoms constituting the above-described aromatic hydrocarbon ring with a hetero atom. Examples of the hetero atom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom. Specific examples of the aromatic heterocyclic ring include a pyridine ring and a thiophene ring.

(290) Specific examples of the aromatic hydrocarbon group include a group (an arylene group or a heteroarylene group) obtained by removing two hydrogen atoms from the above-described aromatic hydrocarbon ring or the above-described aromatic heterocyclic ring; a group obtained by removing two hydrogen atoms from an aromatic compound having two or more aromatic rings (such as biphenyl or fluorene); and a group (for example, a group obtained by further removing one hydrogen atom from an aryl group in arylalkyl groups such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, a 1-naphthylethyl group, or a 2-naphthylethyl group) obtained by substituting one hydrogen atom of a group (an aryl group or a heteroaryl group) obtained by removing one hydrogen atom from the above aromatic hydrocarbon ring or the above aromatic heterocyclic ring, with an alkylene group. The alkylene group bonded to the aryl group or the heteroaryl group preferably has 1 to 4 carbon atoms, more preferably 1 or 2 carbon atoms, and particularly preferably 1 carbon atom.

(291) With respect to the aromatic hydrocarbon group, the hydrogen atom contained in the aromatic hydrocarbon group may be substituted with a substituent. For example, the hydrogen atom bonded to the aromatic ring in the aromatic hydrocarbon group may be substituted with a substituent. Examples of substituents include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, and a hydroxyl group.

(292) The alkyl group as the substituent is preferably an alkyl group having 1 to 5 carbon atoms, and more preferably a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group.

(293) Examples of the alkoxy group, the halogen atom, and the halogenated alkyl group, as the substituent, include those exemplified as the substituent that is substituted for a hydrogen atom contained in the cyclic aliphatic hydrocarbon group.

(294) Divalent Linking Group Containing Hetero Atom

(295) In a case where $Y_{a.sup.x1}$ represents a divalent linking group containing a hetero atom, preferred examples of the linking group include $-O-$, $-C(=O)-O-$, $-O-C(=O)-$, $-C(=O)-$, $-O-C(=O)-O-$, $-C(=O)-NH-$, $-NH-$, $-NH-C(=NH)-$ (H may be substituted with a substituent such as an alkyl group, an acyl group, or the like), $-S-$, $-S(=O).sub.2-$, $-S(=O).sub.2-O-$, and a group represented by General Formula $-Y_{a.sup.21}-O-Y_{a.sup.22}-$, $-Y_{a.sup.21}-O-$, $-Y_{a.sup.21}-C(=O)-O-$, $-C(=O)-O-Y_{a.sup.21}-$, $-[Y_{a.sup.21}-C(=O)-O].sub.m''-Y_{a.sup.22}-$, $-Y_{a.sup.21}-O-C(=O)-Y_{a.sup.22}-$ or $-Y_{a.sup.21}-S(=O).sub.2-O-Y_{a.sup.22}-$ [in the formulae, $Y_{a.sup.21}$ and $Y_{a.sup.22}$ each independently represent a divalent hydrocarbon group which may have a substituent, O represents an oxygen atom, and m'' represents an integer in a range of 0 to 3].

(296) In a case where the divalent linking group containing a hetero atom is $-C(=O)-NH-$, $-C(=O)-NH-C(=O)-$, $-NH-$, or $-NH-C(=NH)-$, H may be substituted with a substituent such as an alkyl group, an acyl group, or the like. The substituent (an alkyl group, an acyl group, or the like) preferably has 1 to 10 carbon atoms, more preferably 1 to 8 carbon atoms, and particularly preferably 1 to 5 carbon atoms.

(297) In General Formulae $-Y_{a.sup.21}-O-Y_{a.sup.22}-$, $-Y_{a.sup.21}-O-$, $-Y_{a.sup.21}-C(=O)-O-$, $-C(=O)-O-Y_{a.sup.21}-$, $-[Y_{a.sup.21}-C(=O)-O].sub.m''-Y_{a.sup.22}-$, $-Y_{a.sup.21}-O-C(=O)-Y_{a.sup.22}-$, and $-Y_{a.sup.21}-S(=O).sub.2-O-Y_{a.sup.22}-$, $Y_{a.sup.21}$, and $Y_{a.sup.22}$ each independently represent a divalent hydrocarbon group

which may have a substituent. Examples of the divalent hydrocarbon group include the same one as the divalent hydrocarbon groups which may have a substituent, mentioned in the explanation of the above-described divalent linking group as Ya.sup.x1.

(298) Y.sup.21 is preferably a linear aliphatic hydrocarbon group, more preferably a linear alkylene group, still more preferably a linear alkylene group having 1 to 5 carbon atoms, and particularly preferably a methylene group or an ethylene group.

(299) Y.sup.22 is preferably a linear or branched aliphatic hydrocarbon group and more preferably a methylene group, an ethylene group, or an alkylmethylene group. The alkyl group in the alkylmethylene group is preferably a linear alkyl group having 1 to 5 carbon atoms, more preferably a linear alkyl group having 1 to 3 carbon atoms, and most preferably a methyl group.

(300) In the group represented by General Formula $\text{---}[\text{Y.sup.21---C(=O)---O}]_{\text{sub.m''}}\text{---Y.sup.22---}$, m'' represents an integer in a range of 0 to 3, preferably an integer in a range of 0 to 2, more preferably 0 or 1, and particularly preferably 1. In other words, it is particularly preferable that the group represented by General Formula $\text{---}[\text{Y.sup.21---C(=O)---O}]_{\text{sub.m''}}\text{---Y.sup.22---}$ represents a group represented by General Formula $\text{---Y.sup.21---C(=O)---O---Y.sup.22---}$. Among these, a group represented by Formula $\text{---}(\text{CH.sub.2})_{\text{sub.a'}}\text{---C(=O)---O---}(\text{CH.sub.2})_{\text{sub.b'}}\text{---}$ is preferable. In the formula, a' represents an integer in a range of 1 to 10, preferably an integer in a range of 1 to 8, more preferably an integer in a range of 1 to 5, still more preferably 1 or 2, and most preferably 1. b' represents an integer in a range of 1 to 10, preferably an integer in a range of 1 to 8, more preferably an integer in a range of 1 to 5, still more preferably 1 or 2, and most preferably 1.

(301) Among the above, Ya.sup.x1 is preferably a single bond, an ester bond ---C(=O)---O--- , ---O---C(=O)--- , an ether bond (---O---), a linear or branched alkylene group, or a combination thereof, and more preferably a single bond or an ester bond ---C(=O)---O--- , ---O---C(=O)--- .

(302) In General Formula (a10-1), Wa.sup.x1 represents an aromatic hydrocarbon group which may have a substituent.

(303) Examples of the aromatic hydrocarbon group as Wa.sup.x1 include a group in which (n.sub.ax1+1) hydrogen atoms have been removed from an aromatic ring which may have a substituent. Here, the aromatic ring is not particularly limited as long as it is a cyclic conjugated system having $(4n+2)\pi$ electrons, and may be monocyclic or polycyclic. The aromatic ring preferably has 5 to 30 carbon atoms, more preferably 5 to 20 carbon atoms, still more preferably 6 to 15 carbon atoms, and particularly preferably 6 to 12 carbon atoms. Specific examples of the aromatic ring include aromatic hydrocarbon rings such as benzene, naphthalene, anthracene, and phenanthrene; and aromatic heterocyclic rings obtained by substituting part of carbon atoms constituting the above-described aromatic hydrocarbon ring with a hetero atom. Examples of the hetero atom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom. Specific examples of the aromatic heterocyclic ring include a pyridine ring and a thiophene ring.

(304) Examples of the aromatic hydrocarbon group as Wa.sup.x1 also include a group obtained by removing (n.sub.ax1+1) hydrogen atoms from an aromatic compound including an aromatic ring (for example, biphenyl and fluorene) which may have two or more substituents.

(305) Among the above, Wa.sup.x1 is preferably a group in which (n.sub.ax1+1) hydrogen atoms have been removed from benzene, naphthalene, anthracene, or biphenyl, more preferably a group in which (n.sub.ax1+1) hydrogen atoms have been removed from benzene or naphthalene, and still more preferably a group in which (n.sub.ax1+1) hydrogen atoms have been removed from benzene.

(306) The aromatic hydrocarbon group as Wa.sup.x1 may or may not have a substituent. Examples of the substituent include an alkyl group, an alkoxy group, a halogen atom, and a halogenated alkyl group. Examples of the alkyl group, the alkoxy group, the halogen atom, and the halogenated alkyl group, as the substituent, include the same ones as those described as the above-described substituent of the cyclic aliphatic hydrocarbon group as Ya.sup.x1. The substituent is preferably a linear or branched alkyl group having 1 to 5 carbon atoms, more preferably a linear or branched alkyl group having 1 to 3 carbon atoms, still more preferably an ethyl group or a methyl group, and particularly preferably a methyl group. The aromatic hydrocarbon group as Wa.sup.x1 preferably has no substituent.

(307) In General Formula (a10-1), n.sub.ax1 represents an integer of 1 or more, preferably an integer in a range of 1 to 10, more preferably an integer in a range of 1 to 5, still more preferably 1, 2, or 3, and particularly preferably 1 or 2.

(308) Specific examples of the constitutional unit (a10) represented by General Formula (a10-1) are shown below.

(309) In each of the formulae shown below, R.sup.α represents a hydrogen atom, a methyl group, or a trifluoromethyl group.

(310) ##STR00050## ##STR00051## ##STR00052## ##STR00053##

(311) The constitutional unit (a10) contained in the component (A1) may be one kind or may be two or more kinds.

(312) In a case where the component (A1) contains the constitutional unit (a10), the proportion of the constitutional unit (a10) in the component (A1) is preferably in a range of 10% to 80% by mole and more preferably in a range of 40% to 60% by mole with respect to the total (100% by mole) of all constitutional units constituting the component (A1).

(313) In a case where the proportion of the constitutional unit (a10) is equal to or larger than the lower limit value, the sensitivity can be more easily increased. On the other hand, in a case where it is equal to or smaller than the upper limit value, the balance with other constitutional units is easily obtained.

(314) In regard to constitutional unit (a2):

(315) The component (A1) may further have a constitutional unit (a2) (provided that a constitutional unit corresponding to the constitutional unit (a01) or the constitutional unit (a1) is excluded) containing a lactone-containing cyclic group, a —SO.sub.2—containing cyclic group, or a carbonate-containing cyclic group.

(316) In a case where the component (A1) is used for forming a resist film, the lactone-containing cyclic group, the —SO.sub.2—containing cyclic group, or the carbonate-containing cyclic group in the constitutional unit (a2) is effective for improving the adhesiveness of the resist film to the substrate. Further, due to having the constitutional unit (a2), lithography characteristics can be improved, for example, by the effects obtained by appropriately adjusting the acid diffusion length, increasing the adhesiveness of the resist film to the substrate, and appropriately adjusting the solubility during development.

(317) The “lactone-containing cyclic group” indicates a cyclic group that contains a ring (lactone ring) containing a —O—C(=O)— in the ring skeleton. In a case where the lactone ring is counted as the first ring and the group contains only the lactone ring, the group is referred to as a monocyclic group. Further, in a case where the group has other ring structures, the group is referred to as a polycyclic group regardless of the structures. The lactone-containing cyclic group may be a monocyclic group or a polycyclic group.

(318) The lactone-containing cyclic group for the constitutional unit (a2) is not particularly limited, and any lactone-containing cyclic group may be used. Specific examples thereof include groups each represented by General Formulae (a2-r-1) to (a2-r-7) shown below.

(319) ##STR00054##

(320) [In the formulae, $Ra'.sup.21S$ each independently represent a hydrogen atom, an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, $-COOR''$, $-OC(=O)R''$, a hydroxyalkyl group, or a cyano group; R'' represents a hydrogen atom, an alkyl group, a lactone-containing cyclic group, a carbonate-containing cyclic group, or a $-SO.sub.2-$ -containing cyclic group; A'' represents an oxygen atom, a sulfur atom, or an alkylene group having 1 to 5 carbon atoms, which may contain an oxygen atom ($-O-$) or a sulfur atom ($-S-$); and n' represents an integer in a range of 0 to 2, and m' is 0 or 1.]

(321) In General Formulae (a2-r-1) to (a2-r-7), the alkyl group as $Ra'.sup.21$ is preferably an alkyl group having 1 to 6 carbon atoms. The alkyl group is preferably a linear alkyl group or a branched alkyl group. Specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, a neopentyl group, and a hexyl group. Among these, a methyl group or ethyl group is preferable, and a methyl group is particularly preferable.

(322) The alkoxy group as $Ra'.sup.21$ is preferably an alkoxy group having 1 to 6 carbon atoms. Further, the alkoxy group is preferably a linear or branched alkoxy group. Specific examples of the alkoxy groups include a group formed by linking the above-described alkyl group mentioned as the alkyl group represented by $Ra'.sup.21$ to an oxygen atom ($-O-$).

(323) The halogen atom as $Ra'.sup.21$ is preferably a fluorine atom.

(324) Examples of the halogenated alkyl group as $Ra'.sup.21$ include group obtained by substituting part or all hydrogen atoms in the above-described alkyl group as $Ra'.sup.21$ with the above-described halogen atoms. The halogenated alkyl group is preferably a fluorinated alkyl group and particularly preferably a perfluoroalkyl group.

(325) In $-COOR''$ and $-OC(=O)R''$ as $Ra'.sup.21$, R'' represents a hydrogen atom, an alkyl group, a lactone-containing cyclic group, a carbonate-containing cyclic group, or a $-SO.sub.2-$ -containing cyclic group.

(326) The alkyl group as R'' may be linear, branched, or cyclic, and preferably has 1 to 15 carbon atoms.

(327) In a case where R'' represents a linear or branched alkyl group, it is preferably an alkyl group having 1 to 10 carbon atoms, more preferably an alkyl group having 1 to 5 carbon atoms, and particularly preferably a methyl group or an ethyl group.

(328) In a case where R'' represents a cyclic alkyl group, the cyclic alkyl group preferably has 3 to 15 carbon atoms, more preferably 4 to 12 carbon atoms, and particularly preferably 5 to 10 carbon atoms. Specific examples thereof include a group obtained by removing one or more hydrogen atoms from a monocycloalkane, which may be or may not be substituted with a fluorine atom or a fluorinated alkyl group; and a group obtained by removing one or more hydrogen atoms from a polycycloalkane such as bicycloalkane, tricycloalkane, or tetracycloalkane. More specific examples thereof include a group obtained by removing one or more hydrogen atoms from a monocycloalkane such as cyclopentane or cyclohexane; and a group obtained by removing one or more hydrogen atoms from a polycycloalkane such as adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane.

(329) Examples of the lactone-containing cyclic group as R'' include the same ones as the groups each represented by General Formulae (a2-r-1) to (a2-r-7).

(330) The carbonate-containing cyclic group as R'' is the same as the carbonate-containing cyclic group described below. Specific examples thereof include groups each represented by General Formulae (ax3-r-1) to (ax3-r-3).

(331) The $-SO.sub.2-$ -containing cyclic group as R'' is the same the $-SO.sub.2-$ -containing cyclic group described below. Specific examples thereof include groups each represented by General Formulae (a5-r-1) to (a5-r-4).

(332) The hydroxyalkyl group as $Ra'.sup.21$ preferably has 1 to 6 carbon atoms, and specific examples thereof include

a group obtained by substituting at least one hydrogen atom in the alkyl group as Ra'.sup.21 with a hydroxyl group.

(333) In General Formulae (a2-r-2), (a2-r-3) and (a2-r-5), as the alkylene group having 1 to 5 carbon atoms as A'', a linear or branched alkylene group is preferable, and examples thereof include a methylene group, an ethylene group, an n-propylene group, and an isopropylene group. Specific examples of the alkylene groups that contain an oxygen atom or a sulfur atom include a group obtained by interposing —O— or —S— in the terminal of the alkylene group or between the carbon atoms of the alkylene group, and examples thereof include —O—CH₂—, —CH₂—O—CH₂—, —S—CH₂—, and —CH₂—S—CH₂—. A'' is preferably an alkylene group having 1 to 5 carbon atoms or —O—, more preferably an alkylene group having 1 to 5 carbon atoms, and most preferably a methylene group.

(334) Specific examples of the groups each represented by General Formulae (a2-r-1) to (a2-r-7) are shown below.

(335) ##STR00055## ##STR00056## ##STR00057## ##STR00058## ##STR00059## ##STR00060##

(336) The “—SO₂—containing cyclic group” indicates a cyclic group having a ring containing —SO₂— in the ring skeleton thereof. Specifically, it is a cyclic group in which the sulfur atom (S) in —SO₂— forms a part of the ring skeleton of the cyclic group. In a case where the ring containing —SO₂— in the ring skeleton thereof is counted as the first ring and the group contains only the ring, the group is referred to as a monocyclic group. In a case where the group further has other ring structures, the group is referred to as a polycyclic group regardless of the ring structures. The —SO₂—containing cyclic group may be a monocyclic group or a polycyclic group.

(337) The —SO₂—containing cyclic group is particularly preferably a cyclic group containing —O—SO₂— in the ring skeleton thereof, in other words, a cyclic group containing a sultone ring in which —O—S— in the —O—SO₂— group forms a part of the ring skeleton thereof.

(338) More specific examples of the —SO₂—containing cyclic group include groups each represented by General Formulae (a5-r-1) to (a5-r-4) shown below.

(339) ##STR00061##

(340) [In the formulae, Ra'.sup.51s each independently represent a hydrogen atom, an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, —COOR'', —OC(=O)R'', a hydroxyalkyl group, or a cyano group; R'' represents a hydrogen atom, an alkyl group, a lactone-containing cyclic group, a carbonate-containing cyclic group, or a —SO₂—containing cyclic group; A'' represents an oxygen atom, a sulfur atom, or an alkylene group having 1 to 5 carbon atoms, which may contain an oxygen atom or a sulfur atom; and n' represents an integer in a range of 0 to 2.]

(341) In General Formulae (a5-r-1) and (a5-r-2), A'' has the same definition as that for A'' in General Formulae (a2-r-2), (a2-r-3) and (a2-r-5).

(342) Examples of the alkyl group, the alkoxy group, the halogen atom, the halogenated alkyl group, —COOR'', —OC(=O)R'', and the hydroxyalkyl group as Ra'.sup.51 each include the same ones as those mentioned in the explanation of Ra'.sup.21 in General Formulae (a2-r-1) to (a2-r-7).

(343) Specific examples of the groups each represented by General Formulae (a5-r-1) to (a5-r-4) are shown below. In the formulae shown below, “Ac” represents an acetyl group.

(344) ##STR00062## ##STR00063## ##STR00064## ##STR00065##

(345) The “carbonate-containing cyclic group” indicates a cyclic group having a ring (a carbonate ring) containing —O—C(=O)—O— in the ring skeleton thereof. In a case where the carbonate ring is counted as the first ring and the group contains only the carbonate ring, the group is referred to as a monocyclic group. Further, in a case where the

group has other ring structures, the group is referred to as a polycyclic group regardless of the structures. The carbonate-containing cyclic group may be a monocyclic group or a polycyclic group.

(346) The carbonate ring-containing cyclic group is not particularly limited, and any carbonate ring-containing cyclic group may be used. Specific examples thereof include groups each represented by General Formulae (ax3-r-1) to (ax3-r-3) shown below.

(347) ##STR00066##

(348) [In the formulae, Ra'.sup.x31s independently represent a hydrogen atom, an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, —COOR'', —OC(=O)R'', a hydroxyalkyl group, or a cyano group; R'' represents a hydrogen atom, an alkyl group, a lactone-containing cyclic group, a carbonate-containing cyclic group, or a —SO.sub.2—containing cyclic group; A'' represents an oxygen atom, a sulfur atom, or an alkylene group having 1 to 5 carbon atoms, which may contain an oxygen atom or a sulfur atom; and p' represents an integer in a range of 0 to 3, and q' is 0 or 1.]

(349) In General Formulae (ax3-r-2) and (ax3-r-3), A'' has the same definition as that for A'' in General Formulae (a2-r-2), (a2-r-3) and (a2-r-5).

(350) Respective examples of the alkyl group, the alkoxy group, the halogen atom, the halogenated alkyl group, —COOR'', —OC(=O)R'', and the hydroxyalkyl group as Ra'.sup.31 include the same ones as those mentioned in the explanation of Ra'.sup.21 in General Formulae (a2-r-1) to (a2-r-7).

(351) Specific examples of the groups each represented by General Formulae (ax3-r-1) to (ax3-r-3) are shown below.

(352) ##STR00067## ##STR00068##

(353) Among them, the constitutional unit (a2) is preferably a constitutional unit derived from acrylic acid ester in which the hydrogen atom bonded to the carbon atom at the α -position may be substituted with a substituent.

(354) The constitutional unit (a2) is preferably a constitutional unit represented by General Formula (a2-1).

(355) ##STR00069##

(356) [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. Ya.sup.21 represents a single bond or a divalent linking group. La.sup.21 represents —O—, —COO—, —CON(R')—, —OCO—, —CONHCO— or —CONHCS—, and R' represents a hydrogen atom or a methyl group. However, in a case where La.sup.21 represents —O—, ya.sup.21 does not represent —CO—.

Ra.sup.21 represents a lactone-containing cyclic group, a carbonate-containing cyclic group, or a —SO.sub.2—containing cyclic group.]

(357) In General Formula (a2-1), R has the same definition as described above. R is preferably a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a fluorinated alkyl group having 1 to 5 carbon atoms, and particularly preferably a hydrogen atom or a methyl group in terms of industrial availability.

(358) In General Formula (a2-1), the divalent linking group as Ya.sup.21 is not particularly limited, and suitable examples thereof include a divalent hydrocarbon group which may have a substituent and a divalent linking group having a hetero atom.

(359) Divalent Hydrocarbon Group which May have Substituent:

(360) In a case where Ya.sup.21 represents a divalent hydrocarbon group which may have a substituent, the hydrocarbon group may be an aliphatic hydrocarbon group or an aromatic hydrocarbon group.

(361) Aliphatic Hydrocarbon Group as Ya.sup.21

(362) The aliphatic hydrocarbon group indicates a hydrocarbon group that has no aromaticity. The aliphatic

hydrocarbon group may be saturated or unsaturated. In general, it is preferable that the aliphatic hydrocarbon group is saturated.

(363) Examples of the aliphatic hydrocarbon group include a linear or branched aliphatic hydrocarbon group, and an aliphatic hydrocarbon group containing a ring in the structure thereof.

(364) Linear or Branched Aliphatic Hydrocarbon Group

(365) The linear aliphatic hydrocarbon group preferably has 1 to 10 carbon atoms, more preferably 1 to 6 carbon atoms, still more preferably 1 to 4 carbon atoms, and most preferably 1 to 3 carbon atoms.

(366) The linear aliphatic hydrocarbon group is preferably a linear alkylene group, and specific examples thereof include a methylene group [$\text{—CH}_{2}\text{—}$], an ethylene group [$\text{—(CH}_{2}\text{)}_{2}\text{—}$], a trimethylene group [$\text{—(CH}_{2}\text{)}_{3}\text{—}$], a tetramethylene group [$\text{—(CH}_{2}\text{)}_{4}\text{—}$], and a pentamethylene group [$\text{—(CH}_{2}\text{)}_{5}\text{—}$].

(367) The branched aliphatic hydrocarbon group has preferably 2 to 10 carbon atoms, more preferably 3 to 6 carbon atoms, still more preferably 3 or 4 carbon atoms, and most preferably 3 carbon atoms.

(368) The branched aliphatic hydrocarbon group is preferably a branched alkylene group, and specific examples thereof include alkylalkylene group groups, for example, alkylmethylene groups such as $\text{—CH(CH}_{3}\text{)—}$, $\text{—CH(CH}_{2}\text{CH}_{3}\text{)—}$, $\text{—C(CH}_{3}\text{)}_{2}\text{—}$, $\text{—C(CH}_{3}\text{)(CH}_{2}\text{CH}_{3}\text{)—}$, $\text{—C(CH}_{3}\text{)(CH}_{2}\text{CH}_{2}\text{CH}_{3}\text{)—}$, and $\text{—C(CH}_{2}\text{CH}_{3}\text{)}_{2}\text{—}$; alkylethylene groups such as $\text{—CH(CH}_{3}\text{)CH}_{2}\text{—}$, $\text{—CH(CH}_{3}\text{)CH(CH}_{3}\text{)—}$, $\text{—C(CH}_{3}\text{)}_{2}\text{CH}_{2}\text{—}$, $\text{—CH(CH}_{2}\text{CH}_{3}\text{)CH}_{2}\text{—}$, and $\text{—C(CH}_{2}\text{CH}_{3}\text{)}_{2}\text{CH}_{2}\text{—}$; alkyltrimethylene groups such as $\text{—CH(CH}_{3}\text{)CH}_{2}\text{CH}_{2}\text{—}$, and $\text{—CH}_{2}\text{CH(CH}_{3}\text{)CH}_{2}\text{—}$; and alkyltetramethylene groups such as $\text{—CH(CH}_{3}\text{)CH}_{2}\text{CH}_{2}\text{CH}_{2}\text{—}$, and $\text{—CH}_{2}\text{CH(CH}_{3}\text{)CH}_{2}\text{CH}_{2}\text{—}$. The alkyl group in the alkylalkylene group is preferably a linear alkyl group having 1 to 5 carbon atoms.

(369) The linear or branched aliphatic hydrocarbon group may have or may not have a substituent. Examples of the substituent include a fluorine atom, a fluorinated alkyl group having 1 to 5 carbon atoms, which has been substituted with a fluorine atom, and a carbonyl group.

(370) Aliphatic Hydrocarbon Group Containing Ring in Structure Thereof

(371) Examples of the aliphatic hydrocarbon group containing a ring in the structure thereof include a cyclic aliphatic hydrocarbon group which may have a substituent containing a hetero atom in the ring structure thereof (a group obtained by removing two hydrogen atoms from an aliphatic hydrocarbon ring), a group obtained by bonding the cyclic aliphatic hydrocarbon group to the terminal of a linear or branched aliphatic hydrocarbon group, and a group obtained by interposing the cyclic aliphatic hydrocarbon group in a linear or branched aliphatic hydrocarbon group. Examples of the linear or branched aliphatic hydrocarbon group include the same ones as those described above.

(372) The cyclic aliphatic hydrocarbon group preferably has 3 to 20 carbon atoms and more preferably 3 to 12 carbon atoms.

(373) The cyclic aliphatic hydrocarbon group may be a polycyclic group or a monocyclic group. The monocyclic alicyclic hydrocarbon group is preferably a group obtained by removing two hydrogen atoms from a monocycloalkane. The monocycloalkane preferably has 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane. The polycyclic alicyclic hydrocarbon group is preferably a group obtained by removing two hydrogen atoms from a polycycloalkane, and the polycycloalkane is preferably a group having 7 to 12 carbon atoms. Specific examples thereof include adamantane, norbornane, isobornane, tricyclodecane, and tetracyclododecane.

(374) The cyclic aliphatic hydrocarbon group may have or may not have a substituent. Examples of the substituent include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, and a carbonyl group.

(375) The alkyl group as the substituent is preferably an alkyl group having 1 to 5 carbon atoms, and more preferably a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group.

(376) The alkoxy group as the substituent is preferably an alkoxy group having 1 to 5 carbon atoms, more preferably a methoxy group, an ethoxy group, an n-propoxy group, an iso-propoxy group, an n-butoxy group, or a tert-butoxy group, and still more preferably a methoxy group or an ethoxy group.

(377) The halogen atom as the substituent is preferably a fluorine atom.

(378) Examples of the halogenated alkyl group as the substituent include groups obtained by substituting part or all hydrogen atoms in the above-described alkyl groups with the above-described halogen atoms.

(379) In the cyclic aliphatic hydrocarbon group, part of carbon atoms constituting the ring structure thereof may be substituted with a substituent containing a hetero atom. The substituent containing a hetero atom is preferably —O—, —C(=O)—O—, —S—, —S(=O).sub.2—, or —S(=O).sub.2—O—.

(380) Aromatic Hydrocarbon Group as Ya.sup.21

(381) The aromatic hydrocarbon group is a hydrocarbon group having at least one aromatic ring.

(382) The aromatic ring is not particularly limited as long as it is a cyclic conjugated system having $(4n+2)$ π electrons, and may be monocyclic or polycyclic. The aromatic ring preferably has 5 to 30 carbon atoms, more preferably 5 to 20 carbon atoms, still more preferably 6 to 15 carbon atoms, and particularly preferably 6 to 12 carbon atoms. Here, the number of carbon atoms in a substituent is not included in the number of carbon atoms.

(383) Specific examples of the aromatic ring include aromatic hydrocarbon rings such as benzene, naphthalene, anthracene, and phenanthrene; and an aromatic heterocyclic ring obtained by substituting part of carbon atoms constituting the above-described aromatic hydrocarbon ring with a hetero atom. Examples of the hetero atom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom. Specific examples of the aromatic heterocyclic ring include a pyridine ring and a thiophene ring.

(384) Specific examples of the aromatic hydrocarbon group include a group (an arylene group or a heteroarylene group) obtained by removing two hydrogen atoms from the above-described aromatic hydrocarbon ring or the above-described aromatic heterocyclic ring; a group obtained by removing two hydrogen atoms from an aromatic compound having two or more aromatic rings (such as biphenyl or fluorene); and a group (for example, a group obtained by further removing one hydrogen atom from an aryl group in arylalkyl groups such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, a 1-naphthylethyl group, or a 2-naphthylethyl group) obtained by substituting one hydrogen atom of a group (an aryl group or a heteroaryl group) obtained by removing one hydrogen atom from the above aromatic hydrocarbon ring or the above aromatic heterocyclic ring, with an alkylene group. The alkylene group bonded to the aryl group or the heteroaryl group preferably has 1 to 4 carbon atoms, more preferably 1 or 2 carbon atoms, and particularly preferably 1 carbon atom.

(385) With respect to the aromatic hydrocarbon group, the hydrogen atom contained in the aromatic hydrocarbon group may be substituted with a substituent. For example, the hydrogen atom bonded to the aromatic ring in the aromatic hydrocarbon group may be substituted with a substituent. Examples of substituents include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, and a hydroxyl group.

(386) The alkyl group as the substituent is preferably an alkyl group having 1 to 5 carbon atoms, and more preferably a

methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group.

(387) Examples of the alkoxy group, the halogen atom, and the halogenated alkyl group, as the substituent, include those exemplified as the substituent that is substituted for a hydrogen atom contained in the cyclic aliphatic hydrocarbon group.

(388) Divalent Linking Group Containing Hetero Atom

(389) In a case where Ya.sup.21 represents a divalent linking group containing a hetero atom, preferred examples of the linking group include —O—, —C(=O)—O—, —O—C(=O)—, —C(=O)—, —O—C(=O)—O—, —C(=O)—NH—, —NH—, —NH—C(=NH)—(H may be substituted with a substituent such as an alkyl group, an acyl group, or the like), —S—, —S(=O).sub.2—, —S(=O).sub.2—O—, and a group represented by General Formula —Y.sup.21—O—Y.sup.22, —Y.sup.21—O—, —Y.sup.21—C(=O)—O—, —C(=O)—O—Y.sup.21—, —[Y.sup.21—C(=O)—O].sub.m''—Y.sup.22—, —Y.sup.21—O—C(=O)—Y.sup.22— or —Y.sup.21—S(=O).sub.2—O—Y.sup.22—[in the formulae, Y.sup.21 and Y.sup.22 each independently represent a divalent hydrocarbon group which may have a substituent, O represents an oxygen atom, and m'' represents an integer in a range of 0 to 3].

(390) In a case where the divalent linking group containing a hetero atom is —C(=O)—NH—, —C(=O)—NH—C(=O)—, —NH—, or —NH—C(=NH)—, H may be substituted with a substituent such as an alkyl group, an acyl group, or the like. The substituent (an alkyl group, an acyl group, or the like) preferably has 1 to 10 carbon atoms, more preferably 1 to 8 carbon atoms, and particularly preferably 1 to 5 carbon atoms.

(391) In General Formulae —Y.sup.21—O—Y.sup.22—, —Y.sup.21—O—, —Y.sup.21—C(=O)—O—, —C(=O)—O—Y.sup.21—, —[Y.sup.21—C(=O)—O].sub.m''—Y.sup.22—, —Y.sup.21—O—C(=O)—Y.sup.22— and —Y.sup.21—S(=O).sub.2—O—Y.sup.22—, and Y.sup.22 each independently represent a divalent hydrocarbon group which may have a substituent. Examples of the divalent hydrocarbon group include the same one as the divalent hydrocarbon groups which may have a substituent, mentioned in the explanation of the above-described divalent linking group as Ya.sup.21.

(392) Y.sup.21 is preferably a linear aliphatic hydrocarbon group, more preferably a linear alkylene group, still more preferably a linear alkylene group having 1 to 5 carbon atoms, and particularly preferably a methylene group or an ethylene group.

(393) Y.sup.22 is preferably a linear or branched aliphatic hydrocarbon group and more preferably a methylene group, an ethylene group, or an alkylmethylene group. The alkyl group in the alkylmethylene group is preferably a linear alkyl group having 1 to 5 carbon atoms, more preferably a linear alkyl group having 1 to 3 carbon atoms, and most preferably a methyl group.

(394) In the group represented by General Formula —[Y.sup.21—C(=O)—O].sub.m''—Y.sup.22—, m'' represents an integer in a range of 0 to 3, preferably an integer in a range of 0 to 2, more preferably 0 or 1, and particularly preferably 1. In other words, it is particularly preferable that the group represented by General Formula —[Y.sup.21—C(=O)—O].sub.m''—Y.sup.22— represents a group represented by General Formula —Y.sup.21—C(=O)—O—Y.sup.22—. Among these, a group represented by Formula —(CH.sub.2).sub.a'—C(=O)—O—(CH.sub.2).sub.b'— is preferable. In the formula, a' represents an integer in a range of 1 to 10, preferably an integer in a range of 1 to 8, more preferably an integer in a range of 1 to 5, still more preferably 1 or 2, and most preferably 1. b' represents an integer in a range of 1 to 10, preferably an integer in a range of 1 to 8, more preferably an integer in a range of 1 to 5, still more preferably 1 or 2, and most preferably 1.

(395) Among the above, Ya.sup.21 is preferably a single bond, an ester bond [—C(=O)—O—], an ether bond (—O—),

a linear or branched alkylene group, or a combination thereof.

(396) In General Formula (a2-1), Ra.sup.21 represents a lactone-containing cyclic group, a —SO.sub.2—containing cyclic group, or a carbonate-containing cyclic group.

(397) Suitable examples of the lactone-containing cyclic group, the —SO.sub.2—containing cyclic group, and the carbonate-containing cyclic group as Ra.sup.21 include groups each represented by General Formulae (a2-r-1) to (a2-r-7), groups each represented by General Formulae (a5-r-1) to (a5-r-4), and groups each represented by General Formulae (ax3-r-1) to (ax3-r-3) described above.

(398) Among them, a lactone-containing cyclic group or a —SO.sub.2—containing cyclic group is preferable, and any one of groups each represented by General Formula (a2-r-1), (a2-r-2), (a2-r-6), or (a5-r-1) is preferable. Specifically, groups each represented by any one of Chemical Formulae (r-1c-1-1) to (r-1c-1-7), (r-1c-2-1) to (r-1c-2-18), (r-1c-6-1), (r-s1-1-1), and (r-s1-1-18) are more preferable.

(399) The constitutional unit (a2) contained in the component (A1) may be one kind or may be two or more kinds.

(400) In a case where the component (A1) has the constitutional unit (a2), the proportion of the constitutional unit (a2) is preferably in a range of 5% to 40% by mole and more preferably in a range of 5% to 30% by mole, with respect to the total (100% by mole) of all constitutional units constituting the component (A1).

(401) In a case where the proportion of the constitutional unit (a2) is equal to or larger than the lower limit value of the preferred range, the effect obtained by allowing the component (A1) to contain the constitutional unit (a2) can be sufficiently achieved by the effect described above. In a case where it is equal to or smaller than the upper limit value of the preferred range, the balance with other constitutional units can be obtained, and various lithography characteristics are improved.

(402) In Regard to Constitutional Unit (a3):

(403) The component (A1) may further have a constitutional unit (a3) (provided that a constitutional unit corresponding to the constitutional unit (a01), the constitutional unit (a1), or the constitutional unit (a2) is excluded) containing a polar group-containing aliphatic hydrocarbon group. In a case where the component (A1) has the constitutional unit (a3), the hydrophilicity of the component (A1) is increased, which contributes to an improvement in resolution. Further, acid diffusion length can be appropriately adjusted.

(404) Examples of the polar group include a hydroxyl group, a cyano group, a carboxy group, or a hydroxyalkyl group obtained by substituting part of hydrogen atoms of the alkyl group with a fluorine atom, and the polar group is particularly preferably a hydroxyl group.

(405) Examples of the aliphatic hydrocarbon group include a linear or branched hydrocarbon group (preferably an alkylene group) having 1 to 10 carbon atoms, and a cyclic aliphatic hydrocarbon group (a cyclic group). The cyclic group may be a monocyclic group or a polycyclic group. For example, these cyclic groups can be appropriately selected from a large number of groups that have been proposed in resins for a resist composition for an ArF excimer laser.

(406) In a case where the cyclic group is a monocyclic group, the monocyclic group preferably has 3 to 10 carbon atoms. Among them, a constitutional unit derived from an acrylic acid ester that includes an aliphatic monocyclic group containing a hydroxyl group, cyano group, carboxy group, or a hydroxyalkyl group obtained by substituting part of hydrogen atoms of the alkyl group with a fluorine atom are particularly preferable. Examples of the monocyclic group include a group obtained by removing two or more hydrogen atoms from a monocycloalkane. Specific examples of the monocyclic group include a group obtained by removing two or more hydrogen atoms from a monocycloalkane such as

cyclopentane, cyclohexane, or cyclooctane. Among these monocyclic groups, a group obtained by removing two or more hydrogen atoms from cyclopentane or a group obtained by removing two or more hydrogen atoms from cyclohexane are industrially preferable.

(407) In a case where the cyclic group is a polycyclic group, the polycyclic group preferably has 7 to 30 carbon atoms. Among them, a constitutional unit derived from an acrylic acid ester that includes an aliphatic polycyclic group containing a hydroxyl group, cyano group, carboxy group, or a hydroxyalkyl group obtained by substituting part of hydrogen atoms of the alkyl group with a fluorine atom is particularly preferable. Examples of the polycyclic group include groups obtained by removing two or more hydrogen atoms from a bicycloalkane, tricycloalkane, tetracycloalkane, or the like. Specific examples thereof include a group obtained by removing two or more hydrogen atoms from a polycycloalkane such as adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane. Among these polycyclic groups, a group obtained by removing two or more hydrogen atoms from adamantane, a group obtained by removing two or more hydrogen atoms from norbornane, or a group obtained by removing two or more hydrogen atoms from tetracyclododecane are industrially preferable.

(408) The constitutional unit (a3) is not particularly limited, and any constitutional unit may be used as long as the constitutional unit contains a polar group-containing aliphatic hydrocarbon group.

(409) The constitutional unit (a3) is preferably a constitutional unit derived from an acrylic acid ester in which the hydrogen atom bonded to the carbon atom at the α -position may be substituted with a substituent, where the constitutional unit contains a polar group-containing aliphatic hydrocarbon group.

(410) In a case where the hydrocarbon group in the polar group-containing aliphatic hydrocarbon group is a linear or branched hydrocarbon group having 1 to 10 carbon atoms, the constitutional unit (a3) is preferably a constitutional unit derived from a hydroxyethyl ester of acrylic acid.

(411) Further, as the constitutional unit (a3), in a case where the hydrocarbon group in the polar group-containing aliphatic hydrocarbon group is a polycyclic group, a constitutional unit represented by General Formula (a3-1), a constitutional unit represented by General Formula (a3-2), or a constitutional unit represented by General Formula (a3-3) is preferable, and in a case where the hydrocarbon group is a monocyclic group, a constitutional unit represented by General Formula (a3-4) is preferable.

(412) ##STR00070##

(413) [In the formulae, R has the same definition as described above, j represents an integer in a range of 1 to 3, k represents an integer in a range of 1 to 3, t' represents an integer in a range of 1 to 3, 1 represents an integer in a range of 0 to 5, and s represents an integer in a range of 1 to 3.]

(414) In General Formula (a3-1), j is 1 or 2 and more preferably 1. In a case where j represents 2, it is preferable that the hydroxyl groups are bonded to the 3-position and the 5-position of the adamantyl group. In a case where j represents 1, it is preferable that the hydroxyl group is bonded to the 3-position of the adamantyl group.

(415) It is preferable that j represents 1, and it is particularly preferable that the hydroxyl group is bonded to the 3-position of the adamantyl group.

(416) In General Formula (a3-2), k is preferably 1. The cyano group is preferably bonded to the 5-position or 6-position of the norbornyl group.

(417) In General Formula (a3-3), it is preferable that t' represents 1. It is preferable that 1 represents 1. It is preferable that s represents 1. Further, it is preferable that a 2-norbornyl group or 3-norbornyl group is bonded to the terminal of the carboxy group of the acrylic acid. It is preferable that the fluorinated alkyl alcohol is bonded to the 5-position or 6-

position of the norbornyl group.

(418) In General Formula (a3-4), it is preferable that t' represents 1 or 2. It is preferable that 1 represents 0 or 1. It is preferable that s represents 1. It is preferable that the fluorinated alkyl alcohol is bonded to the 3-position or 5-position of the cyclohexyl group.

(419) The constitutional unit (a3) contained in the component (A1) may be one kind or may be two or more kinds.

(420) In a case where the component (A1) has the constitutional unit (a3), the proportion of the constitutional unit (a3) is preferably in a range of 1% to 30% by mole, more preferably in a range of 1% to 25% by mole, and still more preferably in a range of 1% to 20% by mole, with respect to the total (100% by mole) of all constitutional units constituting the component (A1).

(421) In a case where the proportion of the constitutional unit (a3) is equal to or larger than the lower limit value of the preferred range, the effect obtained by allowing the constitutional unit (a3) to be contained can be sufficiently achieved by the effect described above. In a case where it is equal to or smaller than the upper limit value of the preferred range, the balance with other constitutional units can be obtained, and various lithography characteristics are improved.

(422) In Regard to Constitutional Unit (a4):

(423) The component (A1) may further have a constitutional unit (a4) containing an acid non-dissociable aliphatic cyclic group.

(424) In a case where the component (A1) has the constitutional unit (a4), the dry etching resistance of the formed resist pattern is improved. Further, the hydrophobicity of the component (A1) increases. The improvement in hydrophobicity contributes to the improvement in resolution, a resist pattern shape, and the like, particularly in the case of a solvent developing process.

(425) The “acid non-dissociable cyclic group” in the constitutional unit (a4) is a cyclic group that remains in the constitutional unit without being dissociated even in a case where an acid acts in a case where the acid is generated in the resist composition by exposure (for example, in a case where an acid is generated from the constitutional unit that generates the acid upon exposure, or the component (B)).

(426) Examples of the constitutional unit (a4) preferably include a constitutional unit derived from an acrylic acid ester including an acid non-dissociable aliphatic cyclic group. As the cyclic group, a large number of known ones in the related art as cyclic groups used as the resin component of a resist composition for ArF excimer laser, KrF excimer laser (preferably ArF excimer laser), or the like can be used.

(427) The cyclic group is particularly preferably at least one selected from a tricyclodecyl group, an adamantyl group, a tetracyclododecyl group, an isobornyl group, and a norbornyl group, from the viewpoint of industrial availability. These polycyclic groups may have, as a substituent, a linear or branched alkyl group having 1 to 5 carbon atoms.

(428) Specific examples of the constitutional unit (a4) include constitutional units each represented by General Formulae (a4-1) to (a4-7).

(429) ##STR00071## ##STR00072##

(430) [In the formula, R.sup.α is the same as above.]

(431) The constitutional unit (a4) contained in the component (A1) may be one kind or may be two or more kinds.

(432) In a case where the component (A1) has the constitutional unit (a4), the proportion of the constitutional unit (a4) is preferably in a range of 1% to 40% by mole and more preferably in a range of 1% to 20% by mole, with respect to the total (100% by mole) of all constitutional units constituting the component (A1).

(433) In a case where the proportion of the constitutional unit (a4) is equal to or larger than the lower limit value of the

preferred range, the effect obtained by allowing the constitutional unit (a4) to be contained can be sufficiently achieved. In a case it is equal to or smaller than the upper limit value of the preferred range, the balance with other constitutional units is easily obtained.

(434) In Regard to Constitutional Unit (St):

(435) The constitutional unit (st) is a constitutional unit derived from styrene or a styrene derivative. The “constitutional unit derived from styrene” means a constitutional unit that is formed by the cleavage of an ethylenic double bond of styrene. The “constitutional unit derived from a styrene derivative” means a constitutional unit (provided that a constitutional unit corresponding to the constitutional unit (a10) is excluded) formed by the cleavage of an ethylenic double bond of a styrene derivative.

(436) The “styrene derivative” means a compound obtained by substituting at least part of hydrogen atoms of styrene with a substituent. Examples of the styrene derivative include a derivative obtained by substituting a hydrogen atom at the α -position of styrene with a substituent, a derivative obtained by substituting one or more hydrogen atoms of the benzene ring of styrene with a substituent, and a derivative obtained by substituting a hydrogen atom at the α -position of styrene and one or more hydrogen atoms of the benzene ring with a substituent.

(437) Examples of the substituent that is substituted for the hydrogen atom at the α -position of styrene include an alkyl group having 1 to 5 carbon atoms or a halogenated alkyl group having 1 to 5 carbon atoms.

(438) The alkyl group having 1 to 5 carbon atoms is preferably a linear or branched alkyl group having 1 to 5 carbon atoms, and specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, and a neopentyl group.

(439) The halogenated alkyl group having 1 to 5 carbon atoms is a group obtained by substituting part or all hydrogen atoms in the alkyl group having 1 to 5 carbon atoms with a halogen atom. The halogen atom is particularly preferably a fluorine atom.

(440) The substituent that is substituted for the hydrogen atom at the α -position of styrene is preferably an alkyl group having 1 to 5 carbon atoms or a fluorinated alkyl group having 1 to 5 carbon atoms, more preferably an alkyl group having 1 to 3 carbon atoms or a fluorinated alkyl group having 1 to 3 carbon atoms, and still more preferably a methyl group from the viewpoint of industrial availability.

(441) Examples of the substituent that is substituted for the hydrogen atom of the benzene ring of styrene include an alkyl group, an alkoxy group, a halogen atom, and a halogenated alkyl group.

(442) The alkyl group as the substituent is preferably an alkyl group having 1 to 5 carbon atoms, and more preferably a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group.

(443) The alkoxy group as the substituent is preferably an alkoxy group having 1 to 5 carbon atoms, more preferably a methoxy group, an ethoxy group, an n-propoxy group, an iso-propoxy group, an n-butoxy group, or a tert-butoxy group, and still more preferably a methoxy group or an ethoxy group.

(444) The halogen atom as the substituent is preferably a fluorine atom.

(445) Examples of the halogenated alkyl group as the substituent include groups obtained by substituting part or all hydrogen atoms in the above-described alkyl groups with the above-described halogen atoms.

(446) The substituent that is substituted for the hydrogen atom of the benzene ring of styrene is preferably an alkyl group having 1 to 5 carbon atoms, more preferably a methyl group or an ethyl group, and still more preferably a methyl group.

(447) The constitutional unit (st) is preferably a constitutional unit derived from styrene or a constitutional unit derived

from a styrene derivative obtained by substituting a hydrogen atom at the α -position of styrene with an alkyl group having 1 to 5 carbon atoms or a halogenated alkyl group having 1 to 5 carbon atoms, more preferably a constitutional unit derived from styrene, or a constitutional unit derived from a styrene derivative obtained by substituting a hydrogen atom at the α -position of styrene with a methyl group, and still more preferably a constitutional unit derived from styrene.

(448) The constitutional unit (st) contained in the component (A1) has may be one kind or may be two or more kinds.

(449) In a case where the component (A1) has the constitutional unit (st), the proportion of the constitutional unit (st) is preferably in a range of 1% to 30% by mole and more preferably in a range of 1% to 20% by mole with respect to the total (100% by mole) of all constitutional units constituting the component (A1).

(450) The component (A1) contained in the resist composition may be used alone or in a combination of two or more kinds thereof.

(451) In the resist composition according to the present embodiment, the component (A1) is a resin component that contains the constitutional unit (a01).

(452) Suitable examples of such a component (A1) include a polymeric compound consisting of only a repeating structure of the constitutional unit (a01); a polymeric compound having a repeating structure of the constitutional unit (a01) and the constitutional unit (a1); a polymeric compound having a repeating structure of the constitutional unit (a01) and the constitutional unit (a10); and a polymeric compound having a repeating structure of the constitutional unit (a01), the constitutional unit (a10), and the constitutional unit (a1). Among them, a polymeric compound having a repeating structure of the constitutional unit (a01) and the constitutional unit (a1) is preferable.

(453) In the polymeric compound having a repeating structure of the constitutional unit (a01) and the constitutional unit (a1), the proportion of the constitutional unit (a01) in the polymeric compound is preferably in a range of 20% to 80% by mole, more preferably in a range of 30% to 70% by mole, and still more preferably in a range of 40% to 60% by mole, with respect to the total (100% by mole) of all constitutional units constituting the polymeric compound.

(454) In addition, the proportion of the constitutional unit (a1) in the polymeric compound is preferably in a range of 20% to 80% by mole, more preferably in a range of 30% to 70% by mole, still more preferably in a range of 40% to 60% by mole, with respect to the total (100% by mole) of all constitutional units constituting the polymeric compound.

(455) In the polymeric compound having a repeating structure of the constitutional unit (a01) and the constitutional unit (a10), the proportion of the constitutional unit (a01) in the polymeric compound is preferably in a range of 20% to 80% by mole, more preferably in a range of 30% to 70% by mole, and still more preferably in a range of 40% to 60% by mole, with respect to the total (100% by mole) of all constitutional units constituting the polymeric compound.

(456) In addition, the proportion of the constitutional unit (a10) in the polymeric compound is preferably in a range of 20% to 80% by mole, more preferably in a range of 30% to 70% by mole, still more preferably in a range of 40% to 60% by mole, with respect to the total (100% by mole) of all constitutional units constituting the polymeric compound.

(457) In the polymeric compound having a repeating structure of the constitutional unit (a01), the constitutional unit (a1), and the constitutional unit (a10), the proportion of the constitutional unit (a01) and the constitutional unit (a1) in the polymeric compound is preferably in a range of 20% to 80% by mole, more preferably in a range of 30% to 70% by mole, and still more preferably in a range of 40% to 60% by mole, with respect to the total (100% by mole) of all constitutional units constituting the polymeric compound.

(458) In addition, the proportion of the constitutional unit (a10) in the polymeric compound is preferably in a range of 20% to 80% by mole, more preferably in a range of 30% to 70% by mole, still more preferably in a range of 40% to

60% by mole, with respect to the total (100% by mole) of all constitutional units constituting the polymeric compound.

(459) The component (A1) can be produced by dissolving, in a polymerization solvent, each monomer from which the constitutional unit is derived, adding thereto a radical polymerization initiator such as azobisisobutyronitrile (AIBN) or dimethyl azobisisobutyrate (for example, V-601) to carry out polymerization.

(460) Alternatively, the component (A1) can be produced by dissolving, in a polymerization solvent, a monomer from which the constitutional unit (a01) is derived and, as necessary, a monomer (for example, a compound in which a hydroxyl group of a monomer from which the constitutional unit (a10) is derived is protected) from which a constitutional unit other than the constitutional unit (a01) is derived, and adding thereto a radical polymerization initiator as described above to carry out polymerization and then carrying out a deprotection reaction.

(461) Further, a $\text{—C(CF.sub.3).sub.2—OH}$ group may be introduced into the terminal of the component (A1) during the polymerization using a chain transfer agent such as $\text{HS—CH.sub.2—CH.sub.2—CH.sub.2—C(CF.sub.3).sub.2—OH}$ in combination. As described above, a copolymer into which a hydroxyalkyl group, formed by substitution of part of hydrogen atoms in the alkyl group with fluorine atoms, has been introduced is effective for reducing development defects and reducing line edge roughness (LER: uneven irregularities of a line side wall).

(462) The weight average molecular weight (M_w) (based on the polystyrene-equivalent value determined by gel permeation chromatography (GPC)) of the component (A1), which is not particularly limited, is preferably in a range of 1,000 to 50,000, more preferably in a range of 2,000 to 30,000, and still more preferably in a range of 3,000 to 20,000.

(463) In a case where M_w of the component (A1) is equal to or smaller than the upper limit value of this preferred range, a resist solvent solubility sufficient to be used as a resist is exhibited. On the other hand, in a case where it is equal to or larger than the lower limit value of this preferred range, dry etching resistance and the cross-sectional shape of the resist pattern become excellent.

(464) Further, the dispersity (M_w/M_n) of the component (A1) is not particularly limited, but is preferably in a range of 1.0 to 4.0, more preferably in a range of 1.0 to 3.0, and particularly preferably in a range of 1.0 to 2.0. M_n represents the number average molecular weight.

(465) In Regard to Component (A2)

(466) In the resist composition according to the present embodiment, a base material component (hereinafter, referred to as a “component (A2)”) exhibiting changed solubility in a developing solution under action of acid, which does not correspond to the component (A1), may be used in combination as the component (A).

(467) The component (A2) is not particularly limited and may be freely selected and used from a large number of known ones in the related art as the base material component for the chemically amplified resist composition.

(468) As the component (A2), one kind of a polymeric compound or low molecular weight compound may be used alone, or a combination of two or more kinds thereof may be used.

(469) The proportion of the component (A1) in the component (A) is preferably 25% by mass or more, more preferably 50% by mass or more, still more preferably 75% by mass or more, and may be 100% by mass with respect to the total mass of the component (A). In a case where the proportion is 25% by mass or more, a resist pattern having various excellent lithography characteristics such as high sensitivity, resolution, and roughness amelioration can be easily formed.

(470) The content of the component (A) in the resist composition according to the present embodiment may be adjusted depending on the resist film thickness to be formed and the like.

(471) <Compound (D0)>

(472) The resist composition according to the present embodiment further contains a compound (D0) in addition to the component (A). The compound (D0) is a compound represented by General Formula (d0), and acts, for example, as a quencher (an acid diffusion controlling agent) that traps acid generated upon exposure in a resist composition.

(473) ##STR00073##

(474) [In the formula, M^{m+} represents an m-valent organic cation. m represents an integer of 1 or more. R^{d0} represents a substituent. p₀ represents an integer in a range of 0 to 3. In a case where p₀ represents 2 or 3, a plurality of R^{d0}'s may be the same or different from each other. q₀ represents an integer in a range of 0 to 3. n₀ represents an integer of 1 or more. However, $n_0 + p_0 \leq (q_0 \times 2) + 5$ is satisfied.]

(475) {Anion Moiety}

(476) In General Formula (d0), R^{d0} represents a substituent. Examples of the substituent include a hydrocarbon group, an alkoxy group, an acyl group, and a hydroxyalkyl group.

(477) Examples of the hydrocarbon group as the substituent include a linear or branched alkyl group, an aliphatic cyclic hydrocarbon group, and an aromatic hydrocarbon group.

(478) The linear or branched alkyl group is preferably a linear or branched alkyl group having 1 to 5 carbon atoms, and specific examples thereof include a methyl group, an ethyl group, a propyl group, an n-butyl group, a tert-butyl group, and a pentyl group.

(479) The aliphatic cyclic hydrocarbon group is preferably an aliphatic cyclic hydrocarbon group having 3 to 6 carbon atoms, and specific examples thereof include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, and a cyclohexyl group.

(480) The aromatic hydrocarbon group is preferably an aromatic hydrocarbon group having 6 to 30 carbon atoms, and specific examples thereof include a group obtained by removing one hydrogen atom from an aromatic hydrocarbon ring such as benzene, biphenyl, fluorene, naphthalene, anthracene, or phenanthrene. Among the above, a group (a phenyl group) obtained by removing one hydrogen atom from benzene is more preferable.

(481) The alkoxy group as the substituent is preferably an alkoxy group having 1 to 5 carbon atoms, and specific examples thereof include a methoxy group, an ethoxy group, a propoxy group, an n-butoxy group, a tert-butoxy group, and a pentyloxy group. Among them, a methoxy group is more preferable.

(482) The acyl group as the substituent is preferably an acyl group having 1 to 3 carbon atoms, and specific examples thereof include a formyl group, an acetyl group, and a propionyl group.

(483) The hydroxyalkyl group as the substituent is preferably a hydroxyalkyl group having 1 to 5 carbon atoms, and specific examples thereof include a hydroxymethyl group, a hydroxyethyl group, a hydroxypropyl group, a hydroxybutyl group, and a hydroxypentyl group.

(484) In General Formula (d0), p₀ represents an integer in a range of 0 to 2, and in a case where p₀ represents 2, the substituents in a plurality of R^{d0}'s may be the same or different from each other. Among the above, p₀ is preferably 0 or 1 and more preferably 0.

(485) In General Formula (d0), q₀ represents an integer in a range of 0 to 3. That is, in the anion moiety in General Formula (d0), a case where q₀ is 0 gives a benzene structure, a case where q₀ is 1 gives a naphthalene structure, a case where q₀ is 2 gives an anthracene structure, and a case where q₀ is 3 gives a tetracene structure.

(486) In addition, $n_0 + p_0 \leq (q_0 \times 2) + 5$ is satisfied. That is, in the benzene structure, the naphthalene

structure, the anthracene structure, and the tetracene structure, all hydrogen atoms other than the hydrogen atom substituted with a carboxylate group may be each substituted with the above-described substituent or a hydroxy group. However, the anion moiety in General Formula (d0) has at least one hydroxy group.

(487) That is, in the benzene structure, the naphthalene structure, the anthracene structure, and the tetracene structure, the substitution positions of the carboxylate group, the above substituent, and the hydroxy group are not particularly limited.

(488) In General Formula (d0), $q_{\text{sub}0}$ is preferably 0 or 1 and more preferably 0.

(489) In General Formula (d0), n_0 is preferably an integer in a range of 1 to 3, more preferably 1 or 2, and still more preferably 2.

(490) Preferred specific examples of the anion moiety of the compound (D0) are shown below.

(491) ##STR00074## ##STR00075## ##STR00076## ##STR00077## ##STR00078##

{Cation Moiety}

(492) In General Formula (d0), $M_{\text{sup}m+}$ represents an m -valent onium cation. Among the above, a sulfonium cation or an iodonium cation is preferable.

(493) m represents an integer of 1 or more.

(494) Preferred examples of the cation moiety $((M_{\text{sup}m+})_{\text{sub}1/m})$ include organic cations each represented by General Formulae (ca-1) to (ca-5).

(495) ##STR00079##

(496) [In the formula, $R_{\text{sup}201}$ to $R_{\text{sup}207}$, $R_{\text{sup}211}$, and $R_{\text{sup}212}$ each independently represent an aryl group, an alkyl group, or an alkenyl group, each of which may have a substituent. $R_{\text{sup}201}$ to $R_{\text{sup}203}$, $R_{\text{sup}206}$ and $R_{\text{sup}207}$, or $R_{\text{sup}211}$ and $R_{\text{sup}212}$ may be bonded to each other to form a ring together with the sulfur atoms in the formulae. $R_{\text{sup}208}$ and $R_{\text{sup}209}$ each independently represent a hydrogen atom or an alkyl group having 1 to 5 carbon atoms. $R_{\text{sup}210}$ represents an aryl group which may have a substituent, an alkyl group which may have a substituent, an alkenyl group which may have a substituent, or a $\text{—SO}_{\text{sub}2}\text{—}$ -containing cyclic group which may have a substituent. $L_{\text{sup}201}$ represents —C(=O)— or —C(=O)— . $Y_{\text{sup}201s}$ each independently represent an arylene group, an alkylene group, or an alkenylene group. x represents 1 or 2. $W_{\text{sup}201}$ represents an $(x+1)$ -valent linking group.]

(497) In General Formulae (ca-1) to (ca-5), examples of the aryl group as $R_{\text{sup}201}$ to $R_{\text{sup}207}$, $R_{\text{sup}211}$, and $R_{\text{sup}212}$ include an unsubstituted aryl group having 6 to 20 carbon atoms, and a phenyl group or a naphthyl group is preferable.

(498) The alkyl group as $R_{\text{sup}201}$ to $R_{\text{sup}207}$, $R_{\text{sup}211}$, and $R_{\text{sup}212}$ is preferably a chain-like or cyclic alkyl group having 1 to 30 carbon atoms.

(499) The alkenyl group as $R_{\text{sup}201}$ to $R_{\text{sup}207}$, $R_{\text{sup}211}$, and $R_{\text{sup}212}$ preferably has 2 to 10 carbon atoms.

(500) Examples of the substituent which may be contained in $R_{\text{sup}201}$ to $R_{\text{sup}207}$ and $R_{\text{sup}210}$ to $R_{\text{sup}212}$ include an alkyl group, a halogen atom, a halogenated alkyl group, a carbonyl group, a cyano group, an amino group, an aryl group, and groups each represented by General Formulae (ca-r-1) to (ca-r-7) shown below.

(501) ##STR00080##

(502) [In the formulae, each $R'_{\text{sup}201}$ independently represents a hydrogen atom, a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent.]

(503) Cyclic Group which May have Substituent:

(504) The cyclic group is preferably a cyclic hydrocarbon group, and the cyclic hydrocarbon group may be an aromatic hydrocarbon group or an aliphatic hydrocarbon group. The aliphatic hydrocarbon group indicates a hydrocarbon group that has no aromaticity. The aliphatic hydrocarbon group may be saturated or unsaturated. In general, it is preferable that the aliphatic hydrocarbon group is saturated.

(505) The aromatic hydrocarbon group as R'.sup.201 is a hydrocarbon group having an aromatic ring. The aromatic hydrocarbon group preferably has 3 to 30 carbon atoms, more preferably 5 to 30 carbon atoms, still more preferably 5 to 20 carbon atoms, particularly preferably 6 to 15 carbon atoms, and most preferably 6 to 10 carbon atoms. Here, the number of carbon atoms in a substituent is not included in the number of carbon atoms.

(506) Specific examples of the aromatic ring contained in the aromatic hydrocarbon group as R'.sup.201 include benzene, fluorene, naphthalene, anthracene, phenanthrene, biphenyl, and an aromatic heterocyclic ring obtained by substituting part of carbon atoms constituting one of these aromatic rings with a hetero atom. Examples of the hetero atom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom.

(507) Specific examples of the aromatic hydrocarbon group as R'.sup.201 include a group (an aryl group such as a phenyl group or a naphthyl group) obtained by removing one hydrogen atom from the above-described aromatic ring and a group (an arylalkyl group such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, 1-naphthylethyl group, or a 2-naphthylethyl group) obtained by substituting one hydrogen atom in the aromatic ring with an alkylene group. The alkylene group (an alkyl chain the arylalkyl group) preferably has 1 to 4 carbon atoms, more preferably 1 or 2 carbon atoms, and particularly preferably 1 carbon atom.

(508) Examples of the cyclic aliphatic hydrocarbon group as R'.sup.201 include aliphatic hydrocarbon groups containing a ring in the structure thereof.

(509) Examples of the aliphatic hydrocarbon group containing a ring in the structure thereof include an alicyclic hydrocarbon group (a group obtained by removing one hydrogen atom from an aliphatic hydrocarbon ring), a group obtained by bonding the alicyclic hydrocarbon group to the terminal of a linear or branched aliphatic hydrocarbon group, and a group obtained by interposing the alicyclic hydrocarbon group is in a linear or branched aliphatic hydrocarbon group.

(510) The alicyclic hydrocarbon group preferably has 3 to 20 carbon atoms and more preferably 3 to 12 carbon atoms.

(511) The alicyclic hydrocarbon group may be a polycyclic group or a monocyclic group. The monocyclic alicyclic hydrocarbon group is preferably a group obtained by removing one or more hydrogen atoms from a monocycloalkane. The monocycloalkane preferably has 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane. The polycyclic alicyclic hydrocarbon group is preferably a group obtained by removing one or more hydrogen atoms from a polycycloalkane, and the polycycloalkane preferably has 7 to 30 carbon atoms. Among the above, a polycycloalkane having a bridged ring-based polycyclic skeleton, such as adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane, or a polycycloalkane having a condensed ring-based polycyclic skeleton, such as a cyclic group having a steroid skeleton is preferable.

(512) Among them, the cyclic aliphatic hydrocarbon group as R'.sup.201 is preferably a group obtained by removing one or more hydrogen atoms from a monocycloalkane or a polycycloalkane, more preferably a group obtained by removing one hydrogen atom from a polycycloalkane, particularly preferably an adamantyl group or a norbornyl group, and most preferably an adamantyl group.

(513) The linear or branched aliphatic hydrocarbon group which may be bonded to the alicyclic hydrocarbon group

preferably has 1 to 10 carbon atoms, more preferably 1 to 6 carbon atoms, still more preferably 1 to 4 carbon atoms, and particularly preferably 1 to 3 carbon atoms.

(514) The linear aliphatic hydrocarbon group is preferably a linear alkylene group, and specific examples thereof include a methylene group [—CH₂—], an ethylene group [—(CH₂)₂—], a trimethylene group [—(CH₂)₃—], a tetramethylene group [—(CH₂)₄—], and a pentamethylene group [—(CH₂)₅—].

(515) The branched aliphatic hydrocarbon group is preferably a branched alkylene group, and specific examples thereof include alkylalkylene group groups, for example, alkylmethylene groups such as —CH(CH₃)—, —CH(CH₂CH₃)—, —C(CH₃)₂—, —C(CH₃)(CH₂CH₃)—, —C(CH₃)(CH₂CH₂CH₃)—, and —C(CH₂CH₃)₂—; alkylethylene groups such as —CH(CH₃)CH₂—, —CH(CH₃)CH(CH₃)—, —C(CH₃)₂CH₂—, —CH(CH₂CH₃)CH₂—, and —C(CH₂CH₃)₂CH₂—; alkyltrimethylene groups such as —CH(CH₃)CH₂CH₂—, and —CH₂CH(CH₃)CH₂—; and alkyltetramethylene groups such as —CH(CH₃)CH₂CH₂CH₂—, and —CH₂CH(CH₃)CH₂CH₂—. The alkyl group in the alkylalkylene group is preferably a linear alkyl group having 1 to 5 carbon atoms.

(516) The cyclic hydrocarbon group as R¹ may contain a hetero atom such as a heterocyclic ring. Specific examples thereof include lactone-containing cyclic groups each represented by General Formulae (a2-r-1) to (a2-r-7), —SO₂—containing cyclic groups each represented by General Formulae (a5-r-1) to (a5-r-4), and other heterocyclic groups each represented by Chemical Formulae (r-hr-1) to (r-hr-16).

(517) Examples of the substituent of the cyclic group as R¹ include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, a carbonyl group, and a nitro group.

(518) The alkyl group as the substituent is preferably an alkyl group having 1 to 5 carbon atoms, and a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group is most preferable.

(519) The alkoxy group as the substituent is preferably an alkoxy group having 1 to 5 carbon atoms, more preferably a methoxy group, an ethoxy group, an n-propoxy group, an iso-propoxy group, an n-butoxy group, or a tert-butoxy group, and most preferably a methoxy group or an ethoxy group.

(520) The halogen atom as the substituent is preferably a fluorine atom.

(521) Examples of the above-described halogenated alkyl group as the substituent include a group obtained by substituting part or all hydrogen atoms in an alkyl group having 1 to 5 carbon atoms such as a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group, with the above-described halogen atom.

(522) The carbonyl group as the substituent is a group that substitutes a methylene group (—CH₂—) constituting the cyclic hydrocarbon group.

(523) Chain-like alkyl group which may have substituent:

(524) The chain-like alkyl group as R¹ may be linear or branched.

(525) The linear alkyl group preferably has 1 to 20 carbon atoms, more preferably 1 to 15 carbon atoms, and most preferably 1 to 10 carbon atoms.

(526) The branched alkyl group preferably has 3 to 20 carbon atoms, more preferably 3 to 15 carbon atoms, and most preferably 3 to 10 carbon atoms. Specific examples thereof include a 1-methylethyl group, a 1-methylpropyl group, a 2-methylpropyl group, a 1-methylbutyl group, a 2-methylbutyl group, a 3-methylbutyl group, a 1-ethylbutyl group, a 2-ethylbutyl group, a 1-methylpentyl group, a 2-methylpentyl group, a 3-methylpentyl group, and a 4-methylpentyl

group.

(527) Chain-Like Alkenyl Group which May have Substituent:

(528) Such a chain-like alkenyl group as R'.sup.201 may be linear or branched, preferably has 2 to 10 carbon atoms, more preferably 2 to 5 carbon atoms, still more preferably 2 to 4 carbon atoms, and particularly preferably 3 carbon atoms. Examples of the linear alkenyl group include a vinyl group, a propenyl group (an allyl group), and a butynyl group. Examples of the branched alkenyl group include a 1-methylvinyl group, a 2-methylvinyl group, a 1-methylpropenyl group, and a 2-methylpropenyl group.

(529) Among the above, the chain-like alkenyl group is preferably a linear alkenyl group, more preferably a vinyl group or a propenyl group, and particularly preferably a vinyl group.

(530) Examples of the substituent in the chain-like alkyl group or alkenyl group as R'.sup.201, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, a carbonyl group, a nitro group, an amino group, a cyclic group as R'.sup.201 or the like may be used.

(531) As the cyclic group which may have a substituent, the chain-like alkyl group which may have a substituent, or the chain-like alkenyl group which may have a substituent, as R'.sup.201, a group that is the same as the acid dissociable group represented by above-described General Formula (a1-r-2) can be mentioned as the cyclic group which may have a substituent or the chain-like alkyl group which may have a substituent, in addition to the groups described above.

(532) Among them, R'.sup.201 is preferably a cyclic group which may have a substituent and more preferably a cyclic hydrocarbon group which may have a substituent. More specific examples thereof preferably include a phenyl group; a naphthyl group; a group obtained by removing one or more hydrogen atoms from a polycycloalkane; any one of lactone-containing cyclic groups each represented by General Formulae (a2-r-1) to (a2-r-7); and any one of —SO.sub.2—containing cyclic groups each represented by General Formulae (a5-r-1) to (a5-r-4).

(533) In General Formulae (ca-1) to (ca-5), in a case where R.sup.201 to R.sup.203, R.sup.206 and R.sup.207, or R.sup.211 and R.sup.212 are bonded to each other to form a ring with a sulfur atom in the formula, these groups may be bonded to each other via a hetero atom such as a sulfur atom, an oxygen atom or a nitrogen atom, or a functional group such as a carbonyl group, —SO—, —SO.sub.2—, —SO.sub.3—, —COO—, —CONH—, or —N(R.sub.N)— (here, R.sub.N represents an alkyl group having 1 to 5 carbon atoms). Regarding the ring to be formed, a ring containing a sulfur atom in a formula in the ring skeleton thereof is preferably a 3-membered to 10-membered ring and particularly preferably a 5-membered to 7-membered ring containing a sulfur atom. Specific examples of the ring to be formed include a thiophene ring, a thiazole ring, a benzothiophene ring, a benzothiophene ring, a thianthrene ring, a dibenzothiophene ring, a 9H-thioxanthene ring, a thioxanthone ring, a thianthrene ring, a phenoxathiin ring, a tetrahydrothiophenium ring, and a tetrahydrothiopyranium ring.

(534) R.sup.208 and R.sup.209 each independently represent a hydrogen atom or an alkyl group having 1 to 5 carbon atoms and are preferably a hydrogen atom or an alkyl group having 1 to 3 carbon atoms. In a case where R.sup.208 and R.sup.209 each independently represent an alkyl group, R.sup.208 and R.sup.209 may be bonded to each other to form a ring.

(535) R.sup.210 represents an aryl group which may have a substituent, an alkyl group which may have a substituent, an alkenyl group which may have a substituent, or a —SO.sub.2—containing cyclic group which may have a substituent.

(536) Examples of the aryl group as R.sup.210 include an unsubstituted aryl group having 6 to 20 carbon atoms, and a

phenyl group or a naphthyl group is preferable.

(537) The alkyl group as R^{sup.210}, a chain-like or cyclic alkyl group having 1 to 30 carbon atoms is preferable.

(538) The alkenyl group as R^{sup.210} preferably has 2 to 10 carbon atoms.

(539) The —SO₂—containing cyclic group which may have a substituent, as R^{sup.210}, is preferably a “—SO₂—containing polycyclic group”, and more preferably a group represented by General Formula (a5-r-1).

(540) Y^{sup.201}s each independently represent an arylene group, an alkylene group, or an alkenylene group.

(541) Examples of the arylene group as Y^{sup.201} include groups obtained by removing one hydrogen atom from an aryl group exemplified as the aromatic hydrocarbon group as Ya^{sup.x0} described above.

(542) Examples of the alkylene group and alkenylene group as Y^{sup.201} include groups obtained by removing one hydrogen atom from the chain-like alkyl group or the chain-like alkenyl group as R'^{sup.201} described above.

(543) In General Formula (ca-4), x represents 1 or 2.

(544) W^{sup.201} represents an (x+1)-valent linking group, that is, a divalent or trivalent linking group.

(545) The divalent linking group as W^{sup.201} is preferably a divalent hydrocarbon group which may have a substituent, and as examples thereof include the same divalent hydrocarbon group, which may have a substituent, as Ya^{sup.21} in General Formula (a2-1) described above. The divalent linking group as W^{sup.201} may be linear, branched, or cyclic and is preferably cyclic. Among these, a group obtained by combining two carbonyl groups at both terminals of an arylene group is preferable. Examples of the arylene group include a phenylene group and a naphthylene group, and a phenylene group is particularly preferable.

(546) Examples of the trivalent linking group as W^{sup.201} include a group obtained by removing one hydrogen atom from the above-described divalent linking group as W^{sup.201} and a group obtained by bonding the divalent linking group to another divalent linking group. The trivalent linking group as W^{sup.201} is preferably a group obtained by bonding two carbonyl groups to an arylene group.

(547) Among the above, the cation moiety of the compound (D0) is preferably a cation represented by General Formula (ca-d0-1).

(548) ##STR00081##

(549) [In the formula, R^{sup.d1} represents an aryl group having a fluorine atom or an aryl group having a fluorinated alkyl group. R^{sup.d2} and R^{sup.d3} each independently represent an aryl group which may have a substituent or are bonded to each other to form a ring together with a sulfur atom in the formula.]

(550) In General Formula (ca-d0-1), R^{sup.d1} represents an aryl group having a fluorine atom or an aryl group having a fluorinated alkyl group. Examples of the aryl group include the same ones as the aryl groups as R^{sup.201} to R^{sup.203} in General Formula (ca-1) described above. In addition, the fluorinated alkyl group is preferably a group obtained by substituting part or all hydrogen atoms of an alkyl group having 1 to 5 carbon atoms, for example, a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group with a fluorine atom, and a trifluoromethyl group is more preferable.

(551) R^{sup.d2} and R^{sup.d3} each independently represent an aryl group which may have a substituent or are bonded to each other to form a ring together with a sulfur atom in the formula. The aryl group as R^{sup.b1} to R^{sup.b3} is the same as the aryl group as R^{sup.201} to R^{sup.203} in General Formula (ca-1) described above.

(552) Examples of the ring formed by bonding R^{sup.b2} and R^{sup.b3} to each other together with a sulfur atom in the formula include the same one as the ring formed by bonding the aryl groups as R^{sup.201} to R^{sup.203} in General Formula (ca-1) describe above to each other together with a sulfur atom in the formula.

(553) Suitable cations represented by General Formula (ca-1) are shown below.

(554) ##STR00082## ##STR00083## ##STR00084## ##STR00085## ##STR00086## ##STR00087##
##STR00088## ##STR00089## ##STR00090## ##STR00091##

(555) [In the formulae, g₂ and g₃ represent the numbers of repetitions, g₂ is an integer in a range of 0 to 20, and g₃ is an integer in a range of 0 to 20.]

(556) ##STR00092## ##STR00093## ##STR00094## ##STR00095## ##STR00096## ##STR00097##
##STR00098## ##STR00099##

(557) [In the formula, R^{sup.201} represents a hydrogen atom or a substituent, and the substituent is the same as the substituent exemplified as the substituent which may be contained in R^{sup.201} to R^{sup.207} and R^{sup.210} to R^{sup.212}.]

(558) ##STR00100## ##STR00101## ##STR00102##

(559) Specific examples of the suitable cation represented by General Formula (ca-2) include a diphenyliodonium cation and a bis(4-tert-butylphenyl)iodonium cation.

(560) Specific examples of the suitable cation represented by General Formula (ca-3) include cations each represented by General Formulae (ca-3-1) to (ca-3-6).

(561) ##STR00103##

(562) Specific examples of the suitable cation represented by General Formula (ca-4) include cations each represented by General Formulae (ca-4-1) and (ca-4-2).

(563) ##STR00104##

(564) Specific examples of the suitable cation represented by General Formula (ca-5) include cations each represented by General Formulae (ca-5-1) to (ca-5-3).

(565) ##STR00105##

(566) In the resist composition according to the present embodiment, the cation moiety of the compound (D0) is, among the above, preferably a cation represented by General Formula (ca-1).

(567) Specifically, a cation represented by any one of Chemical Formulae (ca-1-1) to (ca-1-93) and (ca-1-100) to (ca-1-114) is preferable, a cation represented by any one of Chemical Formulae (ca-1-100) to (ca-1-114) is more preferable, a cation represented by any one of Chemical Formulae (ca-1-100) to (ca-1-112) is still more preferable, and a cation represented by any one of Chemical Formulae (ca-1-100) to (ca-1-103) is particularly preferable.

(568) In the resist composition according to the present embodiment, the compound (D0) is preferably a compound (hereinafter, also referred to as a “compound (D01)”) represented by General Formula (d0-1).

(569) ##STR00106##

(570) [In the formula, R^{sup.d1} represents an aryl group having a fluorine atom or an aryl group having a fluorinated alkyl group. R^{sup.d2} and R^{sup.d3} each independently represent an aryl group which may have a substituent or are bonded to each other to form a ring together with a sulfur atom in the formula. R^{sup.d0} represents a substituent. p_{sub.0} represents an integer in a range of 0 to 3. In a case where p_{sub.0} represents 2 or 3, a plurality of R^{sup.d0}'s may be the same or different from each other. q_{sub.0} represents an integer in a range of 0 to 3. n represents an integer of 1 or more. However, n_{sub.0}+p_{sub.0}≤(q_{sub.0}×2)+5 is satisfied.]

(571) The anion moiety of the compound (D01) is the same as the anion moiety of the compound (D0). The cation moiety of the compound (D01) is the same as the cation represented by General Formula (ca-d0-1) described above.

(572) Preferred specific examples of the compound (D0) are shown below.

(573) ##STR00107##

(574) In the resist composition according to the present embodiment, the compound (D0) may be used alone or in a combination of two or more kinds thereof.

(575) The content of the compound (D0) in the resist composition according to the present embodiment is preferably in a range of 1 to 50 parts by mass, more preferably in a range of 3 to 50 parts by mass, still more preferably in a range of 3 to 20 parts by mass, and particularly preferably in a range of 3 to 10 parts by mass, with respect to 100 parts by mass of the component (A).

(576) In a case where the content of the compound (D0) is set within the above-described preferred range, the solubility of the developing solution in exposed portions of the resist film is enhanced, and thus the sensitivity and the resolution are further improved.

(577) <Other Components>

(578) The resist composition according to the present embodiment may further contain other components in addition to the component (A) and the compound (D0) described above. Examples of the other components include a component (B), a compound (D) (provided that the compound (D0) is excluded), a component (E), a component (F), and a component (S), which are described below.

(579) <<Acid Generator Component (B)>>

(580) The resist composition according to the present embodiment may further contain an acid generator component (B) (hereinafter, referred to as a “component (B)”) that generates acid upon exposure, in addition to the component (A).

(581) The component (B) is not particularly limited, and those which have been proposed so far as an acid generator for a chemically amplified resist composition in the related art can be used.

(582) Examples of such an acid generator are numerous and include an onium salt-based acid generator such as an iodonium salt or a sulfonium salt; an oxime sulfonate-based acid generator; diazomethane-based acid generators such as a bisalkyl or bisaryl sulfonyl diazomethane and a poly(bis-sulfonyl)diazomethane; a nitrobenzylsulfonate-based acid generator; an iminosulfonate-based acid generator; and a disulfone-based acid generator.

(583) Examples of the onium salt-based acid generator include a compound represented by General Formula (b-1) (hereinafter, also referred to as a “component (b-1)”), a compound represented by General Formula (b-2) (hereinafter, also referred to as a “component (b-2)”), and a compound represented by General Formula (b-3) (hereinafter, also referred to as a “component (b-3)”).

(584) ##STR00108##

(585) [In the formulae, R^{sup.101} and R^{sup.104} to R^{sup.108} each independently represent a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent. R^{sup.104} and R^{sup.105} may be bonded to each other to form a ring structure. R^{sup.102} represents a fluorinated alkyl group having 1 to 5 carbon atoms or a fluorine atom. Y^{sup.101} represents a divalent linking group containing an oxygen atom or a single bond. V^{sup.101} to V^{sup.103} each independently represent a single bond, an alkylene group, or a fluorinated alkylene group. L^{sup.101} and L^{sup.102} each independently represent a single bond or an oxygen atom. L^{sup.103} to L^{sup.105} each independently represent a single bond, —CO—, or —SO₂—. m represents an integer of 1 or more, and M'^{sup.m+} represents an m-valent onium cation.]

(586) {Anion Moiety}

(587) Anion in Component (b-1)

(588) [In General Formula (b-1), R^{sup.101} represents a cyclic group which may have a substituent, a chain-like alkyl

group which may have a substituent, or a chain-like alkenyl group which may have a substituent.

(589) Cyclic Group which May have Substituent:

(590) The cyclic group is preferably a cyclic hydrocarbon group, and the cyclic hydrocarbon group may be an aromatic hydrocarbon group or an aliphatic hydrocarbon group. The aliphatic hydrocarbon group indicates a hydrocarbon group that has no aromaticity. The aliphatic hydrocarbon group may be saturated or unsaturated. In general, it is preferable that the aliphatic hydrocarbon group is saturated.

(591) The aromatic hydrocarbon group as R.sup.101 represents a hydrocarbon group having an aromatic ring. The aromatic hydrocarbon group preferably has 3 to 30 carbon atoms, more preferably 5 to 30, still more preferably 5 to 20, particularly preferably 6 to 15, and most preferably 6 to 10. Here, the number of carbon atoms in a substituent is not included in the number of carbon atoms.

(592) Specific examples of the aromatic ring contained in the aromatic hydrocarbon group as R.sup.101 include benzene, fluorene, naphthalene, anthracene, phenanthrene, biphenyl, and an aromatic heterocyclic ring obtained by substituting part of carbon atoms constituting one of these aromatic rings with a hetero atom. Examples of the hetero atom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom.

(593) Specific examples of the aromatic hydrocarbon group as R.sup.101 include a group (an aryl group such as a phenyl group or a naphthyl group) obtained by removing one hydrogen atom from the above-described aromatic ring and a group (an arylalkyl group such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, 1-naphthylethyl group, or a 2-naphthylethyl group) obtained by substituting one hydrogen atom in the aromatic ring with an alkylene group. The alkylene group (an alkyl chain the arylalkyl group) preferably has 1 to 4 carbon atoms, more preferably 1 or 2 carbon atoms, and particularly preferably 1 carbon atom.

(594) Examples of the cyclic aliphatic hydrocarbon group as R.sup.101 include aliphatic hydrocarbon groups containing a ring in the structure thereof.

(595) Examples of the aliphatic hydrocarbon group containing a ring in the structure thereof include an alicyclic hydrocarbon group (a group obtained by removing one hydrogen atom from an aliphatic hydrocarbon ring), a group obtained by bonding the alicyclic hydrocarbon group to the terminal of a linear or branched aliphatic hydrocarbon group, and a group obtained by interposing the alicyclic hydrocarbon group in a linear or branched aliphatic hydrocarbon group.

(596) The alicyclic hydrocarbon group preferably has 3 to 20 carbon atoms and more preferably 3 to 12 carbon atoms.

(597) The alicyclic hydrocarbon group may be a polycyclic group or a monocyclic group. The monocyclic alicyclic hydrocarbon group is preferably a group obtained by removing one or more hydrogen atoms from a monocycloalkane. The monocycloalkane preferably has 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane. The polycyclic alicyclic hydrocarbon group is preferably a group obtained by removing one or more hydrogen atoms from a polycycloalkane, and the polycycloalkane preferably has 7 to 30 carbon atoms. Among the above, a polycycloalkane having a bridged ring-based polycyclic skeleton, such as adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane, or a polycycloalkane having a condensed ring-based polycyclic skeleton, such as a cyclic group having a steroid skeleton is preferable.

(598) Among them, the cyclic aliphatic hydrocarbon group as R.sup.101 is preferably a group obtained by removing one or more hydrogen atoms from a monocycloalkane or a polycycloalkane, more preferably a group obtained by removing one hydrogen atom from a polycycloalkane, particularly preferably an adamantyl group or a norbornyl group, and most preferably an adamantyl group.

(599) The linear aliphatic hydrocarbon group which may be bonded to the alicyclic hydrocarbon group preferably has 1 to 10 carbon atoms, more preferably 1 to 6 carbon atoms, still more preferably 1 to 4 carbon atoms, and most preferably 1 to 3 carbon atoms. The linear aliphatic hydrocarbon group is preferably a linear alkylene group, and specific examples thereof include a methylene group [—CH₂—], an ethylene group [—(CH₂)₂—], a trimethylene group [—(CH₂)₃—], a tetramethylene group [—(CH₂)₄—], and a pentamethylene group [—(CH₂)₅—].

(600) The branched aliphatic hydrocarbon group which may be bonded to the alicyclic hydrocarbon group preferably has 2 to 10 carbon atoms, more preferably 3 to 6 carbon atoms, still more preferably 3 or 4 carbon atoms, and most preferably 3 carbon atoms. The branched aliphatic hydrocarbon group is preferably a branched alkylene group, and specific examples thereof include alkylalkylene group groups, for example, alkylmethylene groups such as —CH(CH₃)—, —CH(CH₂CH₃)—, —C(CH₃)₂—, —C(CH₃)(CH₂CH₃)—, —C(CH₃)(CH₂CH₂CH₃)—, and —C(CH₂CH₃)₂—; alkylethylene groups such as —CH(CH₃)CH₂—, —CH(CH₃)CH(CH₃)—, —C(CH₃)₂CH₂—, —CH(CH₂CH₃)CH₂—, and —C(CH₂CH₃)₂CH₂—; alkyltrimethylene groups such as —CH(CH₃)CH₂CH₂—, and —CH₂CH(CH₃)CH₂—; and alkyltetramethylene groups such as —CH(CH₃)CH₂CH₂CH₂—, and —CH₂CH(CH₃)CH₂CH₂—. The alkyl group in the alkylalkylene group is preferably a linear alkyl group having 1 to 5 carbon atoms.

(601) The cyclic hydrocarbon group as R^{sup.101} may contain a hetero atom such as a heterocyclic ring. Specific examples thereof include lactone-containing cyclic groups each represented by General Formulae (a2-r-1) to (a2-r-7), —SO₂—containing cyclic groups each represented by General Formulae (a5-r-1) to (a5-r-4), and other heterocyclic groups each represented by Chemical Formulae (r-hr-1) to (r-hr-16). In the formulae, * represents a bonding site that is bonded to Y^{sup.101} in General Formula (b-1).

(602) ##STR00109##

(603) Examples of the substituent of the cyclic group as R^{sup.101} include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, a carbonyl group, and a nitro group.

(604) The alkyl group as the substituent is preferably an alkyl group having 1 to 5 carbon atoms, and a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group is most preferable.

(605) The alkoxy group as the substituent is preferably an alkoxy group having 1 to 5 carbon atoms, more preferably a methoxy group, an ethoxy group, an n-propoxy group, an iso-propoxy group, an n-butoxy group, or a tert-butoxy group, and most preferably a methoxy group or an ethoxy group.

(606) The halogen atom as a substituent is preferably a fluorine atom.

(607) Examples of the above-described halogenated alkyl group as the substituent include a group obtained by substituting part or all hydrogen atoms in an alkyl group having 1 to 5 carbon atoms such as a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group, with the above-described halogen atom.

(608) The carbonyl group as the substituent is a group that substitutes a methylene group (—CH₂—) constituting the cyclic hydrocarbon group.

(609) The cyclic hydrocarbon group as R^{sup.101} may be a condensed cyclic group containing a condensed ring in which an aliphatic hydrocarbon ring and an aromatic ring are condensed. Examples of the condensed ring include a condensed ring in which one or more aromatic rings are condensed with a polycycloalkane having a bridged ring-based polycyclic skeleton. Specific examples of the bridged ring-based polycycloalkane include bicycloalkanes such as

bicyclo[2.2.1]heptane (norbornane) and bicyclo[2.2.2]octane. The condensed cyclic group is preferably a group containing a condensed ring, in which two or three aromatic rings are condensed with a bicycloalkane, and more preferably a group containing a condensed ring, in which two or three aromatic rings are condensed with bicyclo[2.2.2]octane. Specific examples of the condensed cyclic group as R.sup.101 include those represented by General Formulae (r-br-1) to (r-br-2). In the formulae, * represents a bonding site that is bonded to Y.sup.101 in General Formula (b-1).

(610) ##STR00110##

(611) Examples of the substituent which may be contained in the condensed cyclic group as R.sup.101 include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, a carbonyl group, a nitro group, an aromatic hydrocarbon group, and an alicyclic hydrocarbon group.

(612) Examples of the alkyl group, the alkoxy group, the halogen atom, and the halogenated alkyl group, as the substituent of the condensed cyclic group, include the same ones as those described as the substituent of the cyclic group as R.sup.101.

(613) Examples of the aromatic hydrocarbon group as the substituent of the condensed cyclic group include a group in which one hydrogen atom has been removed from the above-described aromatic ring (an aryl group; for example, a phenyl group or a naphthyl group), a group in which one hydrogen atom in the aromatic ring has been substituted with an alkylene group (for example, an arylalkyl group such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, 1-naphthylethyl group, or a 2-naphthylethyl group), and heterocyclic groups each represented by General Formulae (r-hr-1) to (r-hr-6).

(614) Examples of the alicyclic hydrocarbon group as the substituent of the condensed cyclic group include a group in which one hydrogen atom has been removed from a monocycloalkane such as cyclopentane or cyclohexane; a group in which one hydrogen atom has been removed from a polycycloalkane such as adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane; lactone-containing cyclic groups each represented by General Formulae (a2-r-1) to (a2-r-7); —SO.sub.2—containing cyclic groups each represented by General Formulae (a5-r-1) to (a5-r-4); and heterocyclic groups each represented by General Formulae (r-hr-7) to (r-hr-16).

(615) Chain-like alkyl group which may have substituent:

(616) The chain-like alkyl group as R.sup.101 may be linear or branched.

(617) The linear alkyl group preferably has 1 to 20 carbon atoms, more preferably 1 to 15 carbon atoms, and most preferably 1 to 10 carbon atoms.

(618) The branched alkyl group preferably has 3 to 20 carbon atoms, more preferably 3 to 15, and most preferably 3 to 10. Specific examples thereof include a 1-methylethyl group, a 1-methylpropyl group, a 2-methylpropyl group, a 1-methylbutyl group, a 2-methylbutyl group, a 3-methylbutyl group, a 1-ethylbutyl group, a 2-ethylbutyl group, a 1-methylpentyl group, a 2-methylpentyl group, a 3-methylpentyl group, and a 4-methylpentyl group.

(619) Chain-like alkenyl group which may have substituent:

(620) A chain-like alkenyl group as R.sup.101 may be linear or branched, and the chain-like alkenyl group preferably has 2 to 10 carbon atoms, more preferably 2 to 5 carbon atoms, still more preferably 2 to 4 carbon atoms, and particularly preferably 3 carbon atoms. Examples of the linear alkenyl group include a vinyl group, a propenyl group (an allyl group), and a butynyl group. Examples of the branched alkenyl group include a 1-methylvinyl group, a 2-methylvinyl group, a 1-methylpropenyl group, and a 2-methylpropenyl group.

(621) Among the above, the chain-like alkenyl group is preferably a linear alkenyl group, more preferably a vinyl

group or a propenyl group, and particularly preferably a vinyl group.

(622) Examples of the substituent in the chain-like alkyl group or alkenyl group as R.sup.101 include an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, a carbonyl group, a nitro group, an amino group, and a cyclic group as R.sup.101.

(623) Among the above, R.sup.101 is preferably a cyclic group which may have a substituent. In addition, from the viewpoint of reducing roughness, R.sup.101 is more preferably a polycyclic hydrocarbon group which may have a substituent, and still more preferably a bridged ring-based polycyclic hydrocarbon group which may have a substituent.

(624) In General Formula (b-1), Y.sup.101 represents a single bond or a divalent linking group containing an oxygen atom.

(625) In a case where Y.sup.101 represents a divalent linking group containing an oxygen atom, Y.sup.101 may contain an atom other than the oxygen atom. Examples of atoms other than the oxygen atom include a carbon atom, a hydrogen atom, a sulfur atom, and a nitrogen atom.

(626) Examples of divalent linking groups containing an oxygen atom include non-hydrocarbon-based oxygen atom-containing linking groups such as an oxygen atom (an ether bond; —O—), an ester bond (—C(=O)—O—), an oxycarbonyl group (—O—C(=O)—), an amide bond (—C(=O)—NH—), a carbonyl group (—C(=O)—), or a carbonate bond (—O—C(=O)—O—); and combinations of the above-described non-hydrocarbon-based oxygen atom-containing linking groups with an alkylene group. Furthermore, a sulfonyl group (—SO.sub.2—) may be linked to the combination. Examples of such a divalent linking group containing an oxygen atom include linking groups each represented by General Formulae (y-a1-1) to (y-a1-7) shown below.

(627) ##STR00111##

(628) [In the formulae, V'.sup.101 represents a single bond or an alkylene group having 1 to 5 carbon atoms, and V'.sup.102 represents a divalent saturated hydrocarbon group having 1 to 30 carbon atoms.]

(629) The divalent saturated hydrocarbon group as V'.sup.102 is preferably an alkylene group having 1 to 30 carbon atoms, more preferably an alkylene group having 1 to 10 carbon atoms, and still more preferably an alkylene group having 1 to 5 carbon atoms.

(630) The alkylene group as V'.sup.101 and V'.sup.102 may be a linear alkylene group or a branched alkylene group, and a linear alkylene group is preferable.

(631) Specific examples of the alkylene group as V'.sup.101 and V'.sup.102 include a methylene group [—CH.sub.2—]; an alkylmethylene group such as —CH(CH.sub.3)—, —CH(CH.sub.2CH.sub.3)—, —C(CH.sub.3).sub.2—, —C(CH.sub.3)(CH.sub.2CH.sub.3)—, —C(CH.sub.3)(CH.sub.2CH.sub.2CH.sub.3)—, or —C(CH.sub.2CH.sub.3).sub.2—; an ethylene group [—CH.sub.2CH.sub.2—]; an alkylethylene group such as —CH(CH.sub.3)CH.sub.2—, —CH(CH.sub.3)CH(CH.sub.3)—, —C(CH.sub.3).sub.2CH.sub.2—, or —CH(CH.sub.2CH.sub.3)CH.sub.2—; a trimethylene group (n-propylene group) [—CH.sub.2CH.sub.2CH.sub.2—]; an alkyltrimethylene group such as —CH(CH.sub.3)CH.sub.2CH.sub.2— or —CH.sub.2CH(CH.sub.3)CH.sub.2—; a tetramethylene group [—CH.sub.2CH.sub.2CH.sub.2CH.sub.2—]; an alkyltetramethylene group such as —CH(CH.sub.3)CH.sub.2CH.sub.2CH.sub.2—, or —CH.sub.2CH(CH.sub.3)CH.sub.2CH.sub.2—; and a pentamethylene group [—CH.sub.2CH.sub.2CH.sub.2CH.sub.2CH.sub.2—].

(632) Further, part of methylene groups in the alkylene group as V'.sup.101 and V'.sup.102 may be substituted with a divalent aliphatic cyclic group having 5 to 10 carbon atoms. The aliphatic cyclic group is preferably a divalent group in which one hydrogen atom has been removed from the cyclic aliphatic hydrocarbon group (a monocyclic aliphatic

hydrocarbon group or a polycyclic aliphatic hydrocarbon group) as $Ra'.sup.3$ in General Formula (a1-r-1), and a cyclohexylene group, a 1,5-adamantylene group, or a 2,6-adamantylene group is more preferable.

(633) $Y.sup.101$ is preferably a divalent linking group containing an ester bond or a divalent linking group containing an ether bond, and more preferably any one of linking groups each represented by General Formulae (y-a1-1) to (y-a1-5).

(634) In General Formula (b-1), $V.sup.101$ represents a single bond, an alkylene group, or a fluorinated alkylene group. The alkylene group and the fluorinated alkylene group as $V.sup.101$ preferably have 1 to 4 carbon atoms. Examples of the fluorinated alkylene group as $V.sup.101$ include a group obtained by substituting part or all hydrogen atoms in the alkylene group as $V.sup.101$ with a fluorine atom. Among them, $V.sup.101$ is preferably a single bond or a fluorinated alkylene group having 1 to 4 carbon atoms.

(635) In General Formula (b-1), $R.sup.102$ represents a fluorine atom or a fluorinated alkyl group having 1 to 5 carbon atoms. $R.sup.102$ is preferably a fluorine atom or a perfluoroalkyl group having 1 to 5 carbon atoms and more preferably a fluorine atom.

(636) In a case where $Y.sup.101$ represents a single bond, specific examples of the anion moiety represented by General Formula (b-1) include a fluorinated alkylsulfonate anion such as a trifluoromethanesulfonate anion or a perfluorobutanesulfonate anion; and in a case where $Y.sup.101$ represents a divalent linking group containing an oxygen atom, specific examples thereof include an anion represented by any one of General Formulae (an-1) to (an-3) shown below.

(637) ##STR00112##

(638) [In the formula, $R''.sup.101$ represents an aliphatic cyclic group which may have a substituent, monovalent heterocyclic groups each represented by Chemical Formulae (r-hr-1) to (r-hr-6), a condensed cyclic group represented by General Formula (r-br-1) or (r-br-2), and a chain-like alkyl group which may have a substituent. $R''.sup.102$ is an aliphatic cyclic group which may have a substituent, a condensed cyclic group represented by General Formula (r-br-1) or (r-br-2), lactone-containing cyclic groups each represented by General Formulae (a2-r-1), (a2-r-3) to (a2-r-7), or — $SO.sub.2$ —containing cyclic groups each represented by General Formulae (a5-r-1) to (a5-r-4). $R''.sup.103$ represents an aromatic cyclic group which may have a substituent, an aliphatic cyclic group which may have a substituent, or a chain-like alkenyl group which may have a substituent. $V''.sup.101$ represents a single bond, an alkylene group having 1 to 4 carbon atoms, or a fluorinated alkylene group having 1 to 4 carbon atoms. $R.sup.102$ represents a fluorine atom or a fluorinated alkyl group having 1 to 5 carbon atoms. Each v'' independently represents an integer in a range of 0 to 3, each q'' independently represents an integer in a range of 0 to 20, and n'' represents 0 or 1.]

(639) The aliphatic cyclic group as $R''.sup.101$ and $R''.sup.102$, $R''.sup.103$ which may have a substituent is preferably the group exemplified as the cyclic aliphatic hydrocarbon group as $R.sup.101$ in General Formula (b-1). Examples of the substituent include the same one as the substituent with which the cyclic aliphatic hydrocarbon group as $R.sup.101$ in General Formula (b-1) may be substituted.

(640) The aromatic cyclic group which may have a substituent, as $R''.sup.103$, is preferably the group exemplified as the aromatic hydrocarbon group for the cyclic hydrocarbon group as $R.sup.101$ in General Formula (b-1). Examples of the substituent include the same one as the substituent with which the aromatic hydrocarbon group as $R.sup.101$ in General Formula (b-1) may be substituted.

(641) The chain-like alkyl group as $R''.sup.101$, which may have a substituent, is preferably the group exemplified as the chain-like alkyl group as $R.sup.101$ in General Formula (b-1).

(642) The chain-like alkenyl group as R^{sup.103}, which may have a substituent, is preferably the group exemplified as the chain-like alkenyl group as R^{sup.101} in General Formula (b-1).

(643) Anion in Component (b-2)

(644) in General Formula (b-2), R^{sup.104} and R^{sup.105} each independently represent a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, and examples thereof each include the same one as R^{sup.101} in General Formula (b-1). However, R^{sup.104} and R^{sup.105} may be bonded to each other to form a ring.

(645) R^{sup.104} and R^{sup.105} are preferably a chain-like alkyl group which may have a substituent and more preferably a linear or branched alkyl group or a linear or branched fluorinated alkyl group.

(646) The chain-like alkyl group preferably has 1 to 10 carbon atoms, more preferably 1 to 7 carbon atoms, and still more preferably 1 to 3 carbon atoms. It is preferable that the number of carbon atoms in the chain-like alkyl group as R^{sup.104} and R^{sup.105} is small since the solubility in a resist solvent is also excellent in this range of the number of carbon atoms. Further, in the chain-like alkyl group as R^{sup.104} and R^{sup.105}, it is preferable that the number of hydrogen atoms substituted with a fluorine atom is large since the acid strength increases and the transparency to high energy radiation of 250 nm or less or an electron beam is improved. The proportion of fluorine atoms in the chain-like alkyl group, that is, the fluorination ratio is preferably in a range of 70% to 100% and more preferably in a range of 90% to 100%, and it is most preferable that the chain-like alkyl group is a perfluoroalkyl group in which all hydrogen atoms is substituted with a fluorine atom.

(647) In General Formula (b-2), V^{sup.102} and V^{sup.103} each independently represent a single bond, an alkylene group, or a fluorinated alkylene group, and examples thereof include the same one as V^{sup.101} in General Formula (b-1).

(648) in General Formula (b-2), L^{sup.101} and L^{sup.102} each independently represent a single bond or an oxygen atom.

(649) Anion in Component (b-3)

(650) in General Formula (b-3), R^{sup.106} to R^{sup.108} each independently represent a cyclic group which may have a substituent, chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, and examples thereof include the same one as R^{sup.101} in General Formula (b-1).

(651) In General Formula (b-3), L^{sup.103} to L^{sup.105} each independently represent a single bond, —CO—, or —SO_{sub.2}—.

(652) Among the above, the anion moiety of the component (B) is preferably an anion of the component (b-1). Among these, an anion represented by any one of General Formulae (an-1) to (an-3) is more preferable, an anion represented by any one of General Formula (an-1) or (an-2) is still more preferable, and an anion represented by General Formula (an-2) is particularly preferable.

(653) {Cation Moiety}

(654) In Formulae (b-1), (b-2), and (b-3), M'^{sup.m+} represents an m-valent onium cation. Among the above, a sulfonium cation or an iodonium cation is preferable. m represents an integer of 1 or more.

(655) Examples of the preferred cation moiety ((M'^{sup.m+}).sub.1/m) include the organic cations each represented by General Formulae (ca-1) to (ca-5) described above, and a cation represented by General Formula (ca-1) is preferable.

(656) In the resist composition according to the present embodiment, the component (B) may be used alone or in a combination of two or more kinds thereof.

(657) In a case where the resist composition contains the component (B), the content of the component (B) in the resist composition is preferably less than 50 parts by mass, more preferably in a range of 5 to 40 parts by mass, and still more preferably in a range of 10 to 30 parts by mass, with respect to 100 parts by mass of the component (A1).

(658) In a case where the content of the component (B) is set to be in the preferred range described above, pattern formation can be satisfactorily carried out. Further, in a case where each component of the resist composition is dissolved in an organic solvent, the above range is preferable since a homogeneous solution is easily obtained and the storage stability of the resist composition is improved.

(659) <<Base Component (D)>>

(660) The resist composition according to the present embodiment may further contain a base component (a component (D)) other than the compound (D0), which traps (that is, controls the acid diffusion) the acid generated from the component (B) upon exposure. The component (D) acts as a quencher (an acid diffusion controlling agent) which traps the acid generated in the resist composition upon exposure.

(661) Examples of the component (D) include a photodecomposable base (D1) having an acid diffusion controllability (hereinafter, referred to as a “component (D1)”) which is lost by the decomposition upon exposure and a nitrogen-containing organic compound (D2) (hereinafter, referred to as a “component (D2)”) which does not correspond to the component (D1). Among these, the photodecomposable base (the component (D1)) is preferable since it is easy to enhance the characteristics of high sensitivity, roughness reduction, and suppression of the occurrence of coating defects.

(662) In Regard to Component (D1)

(663) In a case where a resist composition containing the component (D1) is obtained, the contrast between the exposed portions and unexposed portions of the resist film can be further improved at the time of forming a resist pattern.

(664) The component (D1) is not particularly limited as long as it is decomposed upon exposure and loses the acid diffusion controllability. The component (D1) is preferably one or more compounds selected from the group consisting of a compound represented by General Formula (d1-1) (hereinafter, referred to as a “component (d1-1)”), a compound represented by General Formula (d1-2) (hereinafter, referred to as a “component (d1-2)”), and a compound represented by General Formula (d1-3) (hereinafter, referred to as a “component (d1-3)”).

(665) In the exposed portions of the resist film, the components (d1-1) to (d1-3) are decomposed and then lose the acid diffusion controllability (basicity), and thus they cannot act as a quencher, while acting as a quencher in the unexposed portions of the resist film.

(666) ##STR00113##

(667) [In the formulae, Rd.sup.1 to Rd.sup.4 represent cyclic groups which may have a substituent, chain-like alkyl groups which may have a substituent, or chain-like alkenyl groups which may have a substituent. Here, the carbon atom adjacent to the S atom as Rd.sup.2 in General Formula (d1-2) has no fluorine atom bonded thereto. Yd.sup.1 represents a single bond or a divalent linking group. m represents an integer of 1 or more, and each M.sup.m+ independently represents an m-valent organic cation].

(668) {Component (d1-1)}

(669) Anion Moiety

(670) In General Formula (d1-1), Rd.sup.1 represents a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, and examples thereof

each include the same one as R'.sup.201.

(671) Among these, Rd.sup.1 is preferably an aromatic hydrocarbon group which may have a substituent, an aliphatic cyclic group which may have a substituent, or a chain-like alkyl group which may have a substituent. Examples of the substituent which may be contained in these groups include a hydroxyl group, an oxo group, an alkyl group, an aryl group, a fluorine atom, a fluorinated alkyl group, lactone-containing cyclic groups each represented by any one of General Formulae (a2-r-1) to (a2-r-7), an ether bond, an ester bond, and a combination thereof. In a case where an ether bond or an ester bond is included as the substituent, the substituent may be bonded via an alkylene group, and the substituent in this case is preferably linking groups each represented by General Formulae (y-a1-1) to (y-a1-5).

(672) Suitable examples of the aromatic hydrocarbon group include a phenyl group, a naphthyl group, and a polycyclic structure (a polycyclic structure including a bicyclooctane skeleton and a ring structure other than the bicyclooctane skeleton) including a bicyclooctane skeleton.

(673) The aliphatic cyclic group is preferably a group obtained by removing one or more hydrogen atoms from a polycycloalkane such as adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane.

(674) The chain-like alkyl group preferably has 1 to 10 carbon atoms, and specific examples thereof include a linear alkyl group such as a methyl group, an ethyl group, a propyl group, a butyl group, a pentyl group, a hexyl group, a heptyl group, an octyl group, a nonyl group, or a decyl group, and a branched alkyl group such as a 1-methylethyl group, a 1-methylpropyl group, a 2-methylpropyl group, a 1-methylbutyl group, a 2-methylbutyl group, a 3-methylbutyl group, a 1-ethylbutyl group, a 2-ethylbutyl group, a 1-methylpentyl group, a 2-methylpentyl group, a 3-methylpentyl group, or a 4-methylpentyl group.

(675) In a case where the chain-like alkyl group is a fluorinated alkyl group having a fluorine atom or a fluorinated alkyl group as a substituent, the fluorinated alkyl group preferably has 1 to 11 carbon atoms, more preferably 1 to 8 carbon atoms, and still more preferably 1 to 4 carbon atoms. The fluorinated alkyl group may contain an atom other than the fluorine atom. Examples of the atom other than the fluorine atom include an oxygen atom, a sulfur atom, and a nitrogen atom.

(676) Preferred specific examples of the anion moiety of the component (d1-1) are shown below.

(677) ##STR00114##

Cation Moiety

(678) In General Formula (d1-1), M.sup.m+ represents an m-valent organic cation.

(679) The suitable examples of the organic cation as M.sup.m+ include the same ones as the cations each represented by General Formulae (ca-1) to (ca-5), and the cation represented by General Formula (ca-1) is more preferable.

(680) The component (d1-1) may be used alone or in a combination of two or more kinds thereof.

(681) {Component (d1-2)}

(682) Anion Moiety

(683) In General Formula (d1-2), Rd.sup.2 represents a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, and examples thereof include the same one as R'.sup.201.

(684) Here, the carbon atom adjacent to the S atom in Rd.sup.2 has no fluorine atom bonded thereto (the carbon atom adjacent to the S atom in Rd.sup.2 is not substituted with a fluorine atom). As a result, the anion of the component (d1-2) becomes an appropriately weak acid anion, thereby improving the quenching ability of the component (D).

(685) Rd.sup.2 is preferably a chain-like alkyl group which may have a substituent or an aliphatic cyclic group which

may have a substituent. The chain-like alkyl group preferably has 1 to 10 carbon atoms and more preferably 3 to 10 carbon atoms. The aliphatic cyclic group is more preferably a group (which may have a substituent) in which one or more hydrogen atoms have been removed from adamantane, norbornane, isobornane, tricyclodecane, tetracyclododecane, or the like; and a group in which one or more hydrogen atoms have been removed from camphor or the like.

(686) The hydrocarbon group as Rd.sup.2 may have a substituent. Examples of the substituent include the same one as the substituent which may be contained in the hydrocarbon group (the aromatic hydrocarbon group, the aliphatic cyclic group, or the chain-like alkyl group) as Rd.sup.1 in General Formula (d1-1).

(687) Preferred specific examples of the anion moiety of the component (d1-2) are shown below.

(688) ##STR00115##

Cation Moiety

(689) In General Formula (d1-2), M.sup.m+ represents an m-valent organic cation and is the same as M.sup.m+ in General Formula (d1-1).

(690) The component (d1-2) may be used alone or in a combination of two or more kinds thereof.

(691) {Component (d1-3)}

(692) Anion Moiety

(693) In General Formula (d1-3), Rd.sup.3 represents a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, examples thereof include the same one as R'.sup.201, and a cyclic group containing a fluorine atom, a chain-like alkyl group, or a chain-like alkenyl group is preferable. Among the above, a fluorinated alkyl group is preferable, and the same one as the fluorinated alkyl group as Rd.sup.1 described above is more preferable.

(694) In General Formula (d1-3), Rd.sup.4 represents a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, and examples thereof include the same one as R'.sup.201.

(695) Among them, an alkyl group which may have a substituent, an alkoxy group which may have a substituent, an alkenyl group which may have a substituent, or a cyclic group which may have a substituent is preferable.

(696) The alkyl group as Rd.sup.4 is preferably a linear or branched alkyl group having 1 to 5 carbon atoms, and specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, and a neopentyl group. Part of hydrogen atoms in the alkyl group as Rd.sup.4 may be substituted with a hydroxyl group, a cyano group, or the like.

(697) The alkoxy group as Rd.sup.4 is preferably an alkoxy group having 1 to 5 carbon atoms, and specific examples of the alkoxy group having 1 to 5 carbon atoms include a methoxy group, an ethoxy group, an n-propoxy group, an isopropoxy group, an n-butoxy group, and a tert-butoxy group. Among them, a methoxy group and an ethoxy group are preferable.

(698) Examples of the alkenyl group as Rd.sup.4 include the same one as the alkenyl group as R'.sup.201, and a vinyl group, a propenyl group (an allyl group), a 1-methylpropenyl group, or a 2-methylpropenyl group is preferable. These groups may have an alkyl group having 1 to 5 carbon atoms or a halogenated alkyl group having 1 to 5 carbon atoms as a substituent.

(699) Examples of the cyclic group as Rd.sup.4 include the same one as the cyclic group described above as R'.sup.201, and the cyclic group is preferably an alicyclic group obtained by removing one or more hydrogen atoms

from a cycloalkane such as cyclopentane, cyclohexane, adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane, or an aromatic group such as a phenyl group or a naphthyl group. In a case where Rd.sup.4 represents an alicyclic group, the resist composition can be satisfactorily dissolved in an organic solvent, thereby improving lithography characteristics. In a case where Rd.sup.4 is an aromatic group, the resist composition is excellent in light absorption efficiency and thus has good sensitivity and lithography characteristics in the lithography using EUV or the like as a light source for exposure.

(700) In General Formula (d1-3), Yd.sup.1 represents a single bond or a divalent linking group.

(701) The divalent linking group as Yd.sup.1 is not particularly limited, and examples thereof include a divalent hydrocarbon group (an aliphatic hydrocarbon group or an aromatic hydrocarbon group) which may have a substituent and a divalent linking group containing a hetero atom. Examples of these divalent linking groups each include the same one as the divalent hydrocarbon group which may have a substituent and the divalent linking group containing a hetero atom, which are mentioned in the explanation of the divalent linking group as Ya.sup.21 in General Formula (a2-1).

(702) Yd.sup.1 is preferably a carbonyl group, an ester bond, an amide bond, an alkylene group, or a combination of these. The alkylene group is more preferably a linear or branched alkylene group and still more preferably a methylene group or an ethylene group.

(703) Preferred specific examples of the anion moiety for the component (d1-3) are shown below.

(704) ##STR00116## ##STR00117##

Cation Moiety

(705) In General Formula (d1-3), M.sup.m+ represents an m-valent organic cation and is the same as M.sup.m+ in General Formula (d1-1).

(706) One kind of the component (d1-3) may be used alone, or a combination of two or more kinds thereof may be used.

(707) As the component (D1), any one of the above components (d1-1) to (d1-3) may be used alone, or a combination of two or more thereof may be used.

(708) In a case where the resist composition contains the component (D1), the content of the component (D1) in the resist composition is preferably in a range of 0.5 to 20 parts by mass and more preferably in a range of 1 to 15 parts by mass with respect to 100 parts by mass of the component (A1).

(709) In a case where the content of the component (D1) is equal to or larger than the preferred lower limit value, particularly excellent lithography characteristics and a particularly excellent resist pattern shape are easily obtained. On the other hand, in a case where the content of the component (D1) is equal to or smaller than the upper limit value, the sensitivity can be maintained satisfactorily and the throughput is also excellent.

(710) Method of Producing Component (D1):

(711) The methods of producing the components (d1-1) and (d1-2) are not particularly limited, and the components (d1-1) and (d1-2) can be produced by conventionally known methods.

(712) Further, the method of producing the component (d1-3) is not particularly limited, and the component (d1-3) can be produced in the same manner as disclosed in United States Patent Application, Publication No. 2012-0149916.

(713) In Regard to Component (D2)

(714) The component (D) may contain a nitrogen-containing organic compound component (hereinafter, referred to as a "component (D2)") which does not correspond to the above-mentioned component (D1).

(715) The component (D2) is not particularly limited as long as it acts as an acid diffusion controlling agent and does

not correspond to the component (D1), and any conventionally known component may be used. Among the above, aliphatic amines are preferable, and among the aliphatic amines, a secondary aliphatic amine or a tertiary aliphatic amine is more preferable.

(716) An aliphatic amine is an amine having one or more aliphatic groups, and the aliphatic groups preferably have 1 to 12 carbon atoms.

(717) Examples of the aliphatic amine include an amine obtained by substituting at least one hydrogen atom of ammonia (NH₃) with an alkyl group or hydroxyalkyl group having 12 or fewer carbon atoms (an alkylamine or an alkyl alcohol amine) and a cyclic amine.

(718) Specific examples of alkylamines and alkyl alcohol amines include monoalkylamines such as n-hexylamine, n-heptylamine, n-octylamine, n-nonylamine, and n-decylamine; dialkylamines such as diethylamine, di-n-propylamine, di-n-heptylamine, di-n-octylamine, and dicyclohexylamine; trialkylamines such as trimethylamine, triethylamine, tri-n-propylamine, tri-n-butylamine, tri-n-hexylamine, tri-n-pentylamine, tri-n-heptylamine, tri-n-octylamine, tri-n-nonylamine, tri-n-decylamine, and tri-n-dodecylamine; and alkyl alcohol amines such as diethanol amine, triethanol amine, diisopropanol amine, triisopropanol amine, di-n-octanol amine, and tri-n-octanol amine. Among these, a trialkylamine having 5 to 10 carbon atoms is preferable, and tri-n-pentylamine or tri-n-octylamine is particularly preferable.

(719) Examples of the cyclic amine include heterocyclic compounds containing a nitrogen atom as a hetero atom. The heterocyclic compound may be a monocyclic compound (aliphatic monocyclic amine), or a polycyclic compound (aliphatic polycyclic amine).

(720) Specific examples of the aliphatic monocyclic amine include piperidine and piperazine.

(721) The aliphatic polycyclic amine preferably has 6 to 10 carbon atoms, and specific examples thereof include 1,5-diazabicyclo[4.3.0]-5-nonene, 1,8-diazabicyclo[5.4.0]-7-undecene, hexamethylenetetramine, and 1,4-diazabicyclo[2.2.2]octane.

(722) Examples of other aliphatic amines include tris(2-methoxymethoxyethyl)amine, tris { 2-(2-methoxyethoxy)ethyl}amine, tris { 2-(2-methoxyethoxymethoxy)ethyl}amine, tris { 2-(1-methoxyethoxy)ethyl}amine, tris { 2-(1-ethoxyethoxy)ethyl}amine, tris { 2-(1-ethoxypropoxy)ethyl}amine, tris[2-{ 2-(2-hydroxyethoxy)ethoxy}ethyl]amine and triethanolamine triacetate, and triethanolamine triacetate is preferable.

(723) In addition, as the component (D2), an aromatic amine may be used.

(724) Examples of aromatic amines include 4-dimethylaminopyridine, pyrrole, indole, pyrazole, imidazole, and derivatives thereof, tribenzylamine, 2,6-diisopropylaniline, and N-tert-butoxycarbonylpyrrolidine.

(725) The component (D2) may be used alone or in a combination of two or more kinds thereof.

(726) In a case where the resist composition contains the component (D2), the content of the component (D2) in the resist composition is typically in a range of 0.01 to 5 parts by mass with respect to 100 parts by mass of the component (A1). By setting the content within the above range, the resist pattern shape, the post-exposure temporal stability, and the like are improved.

(727) <<at Least One Compound (E) Selected from Group Consisting of Organic Carboxylic Acid, Phosphorus Oxo Acid, and Derivatives Thereof>>

(728) For the purpose of preventing any deterioration in sensitivity and improving the resist pattern shape and the post-exposure temporal stability, the resist composition according to the present embodiment may contain at least one compound (E) (hereinafter referred to as a component (E)) selected from the group consisting of an organic carboxylic

acid, and a phosphorus oxo acid and a derivative thereof, as an optional component.

(729) Examples of suitable organic carboxylic acids include acetic acid, malonic acid, citric acid, malic acid, succinic acid, benzoic acid, and salicylic acid.

(730) Examples of the phosphorus oxo acid include phosphoric acid, phosphonic acid, and phosphinic acid. Among these, phosphonic acid is particularly preferable.

(731) Examples of phosphorus oxo acid derivative include an ester obtained by substituting a hydrogen atom in the above-described oxo acid with a hydrocarbon group. Examples of the hydrocarbon group include an alkyl group having 1 to 5 carbon atoms and an aryl group having 6 to 15 carbon atoms.

(732) Examples of phosphoric acid derivatives include phosphoric acid esters such as di-n-butyl phosphate and diphenyl phosphate.

(733) Examples of phosphonic acid derivatives include phosphonic acid esters such as dimethyl phosphonate, di-n-butyl phosphonate, phenylphosphonic acid, diphenyl phosphonate, and dibenzyl phosphonate.

(734) Examples of phosphinic acid derivatives include phosphinic acid esters and phenylphosphinic acid.

(735) In the resist composition according to the present embodiment, the component (E) may be used alone or in a combination of two or more kinds thereof.

(736) In a case where the resist composition contains the component (E), the content of the component (E) is typically in a range of 0.01 to 5 parts by mass with respect to 100 parts by mass of the component (A1).

(737) <<Fluorine Additive Component (F)>>

(738) The resist composition according to the present embodiment may further include a fluorine additive component (hereinafter, referred to as a “component (F)”) in order to impart water repellency to the resist film or to improve lithography characteristics.

(739) As the component (F), a fluorine-containing polymeric compound described in Japanese Unexamined Patent Application, First Publication No. 2010-002870, Japanese Unexamined Patent Application, First Publication No. 2010-032994, Japanese Unexamined Patent Application, First Publication No. 2010-277043, Japanese Unexamined Patent Application, First Publication No. 2011-13569, and Japanese Unexamined Patent Application, First Publication No. 2011-128226 can be mentioned.

(740) Specific examples of the component (F) include polymers having a constitutional unit (f1) represented by General Formula (f1-1) shown below. This polymer is preferably a polymer (homopolymer) consisting of only a constitutional unit (f1) represented by General Formula (f1-1) shown below; a copolymer of the constitutional unit (f1) and the constitutional unit (a1); and a copolymer of the constitutional unit (f1), a constitutional unit derived from acrylic acid or methacrylic acid, and the above-described constitutional unit (a1). The constitutional unit (a1) to be copolymerized with the constitutional unit (f1) is preferably a constitutional unit derived from 1-ethyl-1-cyclooctyl(meth)acrylate or a constitutional unit derived from 1-methyl-1-adamantyl (meth)acrylate.

(741) ##STR00118##

(742) [In the formula, R has the same definition as described above. Rf.sup.102 and Rf.sup.103 each independently represent a hydrogen atom, a halogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, and Rf.sup.102 and Rf.sup.103 may be the same or different from each other. nf.sup.1 represents an integer in a range of 0 to 5 and Rf.sup.101 represents an organic group containing a fluorine atom.]

(743) In General Formula (f1-1), R bonded to the carbon atom at the α -position has the same definition as described above. R is preferably a hydrogen atom or a methyl group.

(744) In General Formula (f1-1), the halogen atom as Rf.sup.102 and Rf.sup.103 is preferably a fluorine atom.

Examples of the alkyl group having 1 to 5 carbon atoms as Rf.sup.102 and Rf.sup.103 include the same one as the alkyl group having 1 to 5 carbon atoms as R, and a methyl group or an ethyl group is preferable. Specific examples of the halogenated alkyl group having 1 to 5 carbon atoms as Rf.sup.102 and Rf.sup.103 include a group obtained by substituting part or all hydrogen atoms of an alkyl group having 1 to 5 carbon atoms with a halogen atom. The halogen atom is preferably a fluorine atom. Among the above, Rf.sup.102 and Rf.sup.103 is preferably a hydrogen atom, a fluorine atom, or an alkyl group having 1 to 5 carbon atoms and more preferably a hydrogen atom, a fluorine atom, a methyl group, or an ethyl group.

(745) In General Formula (f1-1), nf.sup.1 represents an integer in a range of 0 to 5, preferably an integer in a range of 0 to 3, and more preferably an integer of 1 or 2.

(746) In General Formula (f1-1), Rf.sup.101 represents an organic group containing a fluorine atom and is preferably a hydrocarbon group containing a fluorine atom.

(747) The hydrocarbon group containing a fluorine atom may be linear, branched, or cyclic, and preferably has 1 to 20 carbon atoms, more preferably 1 to 15 carbon atoms, and particularly preferably 1 to 10 carbon atoms.

(748) In addition, in the hydrocarbon group containing a fluorine atom, 25% or more of the hydrogen atoms in the hydrocarbon group are preferably fluorinated, more preferably 50% or more are fluorinated, and particularly preferably 60% or more are fluorinated since the hydrophobicity of the resist film during immersion exposure increases.

(749) Among them, Rf.sup.101 is preferably a fluorinated hydrocarbon group having 1 to 6 carbon atoms and particularly preferably a trifluoromethyl group, —CH.sub.2—CF.sub.3, —CH.sub.2—CF.sub.2—CF.sub.3, —CH(CF.sub.3).sub.2, —CH.sub.2—CH.sub.2—CF.sub.3, or —CH.sub.2—CH.sub.2—CF.sub.2—CF.sub.2—CF.sub.3.

(750) The weight average molecular weight (Mw) (based on the polystyrene-equivalent value determined by gel permeation chromatography) of the component (F) is preferably in a range of 1,000 to 50,000, more preferably in a range of 5,000 to 40,000, and most preferably in a range of 10,000 to 30,000. In a case where the weight average molecular weight is equal to or smaller than the upper limit value of this range, a resist solvent solubility sufficient to be used as a resist is exhibited. On the other hand, in a case where it is equal to or larger than the lower limit value of this range, the water repellency of the resist film is excellent.

(751) Further, the dispersity (Mw/Mn) of the component (F) is preferably in a range of 1.0 to 5.0, more preferably in a range of 1.0 to 3.0, and most preferably in a range of 1.0 to 2.5.

(752) In the resist composition according to the present embodiment, the component (F) may be used alone or in a combination of two or more kinds thereof.

(753) In a case where the resist composition contains the component (F), the content of the component (F) to be used is typically at a proportion of 0.5 to 10 parts by mass, with respect to 100 parts by mass of the component (A1).

(754) <<Organic Solvent Component (S)>>

(755) The resist composition according to the present embodiment may be produced by dissolving the resist materials in an organic solvent component (hereinafter, referred to as a “component (S)”).

(756) The component (S) may be any organic solvent which can dissolve the respective components to be used to obtain a homogeneous solution, and any organic solvent can be appropriately selected and used from those which are known in the related art as the solvent for a chemically amplified resist composition.

(757) Examples of the component (S) include lactones such as γ -butyrolactone; ketones such as acetone, methyl ethyl

ketone, cyclohexanone, methyl-n-pentyl ketone, methyl isopentyl ketone, and 2-heptanone; polyhydric alcohols, such as ethylene glycol, diethylene glycol, propylene glycol and dipropylene glycol; compounds having an ester bond, such as ethylene glycol monoacetate, diethylene glycol monoacetate, propylene glycol monoacetate, and dipropylene glycol monoacetate, polyhydric alcohol derivatives such as compounds having an ether bond, such as a monoalkyl ether (such as monomethyl ether, monoethyl ether, monopropyl ether or monobutyl ether) or monophenyl ether of any of these polyhydric alcohols or compounds having an ester bond (among these, propylene glycol monomethyl ether acetate (PGMEA) and propylene glycol monomethyl ether (PGME) are preferable); cyclic ethers such as dioxane; esters such as methyl lactate, ethyl lactate (EL), methyl acetate, ethyl acetate, butyl acetate, methyl pyruvate, ethyl pyruvate, methyl methoxypropionate, and ethyl ethoxypropionate; aromatic organic solvents such as anisole, ethylbenzyl ether, cresylmethyl ether, diphenyl ether, dibenzyl ether, phenetole, butylphenyl ether, ethyl benzene, diethyl benzene, pentyl benzene, isopropyl benzene, toluene, xylene, cymene and mesitylene; and dimethylsulfoxide (DMSO).

(758) In the resist composition according to the present embodiment, the component (S) may be used alone or as a mixed solvent of two or more kinds thereof. Among these, PGMEA, PGME, γ -butyrolactone, EL, and cyclohexanone are preferable.

(759) Further, a mixed solvent obtained by mixing PGMEA with a polar solvent is also preferable as the component (S). The blending ratio (mass ratio) of the mixed solvent can be appropriately determined, taking into consideration the compatibility of the PGMEA with the polar solvent; however, it is preferably in a range of 1:9 to 9:1 and more preferably in a range of 2:8 to 8:2.

(760) More specifically, in a case where EL or cyclohexanone is blended as the polar solvent, the PGMEA:EL or cyclohexanone mass ratio is preferably in a range of 1:9 to 9:1 and more preferably in a range of 2:8 to 8:2.

Alternatively, in a case where PGME is blended as the polar solvent, the PGMEA:PGME mass ratio is preferably in a range of 1:9 to 9:1, more preferably in a range of 2:8 to 8:2, and still more preferably in a range of 3:7 to 7:3.

Furthermore, a mixed solvent of PGMEA, PGME, and cyclohexanone is also preferable.

(761) Further, the component (S) is also preferably a mixed solvent of at least one selected from PGMEA and EL and γ -butyrolactone. In this case, as the mixing ratio, the mass ratio of the former to the latter is preferably in a range of 70:30 to 95:5.

(762) The amount of the component (S) to be used is not particularly limited and is appropriately set, depending on a thickness of a film to be coated, to a concentration at which the component (S) can be applied onto a substrate or the like. Generally, the component (S) is used such that the solid content concentration of the resist composition is in a range of 0.1% to 20% by mass and preferably in a range of 0.2% to 15% by mass.

(763) As desired, other miscible additives can also be added to the resist composition according to the present embodiment. For example, for improving the performance of the resist film, an additive resin, a dissolution inhibitor, a plasticizer, a stabilizer, a colorant, a halation prevention agent, and a dye can be appropriately contained therein.

(764) After dissolving the resist material in the component (S), the resist composition according to the present embodiment may be subjected to removal of impurities and the like by using a porous polyimide film, a porous polyamideimide film, or the like. For example, the resist composition may be filtered using a filter made of a porous polyimide film, a filter made of a porous polyamideimide film, or a filter made of a porous polyimide film and a porous polyamideimide film. Examples of the porous polyimide film and the porous polyamideimide film include those described in Japanese Unexamined Patent Application, First Publication No. 2016-155121.

(765) The resist composition according to the present embodiment described above contains a resin component (A1)

having the above-described constitutional unit (a01) and the compound (D0).

(766) The constitutional unit (a01) has both an acid dissociable group and a phenolic hydroxyl group. For this reason, in a case where the constitutional unit (a01) is contained, the content rate of the phenolic hydroxyl group can be increased without reducing the content rate of the acid dissociable group in the resin component (A1). As a result, the acid generation efficiency is improved, and thus sensitivity can be improved without reducing resolution. Further, the increase of the phenolic hydroxyl group in the resin component (A1) improves the affinity with the developing solution, and thus the fine resolution and the roughness reducing property can be improved.

(767) The compound (D0) has a phenolic hydroxyl group. For this reason, in a case where the compound (D0) is contained, the solubility of the developing solution in exposed portions of the resist film can be improved, and thus the sensitivity and the resolution can be further improved. In addition, since the diffusion of the acid generated upon exposure (for example, the acid generated from the component (B)) can be appropriately suppressed, the pattern shape can be further improved.

(768) Due to the synergistic effect of these combinations, according to the resist composition according to the present embodiment, it is possible to improve all of the sensitivity, the resolution, the roughness reducing property, and the rectangularity of the pattern shape.

(769) (Method of Forming Resist Pattern)

(770) A method of forming a resist pattern according to the second aspect according to the present invention is a method including a step of forming a resist film on a support using the resist composition according to the first aspect of the present invention described above, a step of exposing the resist film, and a step of developing the exposed resist film to form a resist pattern.

(771) Examples of one embodiment of such a method of forming a resist pattern include a method of forming a resist pattern carried out as described below.

(772) First, the resist composition of the above-described embodiment is applied onto a support with a spinner or the like, and a baking (post-apply baking (PAB)) treatment is carried out, for example, at a temperature condition of 80° C. to 150° C. for 40 to 120 seconds, preferably for 60 to 90 seconds to form a resist film.

(773) Following the selective exposure carried out on the resist film by, for example, exposure through a mask (mask pattern) having a predetermined pattern formed on the mask by using an exposure apparatus such as an electron beam lithography apparatus or an EUV exposure apparatus, or direct irradiation of the resist film for drawing with an electron beam without using a mask pattern, baking treatment (post-exposure baking (PEB)) is carried out, for example, under a temperature condition in a range of 80° C. to 150° C. for 40 to 120 seconds and preferably 60 to 90 seconds.

(774) Next, the resist film is subjected to a developing treatment. The developing treatment is carried out using an alkali developing solution in a case of an alkali developing process, and a developing solution containing an organic solvent (organic developing solution) in a case of a solvent developing process.

(775) After the developing treatment, it is preferable to carry out a rinse treatment. As the rinse treatment, water rinsing using pure water is preferable in a case of an alkali developing process, and rinsing using a rinse liquid containing an organic solvent is preferable in a case of a solvent developing process.

(776) In a case of a solvent developing process, after the developing treatment or the rinsing, the developing solution or the rinse liquid remaining on the pattern may be removed by a treatment using a supercritical fluid.

(777) After the developing treatment or the rinse treatment, drying is carried out. As desired, baking treatment (post-

baking) may be carried out following the developing treatment.

(778) In this manner, a resist pattern can be formed.

(779) The support is not particularly limited, and a known one in the related art can be used. For example, a substrate for an electronic component, and such a substrate having a predetermined wiring pattern formed thereon can be used. Specific examples of the material of the substrate include metals such as silicon wafer, copper, chromium, iron and aluminum; and glass. Suitable materials for the wiring pattern include copper, aluminum, nickel, and gold.

(780) Further, as the support, any support having the substrate described above, on which an inorganic and/or organic film is provided, may be used. As the inorganic film, an inorganic antireflection film (inorganic BARC) can be used. Examples of the organic film include an organic antireflection film (organic BARC) and an organic film such as a lower-layer organic film used in a multilayer resist method.

(781) Here, the multilayer resist method is a method in which at least one layer of an organic film (lower-layer organic film) and at least one layer of a resist film (upper-layer resist film) are provided on a substrate, and a resist pattern formed on the upper-layer resist film is used as a mask to carry out patterning of the lower-layer organic film.

(782) This method is considered as being capable of forming a pattern with a high aspect ratio. More specifically, in the multilayer resist method, a desired thickness can be ensured by the lower-layer organic film, and as a result, the thickness of the resist film can be reduced, and an extremely fine pattern with a high aspect ratio can be formed.

(783) The multilayer resist method is classified into a method in which a double-layer structure consisting of an upper-layer resist film and a lower-layer organic film is formed (double-layer resist method), and a method in which a multilayer structure having three or more layers consisting of an upper-layer resist film, a lower-layer organic film and one or more intermediate layers (thin metal films or the like) provided between the upper-layer resist film and the lower-layer organic film (triple-layer resist method).

(784) The wavelength to be used for exposure is not particularly limited and the exposure can be carried out using radiation such as an ArF excimer laser, a KrF excimer laser, an F.sub.2 excimer laser, an extreme ultraviolet ray (EUV), a vacuum ultraviolet ray (VUV), an electron beam (EB), an X-ray, or a soft X-ray. The resist composition is highly useful for a KrF excimer laser, an ArF excimer laser, EB, or EUV, more useful for an ArF excimer laser, EB or EUV, and particularly useful for EB or EUV. That is, the method of forming a resist pattern according to the present embodiment is a method particularly useful in a case where the step of exposing the resist film includes an operation of exposing the resist film to an extreme ultraviolet ray (EUV) or an electron beam (EB).

(785) The exposure of the resist film may be a general exposure (dry exposure) carried out in air or an inert gas such as nitrogen, or liquid immersion exposure (liquid immersion lithography).

(786) The liquid immersion lithography is an exposure method in which the region between the resist film and the lens at the lowermost position of the exposure apparatus is pre-filled with a solvent (liquid immersion medium) that has a larger refractive index than the refractive index of air, and the exposure (immersion exposure) is carried out in this state.

(787) The liquid immersion medium is preferably a solvent that exhibits a refractive index larger than the refractive index of air but smaller than the refractive index of the resist film to be exposed. The refractive index of such a solvent is not particularly limited as long as it satisfies the above-described requirements.

(788) Examples of the solvent which exhibits a refractive index that is larger than the refractive index of air but smaller than the refractive index of the resist film include water, fluorine-based inert liquids, silicone-based solvents, and hydrocarbon-based solvents.

(789) Specific examples of the fluorine-based inert liquids include liquids containing a fluorine-based compound such as C.sub.3HCl.sub.2F.sub.5, C.sub.4F.sub.9OCH.sub.3, C.sub.4F.sub.9OC.sub.2H.sub.5, or C.sub.5H.sub.3F.sub.7 as the main component, and the boiling point is preferably in a range of 70° to 180° C. and more preferably in a range of 80° to 160° C. A fluorine-based inert liquid having a boiling point in the above-described range is advantageous in that removing the medium used in the liquid immersion after the exposure can be carried out by a simple method.

(790) A fluorine-based inert liquid is particularly preferably a perfluoroalkyl compound obtained by substituting all hydrogen atoms of the alkyl group with a fluorine atom. Examples of the perfluoroalkyl compound include a perfluoroalkyl ether compound and a perfluoroalkylamine compound.

(791) Further, specific examples of the perfluoroalkyl ether compound include perfluoro(2-butyl-tetrahydrofuran) (boiling point: 102° C.), and examples of the perfluoroalkylamine compound include perfluorotributylamine (boiling point: 174° C.).

(792) As the liquid immersion medium, water is preferable in terms of cost, safety, environment, and versatility.

(793) Examples of the alkali developing solution used for a developing treatment in an alkali developing process include an aqueous solution of 0.1 to 10% by mass of tetramethylammonium hydroxide (TMAH).

(794) The organic solvent contained in the organic developing solution, which is used for a developing treatment in a solvent developing process may be any organic solvent capable of dissolving the component (A) (component (A) prior to exposure), and can be appropriately selected from the conventionally known organic solvents. Specific examples of the organic solvent include polar solvents such as a ketone-based solvent, an ester-based solvent, an alcohol-based solvent, a nitrile-based solvent, an amide-based solvent, and an ether-based solvent, and hydrocarbon-based solvents.

(795) A ketone-based solvent is an organic solvent containing C—C(=O)—C in the structure thereof. An ester-based solvent is an organic solvent containing C—C(=O)—O—C in the structure thereof. An alcohol-based solvent is an organic solvent containing an alcoholic hydroxyl group in the structure thereof. The “alcoholic hydroxyl group” indicates a hydroxyl group bonded to a carbon atom of an aliphatic hydrocarbon group. A nitrile-based solvent is an organic solvent containing a nitrile group in the structure thereof. An amide-based solvent is an organic solvent containing an amide group in the structure thereof. An ether-based solvent is an organic solvent containing C—O—C in the structure thereof.

(796) Some organic solvents have a plurality of the functional groups which characterize the above-described solvents in the structure thereof.

(797) In such a case, the organic solvent can be classified as any type of solvent having a characteristic functional group. For example, diethylene glycol monomethyl ether can be classified as an alcohol-based solvent or an ether-based solvent.

(798) A hydrocarbon-based solvent consists of a hydrocarbon which may be halogenated and does not have any substituent other than the halogen atom. The halogen atom is preferably a fluorine atom.

(799) Among the above, the organic solvent contained in the organic developing solution is preferably a polar solvent and more preferably a ketone-based solvent, an ester-based solvent, or a nitrile-based solvent.

(800) Examples of ketone-based solvent include 1-octanone, 2-octanone, 1-nonanone, 2-nonanone, acetone, 4-heptanone, 1-hexanone, 2-hexanone, diisobutyl ketone, cyclohexanone, methylcyclohexanone, phenylacetone, methyl ethyl ketone, methyl isobutyl ketone, acetylacetone, acetonylacetone, ionone, diacetyl alcohol, acetylcarbinol, acetophenone, methyl naphthyl ketone, isophorone, propylenecarbonate, γ -butyrolactone and methylamyl ketone (2-heptanone). Among these examples, the ketone-based solvent is preferably methylamyl ketone (2-heptanone).

(801) Examples of the ester-based solvent include methyl acetate, butyl acetate, ethyl acetate, isopropyl acetate, amyl acetate, isoamyl acetate, ethyl methoxyacetate, ethyl ethoxyacetate, ethylene glycol monoethyl ether acetate, ethylene glycol monopropyl ether acetate, ethylene glycol monobutyl ether acetate, ethylene glycol monophenyl ether acetate, diethylene glycol monomethyl ether acetate, diethylene glycol monopropyl ether acetate, diethylene glycol monophenyl ether acetate, diethylene glycol monobutyl ether acetate, diethylene glycol monoethyl ether acetate, 2-methoxybutyl acetate, 3-methoxybutyl acetate, 4-methoxybutyl acetate, 3-methyl-3-methoxybutyl acetate, 3-ethyl-3-methoxybutyl acetate, propylene glycol monomethyl ether acetate, propylene glycol monoethyl ether acetate, propylene glycol monopropyl ether acetate, 2-ethoxybutyl acetate, 4-ethoxybutyl acetate, 4-propoxybutyl acetate, 2-methoxypentyl acetate, 3-methoxypentyl acetate, 4-methoxypentyl acetate, 2-methyl-3-methoxypentyl acetate, 3-methyl-3-methoxypentyl acetate, 3-methyl-4-methoxypentyl acetate, 4-methyl-4-methoxypentyl acetate, propylene glycol diacetate, methyl formate, ethyl formate, butyl formate, propyl formate, ethyl lactate, butyl lactate, propyl lactate, ethyl carbonate, propyl carbonate, butyl carbonate, methyl pyruvate, ethyl pyruvate, propyl pyruvate, butyl pyruvate, methyl acetoacetate, ethyl acetoacetate, methyl propionate, ethyl propionate, propyl propionate, isopropyl propionate, methyl 2-hydroxypropionate, ethyl 2-hydroxypropionate, methyl-3-methoxypropionate, ethyl-3-methoxypropionate, ethyl-3-ethoxypropionate, and propyl-3-methoxypropionate. Among these, the ester-based solvent is preferably butyl acetate.

(802) Examples of the nitrile-based solvent include acetonitrile, propionitrile, valeronitrile, and butyronitrile.

(803) As desired, the organic developing solution may have a conventionally known additive blended. Examples of the additive include surfactants. The surfactant is not particularly limited, and for example, an ionic or non-ionic fluorine-based and/or a silicone-based surfactant can be used. The surfactant is preferably a non-ionic surfactant and more preferably a non-ionic fluorine surfactant or a non-ionic silicone-based surfactant.

(804) In a case where a surfactant is blended, the amount of the surfactant to be blended is typically in a range of 0.001% to 5% by mass, preferably in a range of 0.005% to 2% by mass, and more preferably in a range of 0.01% to 0.5% by mass with respect to the total amount of the organic developing solution.

(805) The developing treatment can be carried out by a conventionally known developing method. Examples thereof include a method in which the support is immersed in the developing solution for a predetermined time (a dip method), a method in which the developing solution is cast upon the surface of the support by surface tension and maintained for a predetermined time (a puddle method), a method in which the developing solution is sprayed onto the surface of the support (spray method), and a method in which a developing solution is continuously ejected from a developing solution ejecting nozzle and applied to a support which is scanned at a constant rate while being rotated at a constant rate (dynamic dispense method).

(806) As the organic solvent contained in the rinse liquid used in the rinse treatment after the developing treatment in a case of a solvent developing process, an organic solvent hardly dissolving the resist pattern can be appropriately selected and used, among the organic solvents mentioned as organic solvents that are used for the organic developing solution. In general, at least one kind of solvent selected from the group consisting of a hydrocarbon-based solvent, a ketone-based solvent, an ester-based solvent, an alcohol-based solvent, an amide-based solvent, and an ether-based solvent is used. Among these, at least one kind of solvent selected from the group consisting of a hydrocarbon-based solvent, a ketone-based solvent, an ester-based solvent, an alcohol-based solvent, and an amide-based solvent is preferable, at least one kind of solvent selected from the group consisting of an alcohol-based solvent and an ester-based solvent is more preferable, and an alcohol-based solvent is particularly preferable.

(807) The alcohol-based solvent used for the rinse liquid is preferably a monohydric alcohol of 6 to 8 carbon atoms, and the monohydric alcohol may be linear, branched, or cyclic. Specific examples thereof include 1-hexanol, 1-heptanol, 1-octanol, 2-hexanol, 2-heptanol, 2-octanol, 3-hexanol, 3-heptanol, 3-octanol, 4-octanol, and benzyl alcohol. Among these, 1-hexanol, 2-heptanol, and 2-hexanol are preferable, and 1-hexanol and 2-hexanol are more preferable.

(808) As the organic solvent, one kind of solvent may be used alone, or two or more kinds of solvents may be used in combination. Further, an organic solvent other than the above-described examples or water may be mixed thereto. However, in consideration of the development characteristics, the amount of water to be blended in the rinse liquid is preferably 30% by mass or less, more preferably 10% by mass or less, still more preferably 5% by mass or less, and particularly preferably 3% by mass or less with respect to the total amount of the rinse liquid.

(809) A conventionally known additive can be blended with the rinse liquid as necessary. Examples of the additive include surfactants. Examples of the surfactant include the same ones as those described above, the surfactant is preferably a non-ionic surfactant and more preferably a non-ionic fluorine surfactant or a non-ionic silicone-based surfactant.

(810) In a case where a surfactant is blended, the amount of the surfactant to be blended is typically in a range of 0.001% to 5% by mass, preferably in a range of 0.005% to 2% by mass, and more preferably in a range of 0.01% to 0.5% by mass with respect to the total amount of the rinse liquid.

(811) The rinse treatment (the washing treatment) using a rinse liquid can be carried out by a conventionally known rinse method. Examples of the rinse treatment method include a method (a rotational coating method) in which the rinse liquid is continuously applied to the support while rotating it at a constant rate, a method (dip method) in which the support is immersed in the rinse liquid for a predetermined time, and a method (spray method) in which the rinse liquid is sprayed onto the surface of the support.

(812) According to the method of forming a resist pattern according to the present embodiment described above, since the resist composition according to the embodiment described above is used, it is possible to form a resist pattern that has good sensitivity, good resolution, good roughness reducing property, and high rectangularity.

EXAMPLES

(813) Hereinafter, the present invention will be described in more detail based on Examples, but the present invention is not limited to these Examples.

Synthesis Example 1

Synthesis Example of Polymeric Compound (A1-1)

(814) 5.9 g of a monomer (a01-1m-01) and 0.5 g of azobis (isobutyric acid) dimethyl (V-601) as a polymerization initiator were dissolved in 30 g of methyl ethyl ketone (MEK), and the mixture was heated to 70° C. in a nitrogen atmosphere and stirred for 5 hours. Then, the reaction solution was cooled to room temperature.

(815) After completion of the reaction, the obtained reaction solution was precipitated in 300 g of methanol, followed by washing. The obtained white solid substance was filtered and dried under reduced pressure overnight to obtain the target polymeric compound (A1-1).

(816) ##STR00119##

(817) As a result of a GPC measurement to determine the weight average molecular weight (M_w) in terms of polystyrene equivalent value, the obtained polymeric compound (A1-1) had a weight average molecular weight of 6,600 and a polydispersity (M_w/M_n) of 1.56.

Synthesis Examples 2 to 25 of Polymeric Compounds

(818) Polymeric compounds (A1-2) to (A1-22) and (A2-1) to (A2-3), having compositional ratios shown in Table 1, were synthesized in the same manner as in Synthesis Example 1 of polymeric compound by using compounds (a01-1m-02) to (a01-1m-12) shown below and compounds each represented by General Formulae (a10-1-1 pre) and (a1-1m) to (a1-8m) shown below.

(819) Regarding the obtained polymeric compounds, the copolymerization compositional ratio (the ratio (molar ratio) of the constitutional unit derived from each monomer) of the polymeric compound, which was determined by ¹³C-NMR, the weight average molecular weight (M_w) in terms of polystyrene equivalent value, which was determined by GPC measurement, and the polydispersity (M_w/M_n) are shown together in Table 1.

(820) ##STR00120## ##STR00121## ##STR00122## ##STR00123##

(821) Here, the constitutional unit represented by Chemical Formula (a10-1-1) constituting the above-described copolymer is a constitutional unit derived from the monomer represented by Chemical Formula (a10-1-1 pre).

(822) ##STR00124##

(823) TABLE-US-00001 TABLE 1 Copolymerization compositional Weight-average Poly- Polymeric ratio in polymeric compound molecular dispersity compound (molar ratio) (M_w) (M_w/M_n) Synthesis Example 1 (A1-1) 1 (a01-1m-01) = 100 6600 1.56 Synthesis Example 2 (A1-2) 1 (a01-1m-02) = 100 6400 1.57 Synthesis Example 3 (A1-3) 1 (a01-1m-02)/m (a1-1m) = 50/50 6900 1.66 Synthesis Example 4 (A1-4) 1 (a01-1m-01)/m (a1-2m) = 50/50 6700 1.63 Synthesis Example 5 (A1-5) 1 (a01-1m-02)/m (a1-3m) = 50/50 7000 1.60 Synthesis Example 6 (A1-6) 1 (a01-1m-01)/m (a1-4m) = 50/50 6800 1.67 Synthesis Example 7 (A1-7) 1 (a01-1m-02)/m (a1-5m) = 50/50 6700 1.64 Synthesis Example 8 (A1-8) 1 (a01-1m-01)/m (a1-6m) = 50/50 6600 1.63 Synthesis Example 9 (A1-9) 1 (a01-1m-03)/m (a1-1m) = 50/50 6400 1.62 Synthesis Example 10 (A1-10) 1 (a01-1m-01)/m (a1-1m) = 50/50 6500 1.64 Synthesis Example 11 (A1-11) 1 (a01-1m-04)/m (a1-1m) = 50/50 6800 1.66 Synthesis Example 12 (A1-12) 1 (a01-1m-05)/m (a1-1m) = 50/50 6900 1.61 Synthesis Example 13 (A1-13) 1 (a01-1m-06)/m (a1-1m) = 50/50 6500 1.63 Synthesis Example 14 (A1-14) 1 (a01-1m-07)/m (a1-1m) = 50/50 6800 1.67 Synthesis Example 15 (A1-15) 1 (a01-1m-08)/m (a1-1m) = 50/50 7100 1.65 Synthesis Example 16 (A1-16) 1 (a01-1m-09)/m (a1-1m) = 50/50 6900 1.60 Synthesis Example 17 (A1-17) 1 (a01-1m-10)/m (a1-1m) = 50/50 6800 1.62 Synthesis Example 18 (A1-18) 1 (a01-1m-11)/m (a1-1m) = 50/50 7200 1.68 Synthesis Example 19 (A1-19) 1 (a01-1m-12)/m (a1-1m) = 50/50 6900 1.69 Synthesis Example 20 (A1-20) 1 (a10-1-1pre)/m (a01-1m-01) = 50/50 6600 1.59 Synthesis Example 21 (A1-21) 1 (a10-1-1pre)/m (a01-1m-02) = 50/50 6800 1.58 Synthesis Example 22 (A1-22) 1 (a10-1-1pre)/m (a01-1m-02)/ 6800 1.71 n (a1-1m) = 50/30/20 Synthesis Example 23 (A2-1) 1 (a10-1-1pre)/m (a1-1m) = 50/50 6600 1.66 Synthesis Example 24 (A2-2) 1 (a1-7m)/m (a1-1m) = 50/50 6500 1.63 Synthesis Example 25 (A2-3) 1 (a1-8m)/m (a1-1m) = 50/50 6600 1.61

<Preparation of Resist Composition>

Examples 1 to 31 and Comparative Examples 1 to 7

(824) Each of the components shown in Tables 2 to 5 was mixed and dissolved to prepare a resist composition of each Example.

(825) TABLE-US-00002 TABLE 2 Component Component Compound Component (A) (B) (D0) (S) Example 1 (A1)-1 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 2 (A1)-2 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 3 (A1)-3 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 4 (A1)-4 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 5 (A1)-5 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 6 (A1)-6 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 7 (A1)-7 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 8 (A1)-8 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 9 (A1)-9 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 10 (A1)-10 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 11

(A1)-11 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 12 (A1)-12 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000]

Example 13 (A1)-13 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000]

(826) TABLE-US-00003 TABLE 3 Component Component Compound Component (A) (B) (D0) (S) Example 14

(A1)-14 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 15 (A1)-15 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000]

Example 16 (A1)-16 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 17 (A1)-17 (B)-1 (D0)-1 (S)-1 [100] [15] [5]

[6000] Example 18 (A1)-18 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 19 (A1)-19 (B)-1 (D0)-1 (S)-1 [100]

[15] [5] [6000] Example 20 (A1)-20 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000] Example 21 (A1)-21 (B)-1 (D0)-1 (S)-1

[100] [15] [5] [6000] Example 22 (A1)-22 (B)-1 (D0)-1 (S)-1 [100] [15] [5] [6000]

(827) TABLE-US-00004 TABLE 4 Component Component Compound Component (A) (B) (D0) (S) Comparative

(A2)-1 (B)-1 (D0)-1 (S)-1 Example 1 [100] [15] [5] [6000] Comparative (A2)-2 (B)-1 (D0)-1 (S)-1 Example 2 [100]

[15] [5] [6000] Comparative (A2)-3 (B)-1 (D0)-1 (S)-1 Example 3 [100] [15] [5] [6000]

(828) TABLE-US-00005 TABLE 5 Com- Com- Compound Com- ponent ponent (D0) or ponent (A) (B) Component

(D) (S) Example 23 (A1)-3 (B)-1 (D0)-2 (S)-1 [100] [15] [5] [6000] Example 24 (A1)-3 (B)-1 (D0)-3 (S)-1 [100] [15]

[5] [6000] Example 25 (A1)-3 (B)-1 (D0)-4 (S)-1 [100] [15] [5] [6000] Example 26 (A1)-3 (B)-1 (D0)-5 (S)-1 [100]

[15] [5] [6000] Example 27 (A1)-3 (B)-1 (D0)-6 (S)-1 [100] [15] [5] [6000] Example 28 (A1)-3 (B)-1 (D0)-7 (S)-1

[100] [15] [5] [6000] Example 29 (A1)-3 (B)-1 (D0)-8 (S)-1 [100] [15] [5] [6000] Example 30 (A1)-3 (B)-1 (D0)-9

(S)-1 [100] [15] [5] [6000] Example 31 (A1)-3 (B)-1 (D0)-10 (S)-1 [100] [15] [5] [6000] Comparative (A1)-3 (B)-1

(D)-1 (S)-1 Example 4 [100] [15] [5] [6000] Comparative (A2)-1 (B)-1 (D)-1 (S)-1 Example 5 [100] [15] [5] [6000]

Comparative (A2)-2 (B)-1 (D)-1 (S)-1 Example 6 [100] [15] [5] [6000] Comparative (A2)-3 (B)-1 (D)-1 (S)-1

Example 7 [100] [15] [5] [6000]

(829) In Tables 2 to 5, each abbreviation has the following meaning. The numerical values in the brackets are blending amounts (parts by mass).

(830) (A1)-1 to (A1)-22: the polymeric compounds (A1-1) to (A1-22).

(831) (A2)-1 to (A2)-3: the polymeric compounds (A2-1) to (A2-3).

(832) (B)-1: an acid generator consisting of a compound represented by Chemical Formula (B-1).

(833) ##STR00125##

(834) (D0)-1 to (D0)-10: acid diffusion controlling agents composed of compounds each represented by Chemical Formulae (D0-1) to (D0-10).

(835) (D)-1: an acid diffusion controlling agent consisting of a compound represented by Chemical Formula (D-1).

(836) (S)-1: a mixed solvent of propylene glycol monomethyl ether acetate/propylene glycol monomethyl ether=60/40 (mass ratio)

(837) ##STR00126##

<Formation of Resist Pattern>

(838) The resist composition of each Example was applied onto an 8-inch silicon substrate which had been subjected to a hexamethyldisilazane (HMDS) treatment using a spinner, the coated wafer was subjected to a post-apply baking (PAB) treatment on a hot plate at a temperature of 110° C. for 60 seconds so that the coated wafer was dried to form a resist film having a film thickness of 30 nm.

(839) Next, drawing (exposure) was carried out on the resist film by using an electron beam lithography apparatus JEOL-JBX-9300FS (manufactured by JEOL Ltd.), with the target size being set to a 1:1 line and space pattern (hereinafter, referred to as an “LS pattern”) of a line width of 50 nm, at an acceleration voltage of 100 kV. Thereafter, a

post-exposure baking (PEB) treatment was carried out on the resist film at 90° C. for 60 seconds.

(840) Subsequently, alkali development was carried out at 23° C. for 60 seconds using a 2.38% by mass tetramethylammonium hydroxide (TMAH) aqueous solution “NMD-3” (product name, manufactured by TOKYO OHKA KOGYO CO., LTD.).

(841) Thereafter, rinsing was carried out with pure water for 15 seconds.

(842) As a result of the above, a 1:1 LS pattern having a line width of 50 nm was formed.

(843) [Evaluation of Optimum Exposure Amount (Eop)]

(844) According to <Formation of resist pattern> described above, an optimum exposure amount Eop ($\mu\text{C}/\text{cm}^2$) for forming the LS pattern having the target size was determined. The results are shown in Tables 6 to 8 as “Eop ($\mu\text{C}/\text{cm}^2$)”.

(845) [Evaluation of Linewise Roughness (LWR)]

(846) 3σ of the LS pattern formed in <Formation of resist pattern> described above, which is a scale indicating LWR, was determined. The results are shown in Tables 6 to 8 as “LWR (nm)”.

(847) “ 3σ ” indicates a triple value (unit: nm) of the standard deviation (σ) determined from measurement results obtained by measuring 400 line positions in the longitudinal direction of the line with a scanning electron microscope (acceleration voltage: 800V, product name: S-9380, manufactured by Hitachi High-Tech Corporation)

(848) The smaller the value of 3σ is, the smaller the roughness in the line side wall is, which means an LS pattern having a more uniform width was obtained.

(849) [Evaluation of Limiting Resolution]

(850) LS patterns were formed by gradually increasing the exposure amount from the optimum exposure amount Eop described above, and the limiting resolution at the above Eop, specifically, the minimum size of the pattern that was resolved without being collapsed were determined using a scanning electron microscope S-9380 (manufactured by Hitachi High-Tech Corporation). The results are shown in Tables 6 to 8 as “Limiting resolution (nm)”.

(851) [Evaluation of Pattern Shape]

(852) The cross-sectional shape of each of the obtained LS patterns in <Formation of resist pattern> described above was observed using a scanning electron microscope (acceleration voltage: 800V, product name: SU-8000, manufactured by Hitachi High-Tech Corporation). The shape thereof was evaluated according to the following evaluation criteria. The obtained results are shown in Tables 6 to 8 as “Pattern shape”.

(853) A: The cross-sectional shape of the pattern is rectangular and has high verticality.

(854) B: The verticality of the cross-sectional shape of the pattern is slightly inferior to that of A.

(855) C: The cross-sectional shape of the pattern is top rounded (the top of the pattern is round) or T-top shaped.

(856) TABLE-US-00006 TABLE 6 Limiting PAB PEB Eop LWR resolution Pattern (° C.) (° C.) [$\mu\text{C}/\text{cm}^2$] [nm] [nm] shape Example 1 110 90 80 5.2 28 B Example 2 110 90 84 5.0 26 A Example 3 110 90 90 4.6 26 A Example 4 110 90 94 4.6 26 A Example 5 110 90 88 4.6 26 A Example 6 110 90 92 4.7 28 B Example 7 110 90 88 4.7 26 A Example 8 110 90 94 4.6 30 B Example 9 110 90 86 4.5 26 A Example 10 110 90 83 4.7 26 A Example 11 110 90 90 4.7 28 B Example 12 110 90 88 4.5 26 A Example 13 110 90 84 4.6 26 A Example 14 110 90 90 4.8 28 B Example 15 110 90 92 4.8 28 B Example 16 110 90 86 4.5 26 A Example 17 110 90 83 4.6 26 A Example 18 110 90 82 4.8 26 A Example 19 110 90 91 4.7 30 B Example 20 110 90 82 4.9 28 B Example 21 110 90 84 4.8 26 A Example 22 110 90 86 4.7 26 A

(857) TABLE-US-00007 TABLE 7 Limiting PAB PEB Eop LWR resolution Pattern (° C.) (° C.) [$\mu\text{C}/\text{cm}^2$] [nm]

saturated hydrocarbon group having 3 to 20 carbon atoms, part or all hydrogen atoms contained in the chain saturated hydrocarbon group and the aliphatic cyclic saturated hydrocarbon group may be substituted, two or more of Ra.sup.002 to Ra.sup.004 may be bonded to each other to form a cyclic structure, * represents a bonding site, in General Formula (a0-r2-03), Yac.sup.0 represents a carbon atom, Xac.sup.0 represents a group that forms a cyclic hydrocarbon group together with Yac.sup.0, part or all hydrogen atoms contained in this cyclic hydrocarbon group may be substituted, and part of carbon atoms constituting a ring may be substituted a hetero atom, Ra.sup.005 represents an aromatic hydrocarbon group, part or all hydrogen atoms contained in the aromatic hydrocarbon group may be substituted, provided that the aromatic hydrocarbon group excludes an aromatic group in which part of carbon atoms constituting a ring thereof is substituted with a hetero atom, * represents a bonding site, in General Formula (a0-r2-04), Ra.sup.006 and Ra.sup.007 each independently represent a monovalent chain saturated hydrocarbon group having 1 to 10 carbon atoms, part or all hydrogen atoms contained in the chain saturated hydrocarbon group may be substituted, Ra.sup.008 represents a hydrocarbon group which may have a substituent, * represents a bonding site.

2. The resist composition according to claim 1, wherein the constitutional unit (a01) is a constitutional unit represented by General Formula (a0-1-1): ##STR00130## wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, Ya.sup.001 represents a single bond or a divalent linking group, Ya.sup.01 represents a single bond or a divalent linking group, Rax.sup.01 represents an acid dissociable group represented by General Formula (a0-r2-01), General Formula (a0-r2-02), General Formula (a0-r2-03), or General Formula (a0-r2-04), q represents an integer in a range of 0 to 3 and n represents an integer of 1 or more provided that $n \leq a \times 2 + 4$ is satisfied; ##STR00131## wherein, in General Formula, (a0-r2-01), Ra.sup.001 represents a linear or branched alkyl group which may have a substituent, Yaa.sup.0 represents a carbon atom, Xaa.sup.0 represents a group that forms a cyclic hydrocarbon group together with Yaa.sup.0, part or all hydrogen atoms contained in this cyclic hydrocarbon group may be substituted, and part of carbon atoms constituting a ring may be substituted a hetero atom, provided that the cyclic hydrocarbon group excludes a fused cyclic hydrocarbon group formed of an aliphatic hydrocarbon group and an aromatic heterocyclic group containing an oxygen atom or a sulfur atom, * represents a bonding site, in General Formula (a0-r2-02), Yab.sup.0 represents a carbon atom, Xab.sup.0 represents a group that forms a cyclic hydrocarbon group together with Yab.sup.0, part or all hydrogen atoms contained in this cyclic hydrocarbon group may be substituted, and part of carbon atoms constituting a ring may be substituted a hetero atom, Ra.sup.002 to Ra.sup.004 each independently represent a hydrogen atom, a monovalent chain saturated hydrocarbon group having 1 to 10 carbon atoms, or a monovalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms, part or all hydrogen atoms contained in the chain saturated hydrocarbon group and the aliphatic cyclic saturated hydrocarbon group may be substituted, two or more of Ra.sup.002 to Ra.sup.004 may be bonded to each other to form a cyclic structure, * represents a bonding site, in General Formula (a0-r2-03), Yac.sup.0 represents a carbon atom, Xac.sup.0 represents a group that forms a cyclic hydrocarbon group together with Yac.sup.0, part or all hydrogen atoms contained in this cyclic hydrocarbon group may be substituted, and part of carbon atoms constituting a ring may be substituted a hetero atom, Ra.sup.005 represents an aromatic hydrocarbon group, part or all hydrogen atoms contained in the aromatic hydrocarbon group may be substituted, provided that the aromatic hydrocarbon group excludes an aromatic group in which part of carbon atoms constituting a ring thereof is substituted with a hetero atom, * represents a bonding site, in General Formula (a0-r2-04), Ra.sup.006 and Ra.sup.007 each independently represent a monovalent chain saturated hydrocarbon group having 1 to 10 carbon atoms, part or all hydrogen atoms contained in the chain saturated hydrocarbon group may be substituted, Ra.sup.008 represents a hydrocarbon group which may have a substituent, * represents a bonding site.

3. The resist composition according to claim 1, wherein the compound (D0) is a compound represented by General Formula (d0-1): ##STR00132## wherein R.sup.d1 represents an aryl group having a fluorine atom or an aryl group having a fluorinated alkyl group, R.sup.d2 and R.sup.d3 each independently represent an aryl group which may have a substituent or are bonded to each other to form a ring together with the sulfur atom in the formula, R.sup.d0 represents a substituent, p.sub.0 represents an integer in a range of 0 to 3, in a case where p.sub.0 represents 2 or 3, a plurality of R.sup.d0s may be the same or different from each other, q.sub.0 represents an integer in a range of 0 to 3, and n represents an integer of 1 or more, provided that $n_{\text{sub.0}} + p_{\text{sub.0}} (q_{\text{sub.0}} \times 2) + 5$ is satisfied.

4. The resist composition according to claim 1, wherein a proportion of the constitutional unit (a01) in the resin component (A1) is 20% by mole or more based on 100% by mole of all constitutional units constituting the resin component (A1).

5. The resist composition according to claim 1, wherein a content of the compound (D.sub.0) is in a range of 3 to 50 parts by mass with respect to 100 parts by mass of the resin component (A1).

6. A method of forming a resist pattern, comprising: forming a resist film on a support using the resist composition according to claim 1; exposing the resist film; and developing the exposed resist film to form a resist pattern.

7. The method of forming a resist pattern according to claim 6, wherein the resist film is exposed with an extreme ultraviolet ray (EUV) or an electron beam (EB).

8. A resist composition which generates acid upon exposure and exhibits changed solubility in a developing solution under action of acid, the resist composition comprising: a resin component (A1) which exhibits changed solubility in a developing solution under action of acid; and a compound (D.sub.0) represented by General Formula (d), wherein the resin component (A1) has a constitutional unit (a01) derived from a compound represented by General Formula (a0-1):

##STR00133## wherein M.sup.m+ represents an m-valent organic cation, m represents an integer of 1 or more, R.sup.d0 represents a substituent, p.sub.0 represents an integer in a range of 0 to 3, in a case where p.sub.0 represents 2 or 3, a plurality of R.sup.d0's may be the same or different from each other, q.sub.0 represents an integer in a range of 0 to 3, and n represents an integer of 1 or more, provided that $n_{\text{sub.0}} + p_{\text{sub.0}} \leq (q_{\text{sub.0}} \times 2) + 5$ is satisfied; ##STR00134## wherein W.sup.01 represents a polymerizable group-containing group, Ya.sup.01 represents a single bond or a divalent linking group, Rx.sup.01 represents an acid dissociable group represented by General Formula (a0-r2-01), General Formula (a0-r2-02), General Formula (a0-r2-03), or General Formula (a0-r2-04), q represents an integer in a range of 0 to 3, and n represents an integer of 1 or more, provided that $n \leq q \times 2 + 4$ is satisfied; ##STR00135## wherein, in General Formula, (a0-r2-01), Ra.sup.001 represents a linear or branched alkyl group which may have a substituent, Yaa.sup.0 represents a carbon atom, Xaa.sup.0 represents a group that forms an unsubstituted cyclic aliphatic hydrocarbon group together with * represents a bonding site, in General Formula (a0-r2-02), Yab.sup.0 represents a carbon atom, Xab.sup.0 represents a group that forms a cyclic hydrocarbon group together with Yab.sup.0, part or all hydrogen atoms contained in this cyclic hydrocarbon group may be substituted, and part of carbon atoms constituting a ring may be substituted a hetero atom, Ra.sup.002 to Ra.sup.00 each independently represent a hydrogen atom, a monovalent chain saturated hydrocarbon group having 1 to 10 carbon atoms, or a monovalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms, part or all hydrogen atoms contained in the chain saturated hydrocarbon group and the aliphatic cyclic saturated hydrocarbon group may be substituted, two or more of Ra.sup.002 to Ra.sup.004 may be bonded to each other to form a cyclic structure, * represents a bonding site, in General Formula (a0-r2-03), Yac.sup.0 represents a carbon atom, Xax.sup.0 represents a group that forms an unsubstituted cyclic aliphatic hydrocarbon group together with Yac.sup.0, Ra.sup.005 represents an aromatic hydrocarbon group, part or all hydrogen atoms contained in the aromatic hydrocarbon group may be substituted, and part of carbon atoms constituting a ring may be substituted with a hetero atom, * represents a bonding site,

in General Formula (a0-r2-04), Ra.sup.006 and Ra.sup.007 each independently represent a monovalent chain saturated hydrocarbon group having 1 to 10 carbon atoms, part or all hydrogen atoms contained in the chain saturated hydrocarbon group may be substituted, Ra.sup.008 represents a hydrocarbon group which may have a substituent, * represents a bonding site.
